

Downstream Processing Technologies/ Capturing and Final Purification



Opportunities for Innovation, Change, and Improvement. A Review of Downstream Processing Developments in Protein Purification

Nripen Singh and Sibylle Herzer

Abstract Increased pressure on upstream processes to maximize productivity has been crowned with great success, although at the cost of shifting the bottleneck to purification. As drivers were economical, focus is on now on debottlenecking downstream processes as the main drivers of high manufacturing cost. Devising a holistically efficient and economical process remains a key challenge. Traditional and emerging protein purification strategies with particular emphasis on methodologies implemented for the production of recombinant proteins of biopharmaceutical importance are reviewed. The breadth of innovation is addressed, as well as the challenges the industry faces today, with an eye to remaining impartial, fair, and balanced. In addition, the scope encompasses both chromatographic and non-chromatographic separations directed at the purification of proteins, with a strong emphasis on antibodies. Complete solutions such as integrated USP/DSP strategies (i.e., continuous processing) are discussed as well as gains in data quantity and quality arising from automation and high-throughput screening (HTS). Best practices and advantages through design of experiments (DOE) to access a complex design space such as multi-modal chromatography are reviewed with an outlook on potential future trends. A discussion of single-use technology,

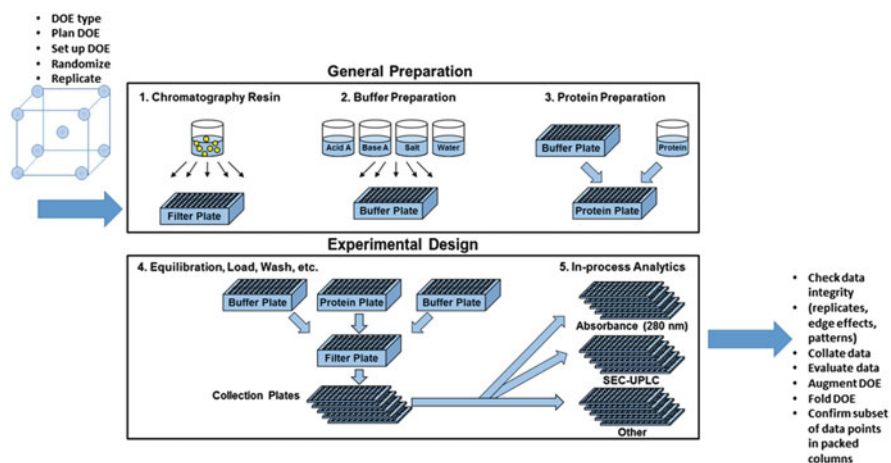
N. Singh (✉)

Bristol-Myers Squibb, Global Manufacturing and Supply, Devens, MA 01434, USA
e-mail: nripen.singh@bms.com

S. Herzer

Bristol-Myers Squibb, Global Manufacturing and Supply, Hopewell, NJ 01434, USA

its impact and opportunities for further growth, and the exciting developments in modeling and simulation of DSP rounds out the overview. Lastly, emerging trends such as 3D printing and nanotechnology are covered.



Graphical Abstract Workflow of high-throughput screening, design of experiments, and high-throughput analytics to understand design space and design space boundaries quickly. (Reproduced with permission from Gregory Barker, Process Development, Bristol-Myers Squibb)

Keywords Bioprocessing, Downstream, High-throughput processing, Modeling, Process improvements, Purification

Contents

1	Introduction	117
2	Traditional Chromatography Methods	120
2.1	Improved Selectivity Through Phase Modulation and Displacement	121
2.2	Use of Traditional Protein A Purification; Improvements in Capacity and Selectivity	123
3	New Purification Strategies	129
3.1	Improved Use of Traditional Ion Exchange Chromatography (IEX) for Downstream Processing	129
3.2	Improved Use of Multimodal Chromatography (MMC) for Downstream Processing	132
3.3	Hybrid Technologies for Downstream Processing	133
3.4	Continuous Processing for Biologics Manufacturing	134
4	Design of Experiments (DoE) and High-Throughput Screening (HTS)	137
4.1	Single-Use Disposable Options	142
4.2	Improved Throughput, Disposable Chromatography Options Such as Membranes and Monoliths	143
4.3	Modeling in Downstream Processing (DSP)	147
5	Concluding Remarks and Future Direction	154
	References	156

1 Introduction

Downstream processing (DSP) is engaged in the separation and refinement of mixtures of components. In its simplest definition, DSP encompasses a tool box of separation techniques designed to achieve mass transfer phenomena, converting mixtures of substances into subsets of mixtures or fractions [1]. At its onset, industrial DSP of proteins considered many of the traditional chemical unit operations, such as aqueous two phase systems (ATPS), precipitation, crystallization, and extraction. Although these chemical techniques still have a stronghold in sister industries such as plasma fractionation [2] and vaccine manufacture [3–7], the lack of general utility, emergence of higher yielding, less harsh techniques, and scale-up limitations in cell culture-based protein DSP stifled popularity of these techniques. Novelty in the context of these unit operations therefore mostly derives from improvements in experimental design, high-throughput screening (HTS), equipment, and systems. A deepened understanding of protein surface mapping going hand-in-hand with improved protein engineering contributes as well. Advances in separation sciences over the last five decades have enabled DSP as it stands today, where process chromatography and filtration have evolved into the pillars of downstream processing of protein biologics for the last three decades. A brief review of their development and capabilities serves here to lay the framework as to what targets need to be surpassed to achieve similar success in DSP for alternate unit operations.

The ubiquity of filtration unit operations in DSP arises from its separation prowess and versatility. From removal of cells and particulates, sterilizing grade filtration, over-concentration and buffer exchange to single-use chromatography alternatives, there are virtually no biologics processes which do not encompass multiple forms of filtration. Industrial research of membrane technology began with the discovery of nitrocellulose in 1846 [8]. Although research continued with the first mention of ultrafiltration by Bechold in 1906 [8], it was the discovery of semi-synthetic and synthetic anisotropic membranes with the invention of the Loeb–Sourirajan membrane in the late 1950s and 1960s that began the success story of industrial membrane applications in the pharmaceutical and eventually biotechnology sectors [8]. As one can imagine, such a long history has driven an extensive portfolio. A profusion of filter materials and filter types are available from simple natural materials, such as ceramics, to synthetics, such as hydrophilic polyether sulfone (PES). Filters are much more amenable to single use, a feature which has ensured an even stronger foothold in the industry in the last decade. In spite of its versatility and simplicity, filtration technologies have not been able to meet the resolution of traditional chromatography steps to date. Significant improvements have been made and a thorough review can be found later in this chapter.

Protein process chromatography's foundation was laid in the 1950s and 1960s with the invention of macroporous base matrices. Typically, base matrices were sufficiently hydrophilic and chemistries such as cellulose [9], dextran [10], polyacrylamide [11], polystyrene [12], and agarose [13, 14] were utilized. Early prototypes were too soft, which severely limited both throughput and scale-up because of flow rate and bed stability limitations, respectively. Although early resins were

initially only modified with charged or hydrophobic moieties, significant improvements in coupling chemistry allowed addition of affinity resins to the portfolio [15]. From the advent of process chromatography, researchers clearly understood that porosity, linker, and coupling chemistry significantly impact the performance of chromatography resins [16]. Affinity steps were and still are hampered by high cost and lack of universal application. Ion exchange chromatography (IEC), on the other hand, emerged as an early favorite in process chromatography, driven by its general applicability and simplicity. Mixed mode (MMC) and hydrophobic interaction chromatography (HIC) were described as early as the 1970s [17–20] without gaining the same popularity as affinity, size exclusion, and IEC. Both MMC and HIC can therefore serve as examples of inventions which have not yet lived up to their innovation potential. Adsorptive resins exploit specific physicochemical properties of a protein to enable separation. Although true affinity resins offer unique selectivity with purification factors in excess of 100- to 1,000-fold, their specific interactions are driven by the same physicochemical properties as the much less selective IEC and HIC. To optimize truly any process, a foundational understanding of each step, including affinity chromatography, should be considered imperative to design in process robustness. As the understanding of main drivers of high selectivity purifications grows, it has become evident that the natural selection of proteinaceous ligands is difficult to match with small, synthetic ligands even if underlying mechanisms are the same. However, the advent of HTS, combined with molecular modeling and improved synthesis of larger ligands to mimic native binding, has enabled identification of more promising chromatography media.

Many of the downstream processing reviews of the last decade have stressed the tremendous improvements in upstream processing although frequently lamenting the lack of corresponding innovation in downstream processing [21–26] with few opposing viewpoints [27]. Review papers have consistently referred to the same set of solutions to overcome the bottleneck of downstream processing of high titer cell culture [28]. To stimulate this discussion on the innovation of bioprocess technologies, one could raise three fundamental questions to determine why they have not taken root in the industry:

1. What are the main barriers for successful implementation of new unit operations or innovations in DSP?
2. At what point in its lifetime should an innovation no longer be considered a viable mainstream option?
3. How well have predictions of success served us in the past and how well do they serve us at present?

Some excellent reviews on innovation barriers are available [29–34] and are not discussed here in any great detail. The intention is to provide a more realistic outlook on new technologies and the potential implications of being either an early adopter of an unsuccessful innovation or a late adopter of a game-changing innovation. The reader is directed to the seminal work of Everett Rogers now in its fifth edition. His model of innovation diffusion and adoption is briefly summarized here (Fig. 1).

Premature implementation of unsuccessful innovation, often referred to as “bleeding edge technology,” is discussed less often, although the impact on

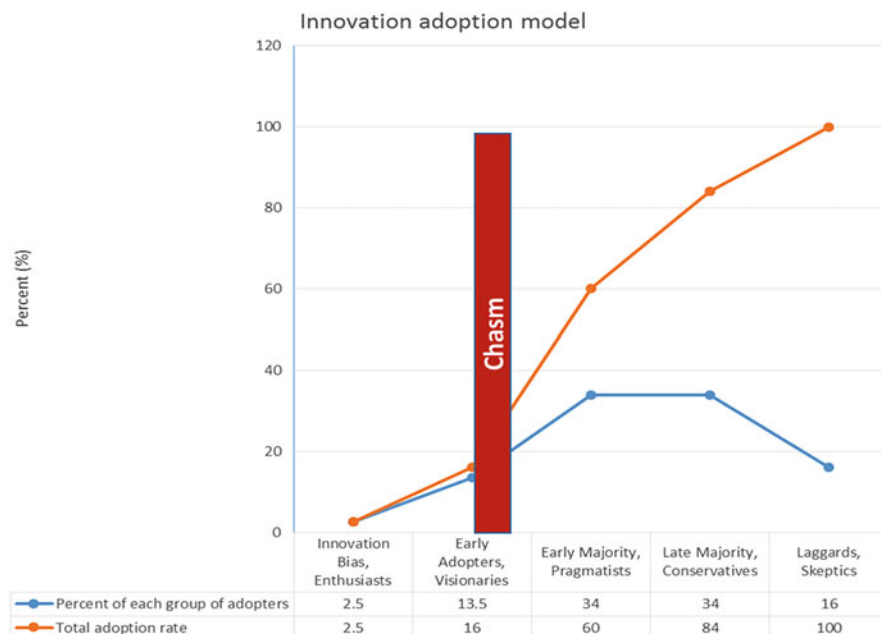


Fig. 1 Overview of the innovation adoption curve: x-axis, time scale, y-axis, percent (%) of group of adopters and adoption rate. The threshold for adoption is shown as a bar (chasm)

corporate culture and resulting reluctance to adopt new technology in the future can be significant. One relevant example to DSP is chromatography itself. Since chromatography's first inception in 1903 [35], more than half a century passed before it started to find utility in the industrial separation of proteins [36]. For an excellent review of the tortuous path the discovery of chromatography took from its outset to first successes please see Guiochon et al. [37]. Another, more recent example is expanded bed adsorption chromatography (EBA). EBA is built upon the concept of fluidized beds, a concept employed industrially since 1922 [38] and successfully used for antibiotics since the 1950s [39]. The utility of fluidized beds for protein adsorption was more limited but expanded in the 1980s and early 1990s with the advent of EBA [40–43]. EBA offers a more stratified bed with very low back mixing through use of particles with a density and size distribution [44]. At its zenith, EBA even warranted its own set of annual meetings [45] as it promised to bundle three unit operations into one, namely centrifugation, filtration, and chromatography. Unfortunately, the technology experienced significant growing pains and failed to deliver for a multitude of reasons. The most prominent hurdle was and is the significant interaction of many feed streams with the resin. EBA's requirements for large bead size to offer sufficient weight led to significant mass transfer challenges impacting both capacity and recovery. Initial hardware designs were costly, lacking robustness with limited scale-up options. Many of these challenges have been overcome. Hardware improvements have been impressive and column diameter is no longer a limitation with columns up to 1.5 m in diameter.

Chromatography media improvements have also been made to minimize biomass interaction and improving density to enable higher flow rates. Tungsten carbide-based beads offer high density, at a cost. The main limitation, however, remains the significant biomass interaction and appropriate bioburden control produced by improper hardware that creates additional problem in GMP operation which has yet to be overcome, leaving EBA as a niche application [46]. With the sobering example of EBA in mind, it may be prudent to ask vendors of any new technology poignant questions about commitment and market share expectations. Ideally, departments focused on assessment of innovations should build models to forecast costs which may reduce or increase over a 10- to 20-year time frame. In the case of EBA, unless some breakthrough discoveries are made on minimizing biomass interactions and significantly reducing media cost and hardware investments, it seems unlikely that this invention can ever enter the realm of disruptive technology.

This brings about the last question as to the general competency to predict success or failure. Unfortunately, research has taught us that human ability to predict future trends and developments accurately is poor regardless of discipline [47]. Non-model-based predictions of future trends in the sciences favor personal preferences and experiences of the author(s), ensuring an unwavering, subjective bias. Any team tasked with assessing the utility of new technologies should therefore consider investing in models to distinguish an interesting academic invention from a true market innovation. Committing to innovations simply based on one's view point or personal preference is unlikely to be successful in a very competitive, highly regulated market.

Having framed expectations around fallacies in innovation prediction and implementation, the reader is advised that this review does not attempt to predict success or failure of a particular innovation. Instead, it tries to cover the breadth of innovations aimed at addressing challenges the industry faces today, remaining impartial, fair, and balanced. In addition, there is more emphasis on chromatographic and non-chromatographic separations directed at the purification of proteins, with a heavier weighting on antibodies. All-encompassing integrated USP/DSP strategies such as continuous processing are discussed as well as gains in data quantity and quality because of automation and high-throughput screening (HTS). Best practices and advantages in experimental design through design of experiments (DOE) to gain access to more complex design spaces such as multi-modal chromatography are reviewed with an outlook on potential future trends. A discussion of single-use technology, its impact and opportunities for further growth, and the exciting developments in modeling and simulation of DSP rounds out the overview. Lastly, an outlook on more interesting forays such as 3D printing [48] and nanotechnology is presented.

2 Traditional Chromatography Methods

Traditional chromatography methods in industrial processes encompass affinity (AC), ion exchange (IEX), and hydrophobic interaction (HIC) chromatography. Size exclusion chromatography (SEC) is found in older and/or smaller scale industrial processes and is not further discussed here because of its throughput

limitations and column packing challenges. Both limitations ensure that size exclusion chromatography remains a poor choice for an industrial process. AC, and IEX in particular, are factotums in the industry because of their robustness, simplicity, and, of course, familiarity [27]. With that said, poor mass transfer and back pressure are inherent limitations of traditional chromatography. Mass transfer in traditional chromatography media is intraparticle pore diffusion controlled, and pore concentration is inversely proportional to bead radius and pressure drop is inversely proportional to the bead radius squared [49]. As improved mass transfer is pitted against an increase in back pressure (cost consideration of smaller bead size aside), traditional bead based chromatography always operates within these constraints.

Downstream processing is challenged with developing robust, fast, productive processes to enable purification high titer upstream processes [46, 50–54]. Traditional chromatography is perceived as a bottleneck and its future has been called into question for over a decade [27, 55, 56]. Juxtaposed to this expectation of the imminent demise of traditional chromatography is the entrenched platform process for antibodies [28, 57–63]. As an antibody platform process shortens process development, simplifies manufacturing, and even enables platform analytics without jeopardizing commercial success, finding an equally robust and easy to use process can be hard to envision, though some approaches are described in other sections of this chapter.

2.1 Improved Selectivity Through Phase Modulation and Displacement

Phase modulators can help to improve selectivity and productivity. Their universal applicability for all modes of chromatography and the availability of economical, compendial displacers have made displacers an attractive option for difficult-to-remove impurities even for platform processes. Neither phase modulators nor displacers are particularly novel, but their accessibility has been bolstered through the use of HTS approaches. The patent landscape should be taken into consideration for both [64–77]. Commonly found phase modulators are buffer and/or pH divalent cations [78], salts [79], organics [80], chelating agents [81], and amino acids [82]. The use of displacers has a longstanding history in chromatography for difficult-to-separate proteins. Either histidine or phosphorylated sugar displacers offered the only means for effective recovery of antibodies from Drimarene Rubine R/K-5BL [83]. Divalent cations can be useful phase modulators for the reduction in electrostatically-bound DNA. More problematic from a scale-up perspective are organics. Alkane diols are preferred over alkyl alcohols. Although alkene diols are inherently more viscous, they do not pose costly operational and safety challenges such as explosion proof operation, and have also been found effective in the reduction of lipopolysaccharides [84]. The number of extensively studied phase modulators and displacers is small as industrial processes require compendial grade materials. The short chain alkane diols (polyethylene glycol and propylene glycol), amino acids, and chaotropes have been studied for IEX, MMC, and AC

[85]. Mechanistically, alkane diols weaken hydrophobic interactions. The chaotropic phase modulator urea weakens hydrophobic and hydrogen bonding interactions and alters protein solvation states, simultaneously slightly reducing electrostatic interactions. Arginine interacts with protein side chains by hydrogen bonding and π -electron interactions. In addition, arginine interacts with aromatic moieties of HIC and MMC ligands. Out of all the modulators, arginine's interactions are the most complex, but its overall impact is a reduction in hydrophobic, hydrogen, and electrostatic interactions. Arginine has been particularly helpful in modulating MMC selectivity [86–88]. Medium chain length fatty acids such as sodium caprylate have also been evaluated [88–90]. Sodium caprylate had a dramatic effect for some proteins but was generally less impactful as compared to the other additives in one study [88]. Whenever inherently “sticky” reagents such as detergents or fatty acids are used, an impact on resin reuse and lifetime must be taken into consideration. This is especially important for displacers where high affinity for the ligand is the means of elution. Development of regeneration and cleaning-in-place protocols to preserve resin lifetime can become very challenging.

Amino acids have been investigated [91] as either phase modulators or buffer systems, as driven by an amino acids pKa and buffering capacity. Both Protein A and IEX have been evaluated and amino acid-based systems compared favorable in HMW and HCP clearance to the baseline buffer system. Histidine-based buffer systems are frequently used as a final formulation buffer, and histidine buffer use especially in the final chromatography step may seem advantageous. However, the use of histidine as a process buffer can be problematic. Diafiltration of histidine can impact final targeted buffer concentration and pH because of preferential sieving and/or retention of some histidine charge variants, also described as the Donnan effect [92–94]. Histidine can be problematic for a CEX step as it binds and hence titrates on the column, leading to undesirable pH transitions and requirements for long step holds to reach a stable pH baseline. The role of arginine in Protein A binding and elution was also studied [95]. Molecular computation served to explore the underlying mechanism in arginine's impact on Protein A purification of IgGs. As arginine weakens hydrophobic, electrostatic, and hydrogen bond interactions, it facilitates the dissociation of IgG from the Protein A ligand. The impact of arginine was compared to the kosmotropic citrate buffer system. The citrate system strengthened the binding of antibodies and led to more formation of aggregate during elution. This coincides well with internal data obtained through HTS pH salt mapping studies of antibodies where aggregation is most pronounced in citrate, followed by succinate, acetate, and glycine (Bristol Myers Squibb (BMS), unpublished results). This is in good agreement with the expanded Hofmeister series. The effect of arginine and citrate was juxtaposed in their solvation behavior as well. Although the number of arginine molecules around the proteins in its dissociated state was higher than in the associated state, citrate had a higher concentration around the protein in its associated state as compared to the dissociated state. It was concluded that this impact was caused by the differences in how the two phase modulators interacted with protein surfaces. Although arginine preferentially interacts with charged, aromatic, and polar amino acids, citrate associates preferentially with basic and aliphatic amino acids.

Both MMC and Protein A [96] steps were recently shown to benefit from high salt wash steps for increased viral clearance at levels of greater than five logs. The impact of this strategy on resin life has not yet been investigated and the duration of the total required wash steps was extensive. However, the benefits of high viral clearance for a Protein A or MMC capture step could very well offset the drawbacks of a potential reduction in resin lifetime and increase in total WFI consumption and processing time. Use of pH modulation for narrow range pH responsive resins can be advantageous. Protein A offers sustained binding at pH 6.0–11 with minimal salt requirements in the alkaline range. The presence of higher concentrations of salt may be required from pH 3.0–6.0 to suppress electrostatic repulsion [96]. Exploration of pH-based washes which break up electrostatic interactions can prove beneficial and have long been exploited for high pH wash steps on Protein A [96, 97]. As indicated, arginine and its analogues are phase modulators of high versatility with a multitude of benefits from suppression of HMW formation over increased viral clearance to desirable shifts in elution behavior. The main drawback of some phase modulators, such as arginine, is the intellectual property landscape [73, 98, 99].

Salt washes can reduce HMW formation during elution if impurities are removed. Depending on a protein's solubility, sometimes addition of moderate amounts of a mildly kosmotropic salt, such as sodium chloride, can also prove beneficial. Lastly, recent years have also seen more and more use of dual salt buffers to improve HIC selectivity and reduce overall salt concentration. For more information on utilizing a dual salt system, please refer to the HIC section of this chapter for more detail. In summary, high-throughput screening and improved understanding of binding mechanisms and selectivity modulation have led to a wealth of publications on simple, effective methodologies to improve impurity clearance, recovery, and capacity. Combinatorial approaches of surface modeling, first principles, and HTS experiments are an avenue to explore further in the future to optimize processes more specifically and quickly. Additional points concerning the modeling aspects of this approach can also be found in subsequent sections of this chapter. Displacers can offer significant advantages over traditional elution techniques. Although there have been few publications concerning use of displacers in industry, patents are still being filed as evidenced by a recent patent application [77]. A comprehensive review of displacers was provided by Srajer Gajdosik et al. [100].

2.2 Use of Traditional Protein A Purification; Improvements in Capacity and Selectivity

The recent patent expiry of Protein A led to the addition of new Protein A vendors to the market. Some of the new Protein A resins are presently offered at reduced cost (\$6,000–7,000/L). This reduction in cost weakens concerns about the viability of the Protein A-based platform process. Nevertheless, even at this reduced cost it is hard to envision a disposable Protein A option, one of several reasons why alternative capture steps are investigated. As it is likely that Protein capture persists

in the industry in at least the near term, a basic understanding of the underlying binding mechanisms and implied opportunities and limitations is warranted.

Antibodies and Fc fusion proteins bind to Protein A predominantly hydrophobically [101], a concept long exploited for weak binders [102]. Studies have shown that a gain in enthalpy, not entropy, drives binding to Protein A [103]. Although the addition of salt to condition a bioreactor harvest for a weak binder is commonly practiced in research, this is not desirable at the large scale. If addition of salt should be required, the addition of kosmotropic salts such as sodium citrate or potassium phosphate should be considered. To date, no comprehensive study has been published regarding the impact of dual salts on capacity and selectivity, a phenomenon studied for HIC [104–106]. Although the addition of salt can increase in capacity, this gain is offset by a commensurate reduction in selectivity and the potential for increased HMW formation during elution if salt concentration during wash steps needs to be maintained to reduce product loss. The use of sodium salts of carboxylic acids to strengthen Protein A affinity was recently described [81]. EDTA and sodium citrate offered comparable levels of binding capacity improvements. The use of EDTA rather than salt during wash steps can circumvent the formation of HMW during elution, simultaneously maintaining good binding strength and capacity [107]. The benefits of pH optimization to maximize wash step efficiency during process development has been described earlier in this chapter. Most antibodies elute from Protein A effectively in a pH range of 3–4, and therefore the elution buffer and pH should also be carefully considered to maintain product quality [108]. Glycosylation pattern [109], IgG class [103, 110], and variable regions all impact not only binding affinity but also elution pH. The impact of the VH3 domain [111] has led to the development of Protein A analogues engineered to eliminate the electrostatically driven interaction, which begins to weaken in the range of pH 2.8–4.0. Current native, recombinant and sequence engineered Protein A ligands have isoelectric points from 4–5. Electrostatic repulsion is the driving mechanism of Protein A elution; the presence of salt during elution can interfere with elution and induce aggregate formation. However, Fc fusion proteins and IgG4 antibodies, which often have lower isoelectric points, may require addition of some salt, or “salting in,” to prevent aggregate formation.

The process-related impurity profile in the Protein A eluate is dominated by the physicochemical properties of the antibody [112–117]. For all adsorptive resins, linker, coupling chemistry, and ligand impact the impurity profile [118, 119]. Selectivity is driven by initial resin choice as well as product surface properties. Bench marking publications, extolling the properties of one particular resin, yet utilizing optimal conditions for just one already established resin, do not offer an unbiased assessment [120]. At a minimum, baseline conditions for each resin should be employed in the comparison as per vendor recommendation unless the goal is specifically to replace the current resin with minimal change. Conversely, changing to a completely different base matrix, linker, or ligand chemistry for a platform process should not be an endeavor taken on lightly as the time investment to optimize mobile phases to ensure robust performance is not trivial, as differences in resin characteristics are significant.

The ligand itself has of course also been subject to intense scrutiny in an effort to maximize its desirable properties and minimize its shortcomings. Protein A, a cell wall

protein of *Staphylococcus aureus*, consists of five, highly homologous domains. Recombinant Protein A ligands often consist of repeats of just one of the five domains. Domains are chosen and/or often mutated to increase alkaline resistance, reduce VH3 domain-based interaction [121] (depending on domain choice) and increase ease of expression. Each of the five highly homologous domains could theoretically bind one IgG antibody [122], although stoichiometries are significantly lower than 5:1. For standard IgGs, ratios of 1.6–2.5 apply [103, 123–125]. As reported previously [123], Fc fusion proteins can offer improved stoichiometry. Internal data for an Fc-fusion protein demonstrated ratios as high as 4.5:1 with a corresponding dynamic binding capacity of 60 mg/mL at 5% breakthrough (DBC5%) at 3–4 min residence time (RT). The VH3 domain of the Fc-fusion protein contributed to the more favorable stoichiometry as the ratio dropped from 4.5 to 3.2 when switching from the recombinant Protein A to the Z-domain-based MabSelect SuRe ligand. Although steric hindrance is clearly an effector in the ratio of antibody to Protein A, one can envision several approaches to boost this ratio. One obvious solution is an increase in the number of domains. A commensurate increase in binding capacity when increasing the number of repeats from four to six has recently been described [126]. This approach seems most simplistic, but as prokaryotic expression is currently used to generate recombinant Protein A, the limitations of the microbial host for large proteins and the potential increase in ligand cost must be considered. Fortunately, recombinant Protein A ligands have already been reduced in size as compared to native Protein A at 42 kDa. However, there is an upper limit of domain multiplicity at which effectiveness and cost of expression becomes prohibitive. Another potential option would be a further reduction in functional domain to the bare minimum and use of an inert binding spacer or linker arm to maximize flexibility and distance between the Protein A domains to increase capacity. This approach has not been utilized for commercial resins, which seems to indicate that its practical application is not trivial. One could also envision a more flexible network of Protein A ligand display through grafted polymers. However, given some of the recovery issues observed for other grafted polymer IEX resins, this also seems less attractive. Lastly, the dual flow loading regime initially proposed by GE Lifesciences [127] has recently been revisited with reported capacities of up to 65 g/L [128].

To enhance selectivity further, pegylation of the Protein A molecule has also been explored [129]. Considering the increase in cost that this approach would likely incur, in combination with the availability of simple wash regimes to accomplish similar impurity levels, it is questionable how attractive this approach would be commercially. In addition, more work with industrially relevant feed streams would be required to demonstrate its utility. Several publications have also explored systematic mutations of Protein A to achieve milder elution conditions [130–133]. Aggregate formation during Protein A elution has been a well-studied phenomenon and recent findings indicate that an antibody's tertiary and partially secondary structure can unravel because of Protein A association, as well as the low pH elution and viral inactivation hold [134–137].

To maximize the value of Protein A, resin re-use is critical. Development of a good regeneration, cleaning, and sanitization protocol can be time consuming.

Recent advances in studying both the mechanism for fouling [117, 138] and ligand degradation [139] should be helpful in devising more effective cleaning-in-place protocols in the future. Differences in resin chemistry and ligand stability must be taken into account to ensure that appropriate buffers are used. It is a current fallacy in the industry to use silica optimal buffers [140] interchangeably for agarose- or cellulose-based resins [141]. The use of low pH for the acid-labile agarose and cellulose-based resins is not recommended. Effective sanitization is of the utmost importance for campaign-based production schedules where intermittent periods of prolonged column storage put columns at an increased risk of bioburden. Resin lifetime studies in plate HTS format have also been described [142]. Although these should be considered excellent starting points for a first read on best cleaning agents, cleaning effectiveness is best studied or at least modeled in a packed bed. The impact of column packing quality on cleaning effectiveness should also be considered. Contact times should be based on the lower end of a column's packing quality and a change in flow distribution also needs to be considered. Although cleaning regimes have been published [141, 143], none of them have harnessed the power of DoE and predictive modeling. The fouling mechanism and exploration for potential remedial actions provided interesting insights [117, 138, 143] and serve well in designing better cleaning protocols. A recent study also showed the greater promise of sodium citrate in Protein A ligand stabilization as compared to sodium sulfate [143].

2.2.1 Advances in Custom Affinity Media

Current understanding of an antibody platform process relies on the use of a primary, universally applicable, high selectivity step. Although some arguments have been made for its replacement, the distinct advantage of purification factors well above 1,000-fold have also led to publications making the case for staying with an affinity-based platform as a sustainable, multi-decade solution [27, 55]. Recent advances in ligand identification and development (ProMetic, Affilogic, Sterogene, SomaLogic, VersaMatrix, BAC, Thermo, Avitide) have considerably dropped the price point and lead times [144]. These developments, in combination with advances in high-throughput screening on both the chromatography and analytical sides, could lead to a shift in the development of difficult-to-purify proteins. One could indeed imagine a workflow where a custom affinity resin may be utilized, which then enables a recovery and polishing step platform. As platform processes for antibodies are almost exclusively built around the use of Protein A, the wide span of potential isoelectric points, and the lower selectivity of alternative capture steps, competing approaches never gained the market share it initially hoped for affinity purification. Ligand stability, cleanability, leaching, cost, and toxicity, as well as linker chemistry, are important consideration for affinity chromatography. For any chromatography resin, to ensure a commercially viable processes, security of supply, single sourcing, and customer support are important considerations.

Detailed risk assessments of commercial viability through security of supply and vendor management are advisable.

2.2.2 Advances in Traditional Ion Exchange Chromatography

Although IEX continues to be ubiquitous in the industry, traditional ion exchange chromatography requires feed adjustments for most upstream feed streams. Either dilution or buffer exchange can be used for feed adjustments, but this is not particularly attractive at large scale. As constraints around buffer holding capacity and dilution have already been anticipated for standard platform processes [145], moving to an IEX-based capture does little to overcome these issues. For most antibody processes, disposable membrane adsorbers have become a first choice for flow-through steps after Protein A processing, although comparison for at least CEX steps did not always indicated better impurity clearance at comparable capacities [146]. Membrane exchangers are discussed further elsewhere (single-use systems). AEX provides robust viral clearance [147–155] with well-known boundaries of pH and conductivity. It is difficult to envision future antibody processes limited to the use of CEX in a two-step platform.

Significant improvements in binding capacity have been made over the last decades. The main driver for increases in binding capacity is pore size distribution as it drives available surface area [156]. Grafted polymer display of ligands and hydrogel coatings of bead pores, as well as convective through-pores, has been a more recent addition to the resin portfolio. Differences in protein uptake driven by changes to pore dimensions and phase ratios were observed early on [157]. Although the available pore space was shown to be reduced, uptake rates were nevertheless increased [158]. Further study of the underlying binding mechanism indicated that binding was different from traditional chromatography media and that a shrinking-core diffusion model should be adopted [159]. It was also proposed that grafted polymer binding surfaces should be considered and modeled as a three- rather than two-dimensional surface [160]. An impact on uptake rates through a “hopping” mechanism which could be enhanced by the choice of counter ion confirmed the three-dimensional nature of polymer grafted ligand surfaces [161]. Although uptake could be accelerated, depending on chosen elution conditions, recovery could be impaired as well [162]. CEX offers distinct advantages compared to AEX in resolving product-related impurities such as high molecular weight species and charge variants [163]. Well-established commercially viable IEX ligands are limited and range from tertiary and quaternary amine ligands for anion exchange to sulfopropyl and carboxymethyl ligands for cation exchange. To overcome some of the conductivity limitations, “salt tolerant” media have been developed. Salt tolerance can be achieved by either changing ligand type [164] or through a higher ligand density. Increases in ligand density are derived from either grafted polymer ligand display, polymeric ligands [165], or simply through an increase in ligand density. Although ligand density alone does not determine binding capacity [156, 166], the use of grafted polymers can increase accessible

surface area and ligand density at the same time. Until recently, there has been little concerted effort to understand better the underlying mechanisms for salt tolerance, as studies as early as 1996 indicated significant differences [167]. Recently, studies have provided insight on the molecular properties of these resin types through exploration of macroscopically observable adsorption equilibria and kinetic properties paired with microscopic evidence of transport phenomena. A mechanism for the formation of high molecular weight species was proposed in extended contact with the highly charged surface which plays a key role in this effect. One of the studies showed that the grafted polymer resin induced HMW for antibodies through denaturation of a region associated with the Fc tail [168]. This observation seems to suggest a high propensity to denature for any antibody depending on processing conditions. Further experiments revealed that this effect could also be observed for resins with a bimodal pore size distribution, in particular resins with large through and smaller binding pores [169]. These observations seem to imply an upper boundary for capacity and/or binding strength which is driven not only by steric hindrance but also by the deleterious side effects such as the formation of high molecular weight species. This effect has been well-known in hydrophobic interaction chromatography, but now seems to be more universally applicable to other modes of chromatography. As with HIC, the severity of the impact relates to an antibody's inherent stability. Similar effects were shown for BSA and effects could be attributed to increased charge density rather than hydrophobic interactions [170].

2.2.3 Traditional Chromatography: Hydrophobic Interaction Chromatography (HIC)

As IEX is suboptimal for capture because of conductivity constraints, HIC may be a viable alternative. HIC, however, frequently requires the addition of a kosmotropic salt. As most feed streams, especially clarified cell lysates, are rich in hydrophobic impurities, these impurities limit both capacity and resin re-use. HIC can be extremely powerful for removal of not only cell culture impurities but also product-related impurities [58, 171–174]. For strongly hydrophobic proteins, even short contact times in flow-through mode can be sufficient to induce HMW formation [170, 175–179]. HIC has therefore been difficult to adopt for a platform process but can be included as part of a platform toolbox. A more recent review of hydrophobic interaction chromatography is provided here [180]. Although no final conclusive mechanistic model for HIC has been established, significant progress has been made in building a better understanding of key metrics. The use of the second virial coefficient and high adiabatic compressibility has enabled more predictive models [181, 182]. Proteins with higher structural flexibility were also found to be more likely to denature on HIC surfaces and required consideration of the adsorptive properties of these denatured subsets to more accurately model HIC binding [183]. As in IEX, the protein-accessible surface was found to be the main driver for HIC binding capacity [184]. A more recent attempt to model the behavior

deterministically employed a genetic algorithm with good success for a single component system model protein [185].

One development in HIC is the use of dual salts to improve capacity and selectivity [186]. The benefits of the use of dual salts were first explored in the purification of plasmid DNA [187], but have since translated to protein purification. The concept was first introduced to enhance binding capacity at increased solubility of a model antibody [188]. The concept was then extended further to include ternary mixtures of salt and a mathematical model built based on the synergistic effects of ions rather than salts [105]. As first proposed by Melander and Horváth [189], surface tension measurements have been used in the past to compare lyo- or kosmotropic salt strength and predict DCB. This concept could not be transferred to dual salt mixtures [104]. Instead, it was found that the aggregation temperature provided a better metric for the degree of hydrophobic forces and correlation to the DBCs. A better foundational understanding of HIC is critical not only for robust processes in the industry but also for the improved adoption of mixed mode chromatography, where the complexity increases further as ionic, hydrogen bonding, hydrophobic, and hydrophilic interactions need to be modeled and predicted accurately.

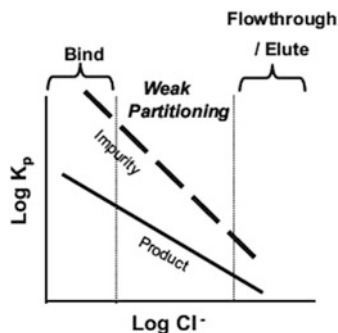
3 New Purification Strategies

As stated above, most commonly used chromatography techniques used in the manufacturing of biological products comprise affinity (AC), ion exchange (IEX), and hydrophobic interaction (HIC) chromatography. However, production costs for drugs generally have shifted from upstream cell culture to downstream recovery and purification, pushing for the development of new separations technologies or improved use of traditional ion exchange chromatography (IEX) for downstream processing.

3.1 *Improved Use of Traditional Ion Exchange Chromatography (IEX) for Downstream Processing*

Incremental improvements can drive efficiency significantly in downstream purification. Prominent examples are overloading chromatography and weak partitioning chromatography (WPC) [190]. WPC, first introduced in 2008 [190], operates by exploiting differences in the distribution or partition coefficient of the product and impurity under weak binding conditions and does not necessarily require high loadings for effective recovery as shown in Fig. 2. Conditions are sought where the partition coefficient is >0 for both products and impurities, and the partition coefficient of impurities is ideally still significantly larger than that of the product.

Fig. 2 Schematic diagram of anion exchange chromatography, showing the log of the distribution coefficient, K_p , for both the product mAb and an impurity as a function of the log of the counterion concentration (Reproduced from Kelley et al. [190])



Overload ion exchange chromatography, on the other hand, was defined as a type of frontal chromatography where overloading eventually leads to preferential displacement of product and retention of impurities as opposed to flow-through frontal chromatography, where the product remains unbound. Overload chromatography therefore implies operation in a stronger partitioning regime where very high loadings are required to ensure product recovery through displacement by impurities. Although overload chromatography claims stronger binding or higher partition coefficients, based on patents filed, it would be hard to distinguish the two techniques as the ranges for effective partition coefficients claimed would also cover strong binding at modest loadings (US Patent No. 9,144,755, 2015; US Patent No. 8,067,182, 2011; USA Patent No. 20070060741, 2007; US Patent No. 20070054390, 2007). Both techniques can be implemented in existing manufacturing facilities without significant capital. In addition, integration of multiple unit operations into a single fluid system can eliminate the need for in-process hold tanks and minimize validation work.

Cation exchange chromatography (CEX) is typically operated in bind and elute mode with capacities <100 g/L. This step can be a bottleneck in downstream processing of high product titer in cell culture fluids. Overload chromatography of a cation exchange steps using a perfusive resin or membrane favorable partition conditions and very high loadings enables product displacement by impurities at high loading, flow rates, and consequently throughput. Liu et al. demonstrated that the performance of diffusive and perfusive cation exchange resins (SP-Sepharose FF (SPSFF) and Poros 50HS) can be operated using isocratic overloaded conditions to reduce HCP, DNA, HMW, leached protein A, and gentamicin, allowing monoclonal antibody to break or flow through the matrices after the resin reaches its DBC for the antibody. Contrasting the different matrices when used in the overloaded mode, the conventional perfusive CEX matrix provides a higher binding capacity than convective matrices and better mass transport than the diffusive matrix; it therefore provides a higher capacity to retain HMW and to remove other process-related impurities. As fewer steps and a smaller volume of buffer are needed for overloaded processing, buffer tank capacity also becomes less of an issue. Finally, the use of CEX in the overloaded mode could enable a more streamlined downstream process through the coupling together of multiple steps by simply

incorporating in-line pH adjustment capabilities. Weak partitioning chromatography (WPC) also operates under isocratic conditions. As mentioned earlier, weaker interactions are expected and loadings are generally more comparable to traditional chromatography. Selectivity can be finely tuned, albeit at the cost of recovery, a frequently observed phenomenon in traditional process chromatography. Similar to overloaded chromatography, yield improves with increased column loading. A short wash at the end of the load stage serves to improve yield further. Under conditions that favor very strong product binding, competitive effects from product binding can give rise to a reduction in column loading capacity. More recent work also studied the impact of different base matrices for anion exchange resins in weak partitioning mode [117]. Fractogel[®] EMD TMAE HiCap, Q Sepharose FF, and POROS 50 HQ were compared in WPC mode by transmission electron microscopy and inverse size exclusion chromatography, along with characterization of WPC-specific protein interactions. The results show that, for each resin, salt and pH conditions can be found for high molecular weight aggregates using the information of product-specific K_p values. However, isocratic elution and adsorption mechanisms are different for each resin and for the different components.

The robust performance of WPC anion exchange chromatography has been demonstrated in multiple cGMP mAb purification processes. Excellent clearance of host cell proteins, leached Protein A, DNA, high molecular weight species, and model virus has been achieved. Optimization of the AEX-WPC step requires tuning of the pH and counterion concentration for each mAb, which can easily be obtained through generating adsorption isotherms in high-throughput batch chromatography. The governing variable used in the design of a WPC step is the product partition coefficient, K_p , which is a fundamental descriptor of the strength of the protein–resin interaction. The improvement in impurity clearance can be correlated to the product K_p using either impurity LRVs or K_p . The product K_p is also used to define the operating window based on allowable ranges of product recovery and load dilution, and the minimum desired column loading. Iskra et al. have also demonstrated the potential of WPC to be included as part of the modular viral clearance approach ([147], p. 73).

Both strategies significantly improve facility utilization through enhancement of throughput for traditional IEX chromatography. Column volume reduction of as much as 90%, reduction in both buffer volumes and total number of buffers, and reduced operational complexity all illustrate the power of both of these options. This can be further enhanced through the use of disposable, pre-packed columns. Single use of traditional chromatography columns is usually challenging because of the relatively low upper limit of available pre-packed columns as well as the cost of single use. In commercial facilities, where columns are re-used up to a pre-established end of life cycle number, a reduction in column volume and hence cost can enable the use of disposable columns. Additional savings are garnered from avoidance of column packing, testing, cleaning, and life cycle studies. Validation cost savings can be significant, as can avoidance of potential investigation of bioburden, a phenomenon more readily preventable for disposable options. This can be a real boon for facilities which operate in campaigns and have older less hygienic column designs.

3.2 Improved Use of Multimodal Chromatography (MMC) for Downstream Processing

Multimodal chromatography consists of stationary phases that combine more than one mechanism of interaction into the design of their ligand, including coulombic interactions, hydrophobic interactions, and hydrogen bonding [106, 191, 192]. Multimodal ligands consist of ionic (amino, carboxyl, or sulfonic groups), hydrogen bonding (hydroxyl, thiol, or amide groups), and hydrophobic (aliphatic chain, thiol, or aromatic ring) groups. Electrostatic interactions play a significant role at low to moderate ionic strength, and hydrophobic interactions along with hydrogen bonding play a significant role at high ionic strength. The additional flexibility in designing purification strategies that employ multimodal materials makes them attractive candidates to face emerging challenges in industry, such as the need for more selective chromatography media, improved resolution, and tolerance of feedstock conditions. MMC resins provide additional flexibility in designing chromatography operations and purification processes because the multifunctional ligand can interact with both protein and aggregate species via multiple, simultaneous, and independent intermolecular mechanisms [193, 194]. Given the significant interest in evolving the downstream process platform for monoclonal antibodies, a number of applications for MMC have emerged, particularly modes involving a combination of hydrophobic and electrostatic interactions. The salt-independent binding behavior of MMC resins has been shown to bind mAb cell culture supernatant directly to resins that combine hydrophobic and electrostatic interactions [194, 195]. The hydrophobic nature of MMC resins has been exploited for aggregate removal in post-Protein A polishing steps [193, 196].

Although MMC offers significant opportunities for enhanced selectivity, methods development in these systems can be more complex than single mode of interaction, such as ion exchange or hydrophobic interaction chromatography. The fundamental understanding of how proteins interact with multimodal resins is still improving [106, 197–201]. A connection is needed between some of the fundamental characterizations of protein binding and the development of practical separation steps for preparative protein purification. Several mechanistic models have also been proposed in the literature. Modeling of the retention profile with a salt-dependent Langmuir isotherm provided promising results [200]. Mechanics of protein adsorption in a multimodal system were studied and similarly good agreement with the Langmuir isotherm in the electrostatic attraction regime was found, although the binding behavior becomes more complex when operated near the pI [197]. It was speculated that a multilayer or rearrangement effect when transitioning from electrostatic to hydrophobic interactions increased complexity near the pI. A thermodynamic model based on steric mass action (SMA) included interaction parameters, and could be applied to both hydrophobic interaction and ion exchange chromatography [198]. Multimodal adsorption isotherms were also developed as an extension of Mollerup's generalized thermodynamic framework to

account for the combined hydrophobic and ion exchange effects in multimodal systems [198, 199, 202].

Recent efforts in the development of multimodal membranes (MMM) have focused on addressing the limitations of traditional chromatography's inverse relationship of binding capacity and throughput for multi-modal chromatography [201, 203, 204]. A newly designed multimodal membrane (MMM) with high protein binding capacities and high load productivity was recently introduced [201]. A thermodynamic model [199] provided insights on the nature of protein–MMM interactions and predicted binding capacities under non-test conditions. This in turn is of importance in limiting the number of experiments. It was shown that the rate-limiting step of mAb adsorption on the MMM is the reaction rate of mAb binding with the multimodal ligands rather than the mass transport of protein molecules. Thus, although high load productivities were achieved, improvements in membrane design leading to faster adsorption kinetics would enable still higher productivities.

Numerous studies have shown the impact of mobile phase modulators on protein binding. An overview of the mechanism, recent reviews, and driving forces has been provided earlier in this chapter. Recent research using protein libraries in both multimodal anion and cation exchange systems has demonstrated that mobile phase modifiers such as arginine also play an important role in creating enhanced selectivity when designing multimodal chromatographic processes.

3.3 Hybrid Technologies for Downstream Processing

Alternate approaches have been evaluated for purification of biologics based on functionalized fibrous media and/or composites. Most rely on incorporating a porous gel into the fiber matrix, providing the necessary surface area gain to ensure reasonable binding capacities. However, in such constructions, poor uniformity in gel location and mass generally leads to poor efficiencies (shallow breakthrough and elution fronts). In addition, resistance to flow can be high even for short bed depths, a problem often aggravated by gel compression under modest pressures. The incorporation of activated carbon or silica gel particulates within the fiber matrix has been pursued as an alternate approach. The particulates are often porous, can offer a large surface area, and possess a native adsorptive functionality that separate proteins based on differences in their size and effective charge [205]. Activated carbon was applied to the purification of larger mAbs, where it was found that lower molecular weight proteinaceous impurities are most efficiently removed at their isoelectric point. The most efficient recovery of a small protein from activated carbon occurs at a solution pH further away from the protein's isoelectric point, where it is strongly charged [205].

Recently, fiber-based chromatography media have been produced for biologics purification applications that utilizes a surface-functionalized winged fiber as the adsorptive media in both bind-elute and flow through modes [206–208]. The winged projections on the fiber surface afford a much higher surface area than ordinary round fibers of similar dimensions. The surface-functionalized fiber media also have a much higher protein binding capacity than similarly functionalized fibers which lack such winged projections. The fiber-based stationary phase is non-porous and displays a convoluted surface structure that provides a sufficient surface area for high binding capacity. As the protein binding occurs only on the surface of the fiber, there are no size exclusion issues with binding as seen in the case of porous bead-based bind/elute systems. Furthermore, because the particles can be transported directly to the ligand site by convection, there are no diffusion limitations in the system and the feed streams may be processed at much higher flow rates or shorter residence times. These fiber-based chromatographic media can be combined with a high frequency simulated moving bed (SMB) design to provide superior productivity in a simple bioseparation. Electrospun polymeric nanofiber adsorbents offer an alternative ligand support surface for bioseparations. Their non-woven fiber structure with diameters in the sub-micron region creates a remarkably high surface area material that allows for rapid convective flow operations. Proof of concept experiments demonstrated the performance of an anion exchange nanofiber adsorbent based on criteria including flow and mass transfer properties, binding capacity, reproducibility, and life-cycle performance [209]. Binding capacities of the DEAE ligand surface modified adsorbents appear low at 10 mg/mL, but in the context of throughput this is a significant achievement. Suitable packing into a flow distribution device has allowed for reproducible bind-elute operations at flow rates of 2,400 cm/h, many times greater than those used in typical beaded systems. These characteristics make them ideal candidates for continuous chromatography systems. An SMB system was developed and optimized to demonstrate the productivity of nanofiber adsorbents through rapid bind and elute cycle times of 7 s which resulted in a 15-fold increase in productivity compared with packed bed resins. The integrity of nanofiber adsorbents used in the system would be an important criterion of the validation process.

3.4 Continuous Processing for Biologics Manufacturing

There is a growing interest in realizing the benefits of continuous processing in biologics manufacturing, which is reflected by the significant number of industrial and academic researchers who are actively involved in the development of continuous bioprocessing systems [210–213]. Although at least some regulatory agencies responded positively to the potential of continuous processing in biologics, regulatory guidance would have to be adapted considerably to allow a clear understanding of lot definition and testing requirements. The advantages of continuous manufacturing include sustained operation with consistent product quality, reduced

equipment size, high-volume productivity, streamlined process flow, low process cycle times, and reduced capital and operating costs. This technology, however, poses challenges that need to be addressed before routine implementation is considered. Historically, conversion from batch to continuous processing has resulted in lean, fully automated, and agile manufacturing, regardless of the nature of the industry [214–220]. This conversion has enabled these industries to overcome key business continuity challenges by reducing high manufacturing capital expenditure (CAPEX), important operational expenses (OPEX), and enhancing portability [213]. The technology has been in use for many years in the chemical industry, originally developed for difficult petrochemical separations [221]. Its use in the pharmaceutical industry quickly grew because of its superior performance in chiral separations. Following the first US Food and Drug Administration's approved drug manufactured by SMB technology reaching the market in 2002, efforts have been focused on systems that yielded poor productivity using batch chromatography. The biologics industry is facing similar challenges following the emergence of competition from biosimilars, the need for decreased COGs, and the desire to gain access to emerging markets through standardization and portability [213]. To leverage the lessons learned from other industries, there has been significant interest in the implementation of continuous bioprocessing [222]. This strategic shift is a result of the convergence of both the business drivers and advances in downstream technology. Although, the biologics industry has extensive experience in continuous (perfusion) upstream processing, there are limited examples of downstream continuous processing because of the lack of enabling disruptive technologies [222]. However, the past few years have seen maturation of separation technologies such as simulated moving bed (SMB), periodic counter-current chromatography (PCC), multi-column chromatography (MCC) [210, 212, 213, 222, 223], continuous countercurrent tangential chromatography (CCTC) [224–226], aqueous two-phase systems (ATPS), single pass TFF [227, 228], high capacity membrane absorbers [229–231], and continuous viral inactivation [232, 233]. At present, the confluence of these emerging technologies and business needs to innovate, driven not least of all by the fact that high titer cell culture is catalyzing the development of continuous bioprocessing.

Several groups have examined the use of multi-column chromatography for fully integrated continuous production of mAbs from a perfusion bioreactor [212, 213, 222, 225, 233]. This integrated system can provide five to ten times higher volumetric productivity than a fed-batch process with comparable yield and purity with a smaller column size [213]. The design and operation of semi-continuous affinity chromatography using a PCC-based chromatography system have been demonstrated for clinical and commercial manufacturing [234], and multi-column counter-current solvent gradient purification (MCSGP) chromatography has been demonstrated with anion and cation exchange resins [212]. Although multi-column chromatography has been successfully integrated with perfusion bioreactors, these systems require column packing and complex valve switching, causing them to

operate in a cyclic rather than true steady-state mode. Recent work has demonstrated that continuous countercurrent tangential chromatography (CCTC) can overcome many of the key limitations of conventional batch and multi-column chromatography, providing opportunities for truly continuous product capture [224, 226, 235]. The CCTC process utilizes the resin in the form of a slurry which flows through a series of static mixers and hollow fiber membrane modules as shown in Fig. 3. The micro-porous hollow fiber membranes retain the large resin particles, simultaneously letting all dissolved and unbound species, including proteins and buffer components, pass through the membrane and into the permeate. The buffers used in the binding, washing, elution, stripping, and equilibration steps flow countercurrent to the resin slurry in a multi-stage configuration, enabling high resolution separations and reducing the amount of buffer needed for protein purification. In contrast to other continuous bioprocessing systems, the CCTC system provides the opportunity to run at steady state without having to switch any valves after system start up, allowing the users to abandon completely all activities related to handling packed beds, including packing, cleaning, validation, and storage.

Although the technical feasibility of a continuously operated platform production process for mAbs has been successfully proven, there are still some challenges to overcome on the way to an industrial application [233, 236–238]. A high potential for simplification and cost savings can be identified in the field of assembly preparation and delivery. Validated sterility testing is also not available

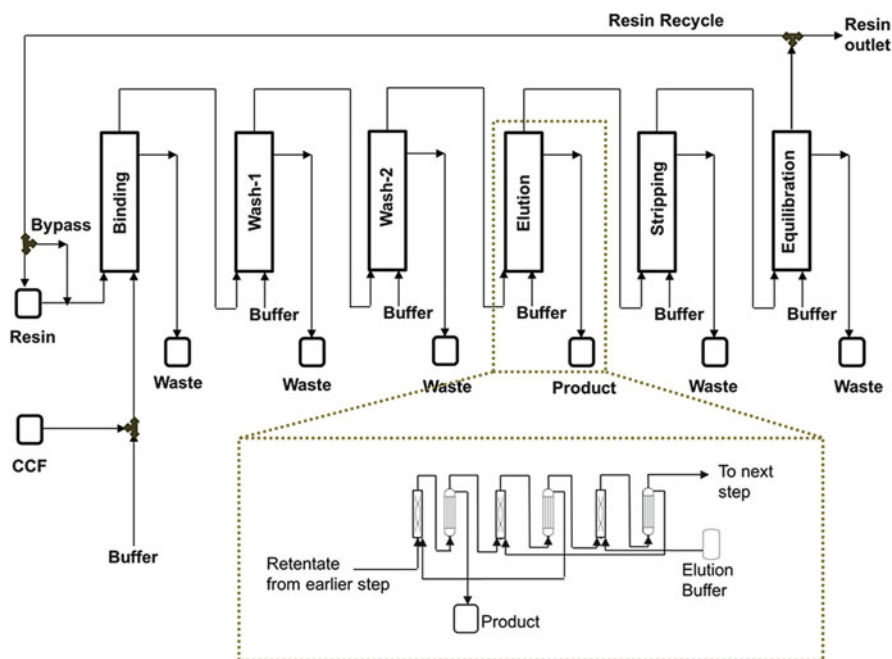


Fig. 3 Schematic of continuous countercurrent tangential chromatography (CCTC)

for chromatography columns to date. Low bioburden equipment in combination with gamma irradiation without validated sterility could be a concept to pass this aspect, although the design of low bioburden equipment is not trivial. Another aspect is the implementation of SUS in continuously operated processes operated for several weeks. The SU equipment, which is available today, is mainly designed for applications in batch processes. Consequently, the question of whether this equipment can be used for weeks instead of days has not been answered yet. Analytics to evaluate extractables and leachables have yet to be applied. Regarding continuous processing of antibodies, additional gaps have been identified. Because of very low flow rates in continuously operated antibody processes, commercial, GMP compliant, production scale equipment is not available. Furthermore, concepts for the implementation of in-line and at-line analytics to improve process understanding and enable pooled batch release have to be found. Assessment of virus safety of a continuously operated process is still a gap as well. Viral safety is always assessed holistically across the entire process. The concept of a continuous viral inactivation at low pH or through detergents still requires a re-thinking of traditional virus reduction studies to prove its effectiveness. Apart from that, solutions for the regulatory acceptance of continuously operated unit operations (UOs) and processes have to be developed as well (Fig. 3).

4 Design of Experiments (DoE) and High-Throughput Screening (HTS)

To provide an overview of the many different types of design of experiments (DoE) or a thorough explanation goes beyond the scope of this chapter. Excellent guidance can be found in the seminal works of Box et al. [239] as well as Montgomery [240]. This brief overview delves into implementation hurdles, potential areas for future growth and expansion, and synergies of DoE and HTS. The pitfalls and frequency of poor statistical experimental set-up and analysis have been reviewed and discussed in other fields [241–243]. One can speculate that a comparable deep dive into set-up and analysis of DSP DoEs would find similar misinterpretation of data caused by improper experimental set-up, execution, and analysis.

Adoption of DoE in biologics, in spite of quality by design (QbD) aspirations, has been slow, although publication output in the last 15 years has risen considerably [244, 245]. Several factors have hindered DoE adoption, not least of all a lack of training and understanding of its utility and knowledge of its practical application, although recent publications have started to fill that gap [59, 246]. The advent of HTS has significantly lessened two obstacles. Biological assays are now available in HTS format, enabling replication to reduce variability and debottlenecking analytics, and high-throughput process development (HTPD) on robotic systems has accelerated process development (PD). Throughput for assay staples in early

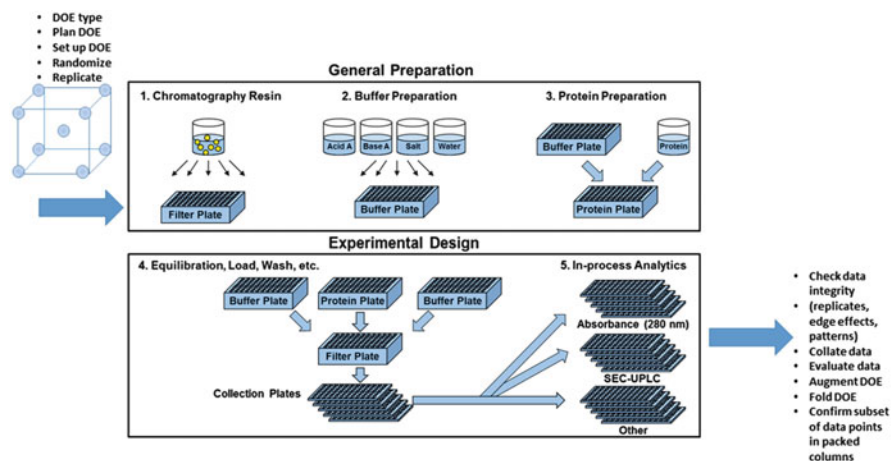


Fig. 4 Workflow of high-throughput screening, design of experiments, and high-throughput analytics to understand design space and design space boundaries quickly. (Reproduced with permission from Gregory Barker, Process Development, Bristol-Myers Squibb)

PD, namely protein concentration, size exclusion chromatography (SEC), and host cell protein (HCP), has been significantly improved. Ultra-performance liquid chromatography (UPLC) offers shortened run times with a minimal loss in resolution [247], whereas high-throughput HCP, DLS, and even residual DNA assays [248, 249] broaden the accessible range further. The low variability of SEC assays [250] as compared to host cell protein assays [251] can make SEC data an excellent internal control of overall experimental variability as long as factor ranges are chosen to show an impact on HMW and LMW levels (Fig. 4).

DoE has distinct advantages over one factor at a time (OFAT) experiments [252] but caution should be used before the untrained delves into the use of statistical design of experiments unassisted. Working with a statistician is beneficial but still requires a solid understanding of the fundamentals of statistics [253]. Even the best statistician is unlikely to interpret data correctly if common ground and language is not available. Ideally, PD scientists and engineers understand both the basics of DoE and how to test designs for pitfalls. It is presumed that the limitations and issues of the experiment are recognized even before executing a design as well as data analysis and the appropriate model selection [254–256]. Model optimization and selection are also not trivial [257–259]. Familiarity of most scientists and engineers with linear regression for one variable can also lead to an over-emphasis on driving the coefficient of determination close to 100%. Models should always be checked for variance inflation factors (VIFs), a measure of overfitting of the data with rules on multicollinearity in mind [260]. Focus should be on the factors that are strong drivers of the model in agreement with the mechanistic understanding of a process step. Recursive partitioning tools, often built into a model, are used to test a model's validity and should be employed to assess different models [261]. This is not to say that data cannot be in disagreement with the current understanding of the

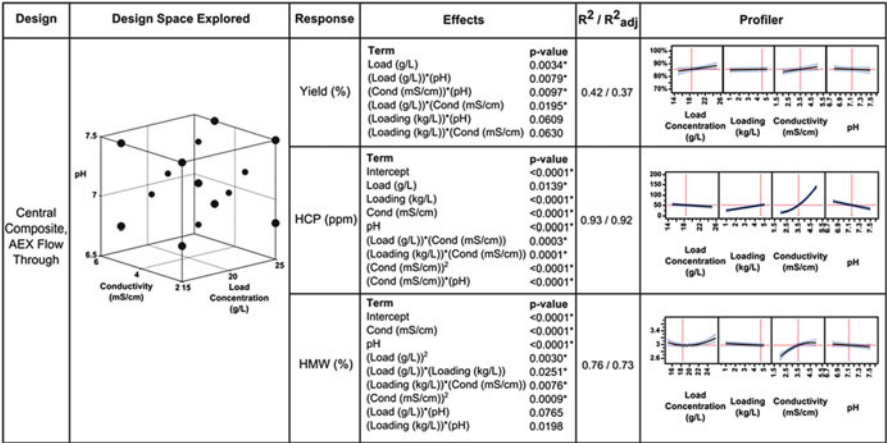


Fig. 5 Design space and observed curvature for both HCP and HMW removal in a flow-through anion exchange membrane chromatography platform process [267]

underlying principles. However, artifacts or overfitting are the more likely root causes of discrepancies. Overfitting of data is very common [262, 263] and this topic is also revisited in the modeling section of this chapter. Many researchers do not consider the significance, of the p-value but are fixated on coefficients of determination [264, 265]. Model cross-validation, built into many of the available software packages, should be used rigorously. Lastly, if fitting for curvature, care must be taken to evaluate the impact of each quadratic term carefully. Assessment of the relative importance of a quadratic term by its contribution to fit through its t-ratio, the fraction of its estimate and standard error, can help to evaluate its overall importance in the model. The prudent researcher also remembers that use of center points in fractional or full-factorial designs, other than definitive screening designs, allows testing for, but not assignment of, curvature. Although most software packages offer automated model analysis, it is always prudent to remove interaction and curvature terms manually to determine how much they actually impact the coefficient of determination. Terms that contribute less than 2% to the overall fit of the model are best eliminated [255, 266].

The utility of a screening or main effects designs should be carefully weighed. A good example is the response variable HCP clearance in an AEX FT step. Curvature in the HCP clearance response is observed in about 80% of cases (unpublished data). An example of assigned curvature in an AEX flow-through design space is shown in Fig. 5.

This curvature can be driven by pH, conductivity, and protein concentration. When screening the design space of an AEX FT step it is therefore prudent to build an augmentable design to assign curvature appropriately in a follow-up augmented DOE if curvature should contribute significantly to fit. This implies that screening designs are only recommended where main effects are expected to be dominant [268]. In the case of HCP curvature for AEX FT steps, curvature in the design space

stems from impurity breakthrough. The breakthrough response of HCP impurities is low at low conductivities (1–3 mS/cm) but begins to slope up from 3–6 mS/cm. Depending on the HCP profile, the response either flattens or sees an even sharper rise in breakthrough above 6 mS/cm. A screening design including main effects pH, conductivity, and protein concentration would have little to offer in gaining understanding of this design space [269]. At a minimum, a fractional factorial with center points must be chosen and care must be taken to avoid confounding of known key interactions. Multicollinearity, as observed, for example, for pH, conductivity (or salt concentration), and protein concentration should be taken into consideration in the planning stages of a DoE as well as its evaluation [260, 270]. The appropriate design based on prior knowledge, minimal confounding of known key factors and interactions, and, not least, a degree of uncertainty one is willing to accept should be chosen. Higher variance inflation factors (VIF) can to some extent be ignored if they are driven by interaction or quadratic terms [271]. Design evaluation which is built into most commercially available DoE software should always be conducted to understand aliasing and bias in the design [272].

Definitive screening designs are a more recent addition to DoEs and offer some distinct advantages [273–275]. Definitive screening designs minimize experimental effort yet still allow exploration of quadratic terms and interactions at minimal confounding. In addition, recent advances have also made it possible to augment these designs. In the past, a DoE was executed over the course of days or weeks using small-scale equipment [276]. Most adsorptive techniques require 4–8 h per experiment on small-scale chromatography equipment. Aside from ensuring correct set-up where lines and columns are returned to the same state between experiments, it is critical to block for equipment (if more than one system is used), day of execution, and potential starting material [277]. Although degrees of freedom (DF) are lost [278], they may be regained if they are shown not to impact both model fit and interpretability. Omitting blocking where necessary can waste time and resources [279]. In its worst outcome, over-interpretation of data can lead to selecting conditions far less robust than assumed.

Although traditional chromatography or filtration experiments necessitate blocking for day, equipment used, or both, screening in a 96-well format allows parallel rather than sequential execution of the experiments. Randomization across the plate is still critical to avoid impact of edge effects for both separation and assays [280]. Experiments in a 96-well format have distinct disconnects from a column-based separation. The ability to replicate and avoid blocking in combination with an increased understanding of the disconnects has made this an attractive, rapid, and extremely useful option as a first step toward design space exploration. Although some publications have found this approach material prohibitive [281], this is rarely the case in the industrial setting.

Currently, HTS chromatography spans several formats, including 96-well batch chromatography, 96-tip packed filter tips, and miniature columns. It is well understood that each of these systems has its own benefits and shortcomings in terms of applications, and this has already been well-summarized [282–284]. With respect to throughput and versatility, batch chromatography offers some distinct advantages

as long as a subset of resin types is excluded [284]. For many of the resins utilized in platform antibody processes, batch chromatography can serve for both foundational work, such as batch uptake and isotherm studies, and design space assessment for batch chromatography. HTS can reduce variability through replication and increased precision by means of automation and robotics. HTS is also critical in reducing material [285] and resource and time requirements. Just as in traditional column chromatography on small-scale chromatography equipment, highly material consumptive HTS experiments can be planned to minimize sample consumption. A typical AEX FT DoE could consume up to 10 g of purified antibody in a 96-well DoE. When set up properly though, only a small fraction of each sample or product pool is used for analysis. This in turn allows use of the remaining material for optimization of subsequent unit operations, such as viral filtration or TFF. Another benefit of HTS is its availability for prediction of an entire dynamic binding capacity design space [267], collecting foundational information for prediction of elution behavior [286], thus significantly narrowing the design space to explore in a DoE [89]. HTS serves to lay the groundwork for detailed process understanding, overcoming resource constraints on the analytical side. Lastly, the advantage of in-house custom resin plates should not be underestimated. The ability to analyze resins from different vendors in the same plate allows the narrowing of resin choices very quickly. In addition, it ensures interpretable results through eliminating the need for blocking between plates, which makes comparison of resins difficult even in HTS mode.

One distinct disconnect in batch chromatography is found in the longer incubation times needed to mimic column residence times [287], especially as loading increases. Another offset stems from the much lower concentration of elution pools. Both effects can lead to disconnects in the design space to understand impurity clearance and yield. Although trends are comparable, confirmation of DoE center and star and/or corner points with a subset of column experiments is not only prudent but highly recommended. Worst case studies are routinely conducted to confirm process robustness. Execution of confirmatory experiments can serve to fulfill this role as well if points are selected appropriately. The combinatorial approach of HTS and DoE enables easy augmentation [288, 289] if a design should fracture because of factor settings that lead to complete failure of the experiment. A fractured DoE becomes uninterpretable and always leads to additional experiments, as a subset of the data drives fits entirely. This is true for all responses, even if the design should only be fractured for one response. For example, if yield at very high loading drops to very low values because of either breakthrough during loading or precipitation during elution, impurity levels at high loadings are also no longer interpretable. Part of the design space used can still be assessed in combination with the newly accessed folded or augmented design which needs to have sufficient overlap to ensure interpretability. At this point, blocking is again necessary. A visual check of the newly generated data points in the previously executed experiment and analysis of a shift in the fit is recommended as well.

The data gap seen between 10 mL and 100 or 200 L process columns widens further when utilizing batch and miniature chromatography. In batch

chromatography, where resin is packed in plates during phase exchange, resolution is lost as each separation is based on one equilibrium or one plate during each stage from adsorption, wash, and elution. Models and bridging to packed columns have, however, been described [287]. Miniaturized columns are dominated by wall and capillary effects and are unlikely to represent flow distribution systems of process columns. Caution and prudence should go hand in hand with careful monitoring and comparison of scale-up data. In addition, the use of at least some miniature column on HTS equipment is limited to execution of eight experiments at a time. This in turn requires blocking for each set of experiments. In spite of some of these limitations, the benefits far outweigh the drawbacks. Automation continues to replace traditional small-scale chromatography (1–5 mL column volumes). Time, cost, material, and resource savings, in combination with improved data interpretability for DoE, are necessary to remain competitive. DoE is likely to be utilized more as it becomes more widely taught at Universities and in companies. Recent years have seen a wealth of publications using HTS for both chromatographic experiments and analytics [58, 172, 246, 285, 290–292]. Improvements have been made in the understanding of translation of residence times to incubation time [293], use of a combination of batch uptake data and static capacity to predict dynamic binding capacities over a wide range of residence times [294], and combination of factors previously difficult to explore [267]. With better understanding of similarities and differences between batch and small column chromatography, the translation to large scale model predictions should improve. The understanding of the complete design space could take just a matter of days. Ideally, a combination of computational fluid dynamics with empirical data, taking advantage of recent exploration of the use of 3D printing [295] to understand better the impact of bed and column geometry on macroscopic flow properties in the bed and HTS data, can enable process development in the future.

4.1 Single-Use Disposable Options

Multi-purpose facilities, the high cost of validation for resin re-use, and steam in place (SIP) systems have all driven quick adoption of single-use systems (SUS). Although existing, older facilities may struggle with adopting SUS completely, new facilities can be designed for the flexibility of SUS. Although not yet available [296], recycling of single-use materials may also eventually make this an even more attractive solution, cost savings for WFI for SIP and CIP processes aside. It is beyond the scope of this chapter to highlight all the advantages of single-use technologies which have led to their pervasive use. For two recent reviews, please see [297, 298]. Disposable tubing and filters as well as buffer and media storage bags have been the most readily adopted, although other disposable options, such as equipment, are catching on more slowly. Fully disposable manufacturing suites are also available [299], which could be an attractive option for a global footprint with minimal rework. The implementation of fully disposable manufacturing does not

take into account any regional differences and regulations that one may encounter. Although different options for disposable mixing units, chromatography, and tangential flow filtration skids are commercially available, centrifugation, though available, suffers from a sparsity of options. In this context, although not a truly disposable option, acoustic wave separation for the clarification of high density cell cultures may also be mentioned. Under the paradigm of high titers and 2,000- or 5,000-L bioreactors, microfiltration or microfiltration in combination with precipitation or flocculation may become the predominant recovery option [61, 300]. If 5,000-L disposable bioreactors become a reality in the future, stainless facilities could become obsolete, which should incentivize more vendors to widen or build a disposable equipment inventory. One main drawback of disposable technologies compared to stainless facilities is the lack of interchangeability of connections and technologies. Unfortunately, most vendors have taken it upon themselves to develop their own flavor of patented, aseptic connectors and connections. The lack of compatibility is not desirable from a security of supply perspective and is not as practical as traditional and interchangeable stainless connections which have been standardized. Disposable sensors also offer opportunities for both growth and improvement. Although some gaps were filled very quickly, such as disposable conductivity probes and UV cells, no truly disposable, accurate pH probes are available. The high cost of disposable flow paths, supply delays, and lack of alternative sources of supply may limit their utility. Although it may be painful for a pilot-scale facility to experience delays in the delivery of disposable components, it would not only jeopardize plant operation for commercial facilities but also potentially put patient lives at risk if the necessary disposable parts cannot be delivered for extended periods of time. Although inventory stockpiling can help to offset these issues to some extent, this in turn increases costs and overheads. Another consideration is the long term impact of a fully disposable biologics process. As all of the synthetic process components contribute leachates and extractables to varying degrees, more holistic studies are needed to assess quantities and impacts of leachables and extractables, according to ICH Q7 guidelines, in the final drug product. It seems unlikely that the overall burden of trace synthetics would not be increased and that chronic effects in at least part of the patient populations would emerge over time. The potential, long term side effects implications are still to be evaluated, especially for biologics which are not curative, but sustaining, lifelong therapies.

4.2 Improved Throughput, Disposable Chromatography Options Such as Membranes and Monoliths

The use of membrane and monolith chromatography is now ubiquitous in the industry, yet to date both technologies have only gained a foothold in niche applications such as anion exchange flow-through polishing steps [301, 302],

purification of large biologics such as DNA [303–305], virus-like particles (VLP), viruses [290, 306, 307], and vaccines [308, 309]. In spite of their superior mass transfer properties, the lack of comparable capacities for proteins in adsorptive techniques [36] has led to slow universal adoption. This section provides a brief overview of some of the approaches taken to address these drawbacks and reviews more recent findings in the field.

Membrane chromatography was first introduced as a means to overcome pore diffusion, the main limitation in mass transfer of traditional chromatography resins [310]. The surface of microfiltration membranes is used to attach ligands so that mass transport was driven through convective flow. As microfiltration membranes offer very little pressure drop because of short bed heights, they also offer much better pressure flow properties. Resolution is often comparable to traditional chromatography, albeit at the cost of dilution, which has been mostly attributed to inlet and outlet manifold design [311].

Scale-down models for membrane adsorbers are available, although some disconnects should be taken into consideration [312]. More recent models indicated high confidence in scale-up based on small scale data bolstered with computational fluid dynamics (CFD) [313] where the combination of CFD and scale-down modelling allowed a 15,000-fold scale-up. An in-depth study of membrane pore structure has also been published for at least some of the currently commercially available membrane chromatography media [314]. By design, membrane chromatography always outperforms traditional chromatography media in throughput for flow-through steps in terms of mass of product processed per liter of resin per hour. In addition, membranes do not require packing and testing prior to use, nor do they need to be regenerated or sanitized at the end of processing. At flow rates of 10–15 membrane volumes per minute (MV/min) for membrane exchangers compared to 0.25–0.33 column volumes per minute for traditional chromatography media, the productivity difference in flow rate is as high as 60-fold, not accounting for WFI/buffer, time, and resource savings. Simplicity and robustness drive continued growth in market share. Although membrane adsorbers were initially only available as anion and cation exchangers, HIC [58], MMC [164], immobilized metal chelating (IMC) [315, 316], and AC [317–320] options have also become available. HIC steps frequently offer higher capacity and better selectivity for removal of aggregate. The use of a phenyl-based HIC membrane showed moderate capacity [321], which was offset by much higher throughput. HTS approaches to assess capacity and design space have also been described as being comparable to traditional chromatography techniques [322, 323].

The cost savings of a disposable step which requires neither cleaning nor resin lifetime validation ensure AEX membranes remain a staple in many platform processes. This is advantageous as two-step platform processes often suffer from shortened lifetimes of traditional AEX steps [324, 325]. Most membrane anion exchangers offer robust performance at loadings up to 5 kg of product per liter of membrane and 10 kg/L of membrane for two- and three-step platform processes, respectively. Flow-through anion exchange steps are considered to be a robust viral clearance step. A set of publications [301] explored the impact of ligand type, pH,

and conductivity on viral and impurity clearance. It was found that the quaternary amine ligand offered a narrower design space in terms of pH and conductivity as compared to a primary amine which provided sufficient viral and impurity clearance at conductivities up to 20 mS/cm. The primary amine studies demonstrated that buffer species impacted clearance and phosphate buffer should be avoided. Improved clearance over a wider pH and conductivity range was attributed to secondary, hydrophilic interactions, which were not ablated by competing electrostatic interactions, the primary drivers for binding to quaternary amines. However, a trade-off in antibody yield was also found as product could be more strongly retained. With the slow yet constant improvements made in molecular imprinting, it is also feasible to imagine advancements for new membrane adsorbers [326]. A recent publication assessed the potential of a CEX membrane capture followed by an AEX polishing step for an antibody process [319]. Universal applicability aside, because of the wider range of isoelectric points in antibodies, the advantages in throughput, cost, and disposability would still make this an attractive option for a subset of antibodies. Capacity improvements as demonstrated in this recent publication can be assigned to improvements made in membrane design. Impregnation of membranes with different hydrogels, grafted polymers, or nanoparticles have been assessed to improve capacity [314, 327–332]. Similar approaches have been taken to boost the capacity of monoliths.

The concept of monoliths dates back to the 1950s [333]. The concept stagnated after less successful attempts until a renewed interest in the late 1980s revived the field. Today, monoliths are successful in both the analytical and the process arenas. Monoliths do not contain interparticular voids as they are cast as one block of separation media [334, 335]. This forces all mobile phases to flow through the pores, resulting in convective flow which in turn greatly improves mass transfer. Monoliths also offer distinct advantages over traditional chromatography resins in pressure flow properties. However, as with traditional chromatography columns, increased bed height results in pressure increases. Monoliths, such as membrane and traditional chromatography media, are available in a wide variety of base matrices and ligand and linker chemistries. For recent reviews of monoliths please see [336, 337]. Unlike diffusion-limited traditional bead-based chromatography media, monoliths offer capacity independent of flow rate. As larger through-pores are required to enable convective flow, the available binding surface area per unit volume reduces. This does not come into play for larger biomolecules such as DNA, virions, and IgM, which are diffusion limited [338, 339]. For most proteins, however, capacities are significantly lower. The lower capacities are a distinct disadvantage for binding steps, but of lesser concern for flow-through steps where impurity levels are low after affinity-based capture steps. Recent studies have revisited the use of monoliths for Protein A purification [340]. Although capacities are at least four- to eightfold lower as compared to modern traditional chromatography media, this lack of capacity may be overcome through similar, albeit more costly, solutions explored for traditional chromatography media. Optimization of the linker chemistry, ligand structure, and domain repeat number could serve to close the gap. The use of synthetic biomimetic ligands for a more cost-effective

disposable solution was also re-evaluated [341]. Although dye-based affinity chromatography has a longstanding history in the purification of biologics [342], considerations of ligand leachate [343] and lower affinity and selectivity have resulted in a lower market share of this type of ligand.

As discussed earlier, monoliths and membrane chromatography media have had greater success in establishing a presence in the domain of large proteins, large protein complexes, DNA, virus-like particles (VLP), and virion purification. A distinct advantage of membranes and monoliths is the low shear rate in spite of high velocities. As vaccines and viruses are often less stable, the improvement in throughput is also advantageous. Lastly, the manufacturing scale for large biologics is often smaller as well. Some of the concerns and limitations around scale-up and reproducibility of monolithic stationary phases are lessened if the final scale is relatively small because of the high potency of these products. Scale-up should not be taken lightly when considering a monolith process step. Parallel operation of monoliths has been described as a means to overcome this limitation [336]. Although capacities in flow-through mode are satisfactory, large-scale processes require bed volumes of 20–50 L. To obtain well-defined, interconnected pores within the monolithic structure and reproduce the same structure at large scale are far from trivial [344]. Although many studies have been conducted to manufacture large-volume monolithic adsorbents effectively, to date none have been found to be robust and economical. Aside from concerns about pore size distribution and high production cost, the risk of fractures should also be a consideration. The largest commercially available monolith currently produced is an 8-L tubelike monolith with a thickness of less than 5 cm produced by BIA separation. Tube-type monoliths are commercially available whereas rod-type monoliths are not, though they offer distinct advantages. A rod shape enables longer retention and enhanced interactions as compared to short or thin monoliths. The largest rod-shaped monolith described was less than 1 L in volume. Improvements in analytical techniques and real-time monitoring of polymer casting may eventually improve upon the sparsity of studies on the structural homogeneity of the monolithic bed [345]. In addition, the previously cited work on 3D printing could also prove helpful in overcoming some of these issues [48]. Such 3D printing has not been as extensively explored in DSP as in other industries. Applications have been explored in sensor technology [346–349], scaffolding [350–354], material property studies [355–359], and enrichment and purification of cells and viruses [353, 360–362]. Improvements to the throughput, resolution, and capabilities of 3D printing [355] have been made, and cost continues to decrease [363–365]. As significant improvements lead to cost reduction, a better understanding of scale-up and resolution improvements are needed to envision a future state where monoliths are simply printed rather than cast.

Integration of nanoparticles into monoliths and membranes could overcome some of the capacity limitations [345, 366, 367]. Because nanoparticles can move more freely in either monoliths or membranes, their small bead size does not cause back pressure constraints. Polymer grafting may offer another attractive approach to overcome capacity limitations [368–373]. Lastly, merging of the two

technologies may be even more advantageous, although since its prediction [374], prototypes have yet to emerge. One obvious advantage of both membranes and monoliths is that they do not require column packing and packing quality verification. Minimal preparation time to turn around a production suite between campaigns is of distinct advantage to improve time and throughput. Another potential solution to overcome the limitations of monoliths, membranes, and traditional bead-based chromatography media has also been described [375]. Near-colloidal suspensions of loose particles could be flow-packed in traditional chromatography columns. This approach offered much higher hydraulic permeability with porosity values similar to monoliths.

Lastly, large polymer based systems have also been explored [376–378]. Although many large pharmaceutical companies have shifted their focus from traditional small molecules to biologics [379], the difficulties of producing a biological consistently as compared to a synthetic molecule are far from trivial. Similarly, in the purification of biologics, true affinity ligands with high selectivity and low dissociation constants are usually biologics. Similar to biopharmaceuticals, biological ligands suffer from some of the same issues around production complexity, robustness, and cost. Ever since the eventual displacement of small molecules with biologics has been predicted, organic chemists have sought solutions to overcome some of the limitations of small molecules. Assessment of “drugability” of proteins with small molecules is a similar challenge to the concept of high affinity ligands for a specific class of proteins, as in both instances it is desirable to target cavities small enough to be accessible to small ligands, yet ideally surface bound with minimal or only temporary impact to the tertiary structure of the product [380–382]. The use of large polymers has been considered as a solution to mimic better the complex interaction of biologics. One prominent example is the synthesis and imprint of an anti-bee venom fully synthetic polymer antibody analogue [376]. This approach was effective in the neutralization of bee venom in mice. Polymeric binding systems better suited to cover larger surface areas of protein targets have been employed for products larger than bee venom. In one instance, a flexible polymeric system with high affinity for IgG was described [383]. One distinct advantage of the development of this system was the capture at slightly acidic pH and release at neutral pH. Initial variants were only available as free polymer and two-phase systems did show some promise (BMS internal results). Recent developments with polymer immobilized on chromatography beads also showed promise for more traditional purification approaches [384]. This opens the door to a fully disposable affinity capture with polymeric ligands either linked to or contained in monoliths.

4.3 Modeling in Downstream Processing (DSP)

Modeling can be an incredibly useful tool if used appropriately. As stated by George Box [385]: “Since all models are wrong the scientist cannot obtain a

‘correct’ one by excessive elaboration. On the contrary, following William of Occam, he should seek an economical description of natural phenomena. Just as the ability to devise simple but evocative models is the signature of the great scientist so overelaboration and over parameterization is often the mark of mediocrity.” The reverse of the coin is, of course, oversimplification. As stated by Alfred North Whitehead “Seek simplicity but distrust it” [386]. A good model, whether statistical (empirical) or mechanistic, seeks balance and critical review to ensure relevance.

This section attempts to summarize the current state of modeling and its various applications in downstream processing, referencing several excellent resources for further reading. A few brief definitions are provided for clarity. A mechanistic model is a set of equations, based on known physical and chemical relationships. A mechanistic model is used to predict the behavior of a system. This approach is also sometimes referred to as a white box or first principles approach [387]. Mechanistic models are superior to statistical models, as the latter do not attempt to explain the why or how of the relationship of an input with an output (black box approach) [388]. Statistical models are more adaptable and can be foundational in understanding an underlying mechanism. In addition, because of their wide use, vast improvements in robustness have been made. Two points to consider when using statistical models: (1) large amount of data is needed for neural networks ($n > 100$) and (2) model predictions may be restricted to the observation space of the training data (i.e., extrapolation can be challenging, depending on the variation in the data used to calibrate the model). An example is the Bayesian regularization of neural networks (BRANN) [389]. This approach improves robustness while reducing the need for lengthy cross-validation. This approach can be used in QSAR modeling. It is difficult both to overfit and to overtrain when using BRANN. Models must be useful and practical; it is most practical to think of model types as tool sets in a tool box when it comes to problem solving. Recent approaches have made strides in combining both approaches with success which indicates that a tool box approach is warranted [390–393]. An excellent perspective on a knowledge-based approach and the integration of both mechanistic and deterministic approaches in process development has recently been published [394]. Although modeling has a longstanding history in industry, use of predictive modeling in process development is still the exception rather than the rule. It has been a work of decades to close the gap from primary protein sequence to predictive modeling of both stability profiles and process behavior and, as of yet, there is no closure. Black box approaches, such as evolutionary computing algorithms, have been successfully employed in chromatography optimization [50, 395, 396]. The implementation of genetic algorithms and neural networks, tools which are part of the evolutionary computing algorithm repertoire [397], mimic natural evolution to grasp trends and patterns in complex information.

Models focusing on the optimization and prediction of individual unit operations are most abundant. In chromatography, models are system-dependent but go back to a common theme. Current models focus on describing concentration in three phases: interstitial, pore, and stationary. To simplify, independence of

concentration transport from its cross-sectional position is assumed. The interstitial concentration is then dependent on time and axial position alone and affected by convection, diffusion, and interaction with the pore or stationary phase. The pore phase concentration is defined through an induced velocity, an axial dispersion coefficient, bead porosity, and the chromatography bead radius. An effective mass transfer coefficient can be derived from these two terms which models both pore and film mass transfer. A lumped model distributes concentration depending on porosity, although a general rate model splits the mass transfer coefficient into a pore diffusion coefficient and a film transfer coefficient. Lastly, to complete the model, an isotherm is added to model adsorption or desorption. Different models may be used, such as kinetic, multi-component Langmuir isotherms, or kinetic steric mass action (SMA) isotherms. For greater detail on these concepts, please refer to previously published work [398–401].

Because of their simplicity, the earliest publications focus on model protein systems. In one of the earlier publications, the impact of column regeneration on equilibrium and kinetic characteristics of the adsorption of albumin on AEX was studied [402]. Early on, it was clear that models require experimental calibration [403]. Productivity factors, such as yield and throughput, were often targeted, although conditions considered at the times were not as relevant to preparative chromatography because of small bead size and low titer feed streams. Principal component analysis (PCA), a deterministic statistical model, was found to be useful for the modeling of the separation of four model proteins by CEX [404]. A recent publication compared mechanistic and deterministic models for a three-model protein system [393]. The statistical model was not necessarily set up for success. Both the mechanistic and deterministic approaches showed distinct advantages and disadvantages. Speed and good prediction for simple correlations were the hallmark of the deterministic approach. In addition, the relative importance of each input for the response could be determined. The statistical model failed, however, in its prediction of linear gradient elution (LGE). However, some of the perceived failure is a misconception around the capabilities of deterministic statistical models. It is well understood that predictions outside the explored design space are not feasible with DoEs. Even at the edges of the design space, predictability decreases as uncertainty increases. The mechanistic model was predictive outside the training data set, albeit at the cost of much higher experimental input. As a design space can easily be augmented, a bias in the comparison is evident. The authors proposed a combinatorial approach moving forward as the empirical data were useful inputs of the mechanistic model validation as well. However, the authors believe that, if you choose a good set of informative experiments, the experimental effort can be very low using mechanistic models.

A desire for both robustness and a better understanding of HIC steps has led to several publications. The HIC step was tasked with the separation of HMW species from the monomeric product [173, 174, 405]. It was necessary to apply a binary, irreversible adsorption isotherm as a subset of the aggregate species bound irreversibly to the chromatography medium. Inclusion of this irreversible binding was critical in obtaining a predictive model as the irreversible adsorption strongly

affected selectivity and separation. One of the distinct advantages of this model was the ability to extend important relevant parameters, such as column packing quality, column scale, and adsorbent ligand density. Ligand density effects were only evident when explored in a weak binding regime at the point of product elution. Adsorption isotherm measurements were proposed as a tool to predict performance of new batches of chromatography resin tested under weakly bound conditions. Different models have been employed to describe and predict IEX. A series of publications focused on building a simple model to allow scale-up of linear gradients in CEX [406–408]. A lumped kinetic model was used to evaluate different CEX stationary phases [409]. The model was developed from measured retention factors, mass transfer parameters, and peak fitting under overloaded conditions to determine saturation capacity. The model proved predictive for yield and elution product concentration for a continuous multi-column process. The steric mass action (SMA) model was evaluated for either a set of gradient and frontal experiments or an inverse method for parameter estimation [410]. In this instance, the inverse method was found to be more predictive and to provide the best fit of model response and data. Modeling has also been used to determine best pooling strategies where the separation of monomeric antibody from high molecular weight (HMW) species was modeled with a general rate model and temperature dependent diffusion coupled to a SMA model [411]. This was found to be predictive at low loadings, but not high loadings and pH. However, it served sufficiently well to determine a robust pooling strategy to enrich for monomer. An AEX model determined the impact of electrostatic surface charge for deamidated product variants [412]. Metropolis Monte Carlo (MC) simulations provided a methodological framework for interaction analysis where electrostatic surface properties alone did not suffice. It was found that small distribution shifts had a big impact on retention times for different charge variants. The impact of salt and pH for IEX was explored in a mechanistic model [413]. Here the characteristic protein charge as a response variable of pH and the equilibrium constant was applied to the SMA model. The separation of a monoclonal antibody from product related HMW impurities was determined through a set of pH gradient experiments. The lumped rate model predicted separation of loadings up to 48 g/L. At preparative scale and with modern, high-capacity CEX resins, loadings of up to 120–150 g/L are feasible, indicating limitations to this approach. A deterministic model proved to be more predictive for higher loadings [286]. This model was capable of describing protein binding over broad ranges of protein and salt concentrations, which offers a suitable compromise until current, approximate mechanistic models are precise enough to predict industrial (high loading) scenarios reliably (Fig. 6).

One of the distinct advantages of this model was the ease of data generation. Batch isotherm data in combination with a systematic empirical interpolation (EI) scheme, a lumped kinetic model with rate parameters determined from HETP measurements for non-binding conditions, could be used to predict column behavior numerically. Two different modes of chromatography were assessed and successfully predicted. Membrane exchangers have also been modeled and a recent publication determined optimal parameters for purification of virus-like particles

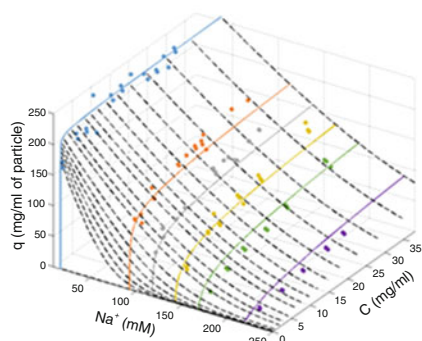
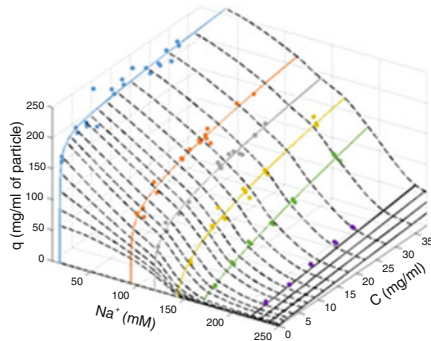
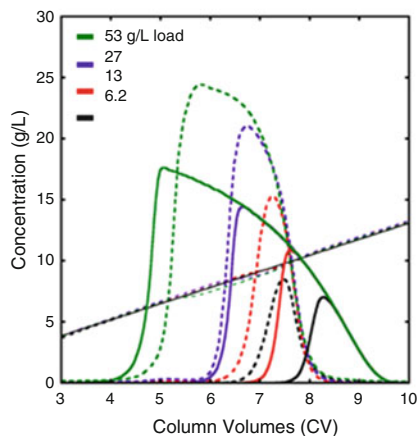
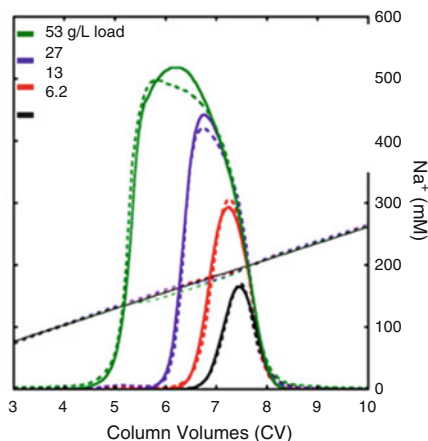
a Isotherm description: Global fit with SMA model**b** Isotherm description: SI Method**c** Salt gradient elution (SMA model)**d** Salt gradient elution (SI model)

Fig. 6 (a) Isotherm prediction based on the steric mass action (SMA) model. (b) Isotherm prediction based on a systematic interpolation (SI) model. (c) Observed (*dotted lines*) and predicted (*solid lines*) elution behavior in a linear gradient (SMA model). (d) Observed (*dotted lines*) and predicted (*solid lines*) elution behavior in a linear gradient (SI model)

(VLPs) from Sf9 insect cells [306]. A radial lumped model in combination with a stoichiometric displacement model (SDM) isotherm was employed. SDM, which adds a solvent release term, and radial lumped models are better suited for modeling of membrane chromatographers and large molecules such as VLPs. A combinatorial approach of HTS and in-silico process modeling, simulation, and optimization allowed one to minimize the consumption of feed material. Although acceptable impurity levels were reached, recoveries were not as high as desired. This publication suggested that a better understanding of mass transport phenomena in membranes would be needed. Continuous purification processes have also been modeled [414, 415]. In one approach [223], a model was built of breakthrough experiments and then validated on a lab-scale multi-column system. A distinct advantage of this

approach was that the model lacked mathematical complexity. So its wider adoption may be more straightforward. Protein A performance has also been modeled targeting wash and elution pH specifically to minimize HMW formation [416]. An effective model required the foundational understanding of the DBC design space, which could be obtained either empirically or through modeling within process-relevant pressure flow constraints. Use of HTS was proposed to minimize column work and maximize data density. A lumped desorption-kinetic limiting model was able to simulate and predict packed column experimental outcomes and identify robust operating conditions. A pilot-scale confirmatory experiment served to ensure that larger-scale results matched predicted and small-scale results. Unfortunately, it was found that the design space for one antibody was not robust as it failed target impurity and recovery values at pH and conductivity control ranges that are impractical at preparative scale (± 0.1 units in pH and ± 1 S/cm).

Ideally, process development would shorten because of predicting chromatographic behavior based on physicochemical properties of a protein. First attempts at predicting behaviors based on properties have been published, although approaches taken vary. Quantitative structure–property relationships (QSPR) and quantitative structure–activity relationship (QSAR) models have been proposed early on [417]. QSPR and QSAR models of protein SMA parameters and free energy changes in IEC were modeled through a support vector machine (SVM), a regression technique for supervised learning models. Model proteins served to build molecular descriptors around physicochemical protein properties to describe stoichiometry, equilibrium, steric effects, and binding affinity in IEC. As a proof of concept, multiscale modeling was used to predict the chromatographic separation through use of isotherm parameters obtained from the QSPR models. Although this approach focused on single gradient separation, it could serve in theory to predict any preparative chromatographic separation. As an early attempt to simulate chromatographic behavior of proteins directly from their crystal structures, this was a first step toward going from primary sequence to purification with minimal process development investment. Different approaches, employing different models, were also described. One publication aimed at predicting IEX retention for different IEX resins [418]. A simplified colloidal model was employed, and insights on the advantages and disadvantages of this approach are provided elsewhere [419]. The distinct advantage of this approach was that protein sequence and stationary phase structural properties suffice to predict mAb retention factors as a function of pH and salt concentration. The effect of high mass loading was not studied here and it is not expected that this model would work well for high loading IEX steps because of limitations of this model at high concentrations [419]. With the advent of HTS, however, the same information can be obtained empirically within days and without the limitations of pH and concentration boundaries. Another publication employed molecular dynamics (MD) to model the retention of proteins in IEX [420]. The developed MD tool simulated interactions of proteins and adsorbents for a spectrum of protein and confirmed the validity of the model empirical with LGE experiments. Good overlap of model and experimental results was found. In addition, the main contributors to CEX binding consisted of arginine

and for AEX binding of aspartic acid residues. In another publication, protein sequence, structure, and dynamics were described by MD, not only to predict chromatographic separation but also with the goal of maximizing recovery of biological activity [390]. As seen before, sufficient granularity or resolution in the model was critical for predictive modeling of protein–ligand interactions. Here, modeling was simplified by studying insulin variants. As insulin is very small at 5.8 kDa, less computing power has to be invested to model the peptides surface properties. Affinity chromatography for mAbs was also modeled based on the structure and the free energy of interaction of the antibody and affinity ligand [421]. However, the effect of the environment (buffer solution, spacer, support matrix) has not yet been taken into consideration, although a procedure to do so has been outlined.

Interesting empirical approaches to gain a more fundamental understanding of protein adsorption were also recently described [422, 423]. Single molecules could be tracked to determine the role of pH for protein separations. This allows in situ study of molecular dynamics at the liquid–solid interface that determines the ensemble chromatographic elution performance. Translational surface diffusion was observed and it was recommended that good models would also consider translational surface diffusion for silica-based adsorption. Further studies elucidating differences in protein interactions driven by secondary and tertiary structure were also proposed.

Holistic process modeling in the spirit of quality by design (QbD) has also been published. One publication analyzed the impact of an IEX and HIC step and feed adjustments between the two steps [424]. A CHO-expressed IgG1 product was analyzed in terms of pH and salt environment impact on product and impurities. A mathematical model encompassing kinetic and thermodynamic parameters was used to optimize operating conditions for column loading and chromatographic elution in the integrated process. Baumann et al. [425] integrated cell culture and downstream processing through a set of micro-scale cultivation experiments and chromatography modeling. This was served to optimize the entire process as a whole, which ensured that overall process output rather than separate optimization of titer and process was considered. It was found that the highest titer did not correspond to the most favorable overall outcome, stressing the importance of an integrated optimization approach. Although a model product expressed in *Escherichia coli* was employed, its approach is universal enough to foresee its implementation in industrial processes. A second paper considered integration of IEX downstream process optimization [426]. A flow sheet approach was taken based on salt gradient elution profiles and peak cutting for two IEX steps. The concerted approach performed better with the same time and effort investment, although a drawback seen was the inability to compute for the two columns in parallel. Feed adjustment between steps was not taken into consideration, which caused difficulties, and it was recommended to integrate ultra/diafiltration, dilution, and mixing unit operations through, for example, a black box model. Issues of suboptimal feed adjustments could however be modeled. An in-house software toolbox was developed to accomplish this. Modeling of the entire DSP is feasible in

theory, as the software can adapt to larger numbers of proteins and scale-up, so that it offers general utility in process modeling.

Modelling of plant utilization and its optimization has found applications as well. One group has published different approaches [53, 145] offering several different approaches to maximize facility usage. De-bottlenecking and cost considerations are stressed as well as visual aids to understand process robustness better. A decision-making tool also indicated that both affinity capture and buffer hold capacity would become bottlenecks first in legacy facilities. One of the main drawbacks of most commercially available as well as in-house facility modeling tools are accurate figures of cost. It can be quite difficult to obtain accurate figures which, in turn, lessens the accuracy of modeling where process economy is concerned. Computational fluid dynamics (CFD) is another useful tool in the decision-making process for biological processes. In downstream processing, temperature profiles, column and pump hardware impact, packed resin permeability impact, shear stress and channeling and vortexing could be esteemed and predicted by CFD [427]. Lastly, modeling has been utilized to gain a better understanding of in-process stability. Empirical models have been proposed to predict in-process hold times better [428]. Molecular simulations have also been employed [429]. The free energy of pairwise mABs association was studied with a conformational sampling algorithm and a scoring function. Weak interactions were explored with a computational method. The free energy landscape was explored with a steric interaction component, electrostatic interactions, and a non-polar component. Two model antibodies with the similar IgG1 backbones but small sequence differences were evaluated for differences in viscosity and propensity to aggregate. The simulations were found to be in good agreement with empirical observations and this model mAb can provide insights on the self-association responsible for bulk solution behavior and aggregation, important concerns in the manufacture of mAbs.

5 Concluding Remarks and Future Direction

Revisiting the theme of three key questions in understanding the intent of this review, the reader is cautioned that viability of a technology is not only dependent on the actual utility of a technology, but, as highlighted previously, a certain critical mass of early adopters or innovators is also needed to move a technology past the initial chasm. Once moved past its critical mass, the merits of a technology and the advent of competing, emerging technologies are the main drivers for success or failure of disruptive technologies. With that said, demand for high quality biologics should continue to increase in the coming decades and disruptive technologies are needed to help overcome some of the challenges. The boundary conditions, however, can change substantially. Amounts to be produced and the number of products will increase. Additionally, there will be fewer or even no blockbuster biologics because of a shift toward stratified medicine. Many of the current blockbusters will be “running out of patent” during the next few years. The trend toward stratified

therapeutics will support a change in plant design, aiming for highly flexible multi-purpose facilities for small production volumes. Such a development will push innovation not only in the development of single-use technologies but also in the development of non-chromatographic, continuous, and flexible downstream operations. The implementation of new separation technologies on an industrial scale require significant investments in development, scale-up, and validation, including associated risks. To bridge the gap, innovations need to announce themselves by distinct increases in efficiency and cost reduction. Careful evaluation to avoid design-in of a non-sustainable technology continues to be critical. A successful, balanced evaluation requires a multidisciplinary team and an inclusive approach. To reduce production volumes, the need for facility flexibility and faster turn-arounds increases. This can lead toward implementation of disposables in manufacturing, continuous processing, and dedicated but decentralized manufacturing concepts in containers.

Consequently, process development needs to adapt and truly integrate USP and DSP development. Cost increases of this approach toward integrated process optimization are offset by gains in overall process improvements. Integration of downstream unit operations reduces feedstream and processing adjustments. An integration of ion exchange and hydrophobic interaction chromatography would, for example, permit the use of only one buffer system for both operations [430]. Further trends in purification technology include other non-chromatographic processes such as membrane adsorption, crystallization, precipitation, or aqueous two-phase separation. Utilization of these previously difficult-to-optimize technologies is bolstered through high-throughput screening, modeling and in-depth understanding of the design space. Predictive biopharmaceutical process design will gain importance in process development as well through improved commercial viability. Furthermore, a strategic shift in biomanufacturing toward continuous processing along with SUS is gaining popularity, enabling companies to produce lower cost drugs with increased flexibility and to enable on-demand manufacturing of multiple products. This has created increased interest in the use of membrane adsorbers but these systems currently have much lower binding capacities than traditional resins and need significant improvement before commercialization. Advances in SUS can lead to its increased implementation in biopharmaceutical production. At present, single-use product lines already encompass a complete facility from storage bags to chromatographic columns, enabling a holistic, single-use, downstream process. Current and future research continues to address limits in scale, standardization, validation, and interactions of disposables and process, a major bottleneck in the implementation of a fully single-use process up to now [298]. Establishment of single-use membrane adsorbers, moving bed chromatography, countercurrent chromatography, two-phase systems, and novel chromatography resins is already on the horizon. Another new approach that will play a major role in DSP is quantitative structure–activity relationship (QSAR) modeling. Data analysis methods and statistics are applied to develop models that can accurately predict biological activities or predict the chromatographic separation of proteins based on their structures [431, 432].

Economic pressures coupled with the desire to gain access to the emerging markets make process innovation imperative [298]. The convergence of a growing interest in academia and industry, maturing of enabling commercially available technologies, and encouragement from regulatory bodies continue to drive bioprocessing toward process innovation and conversion of bioproduction from batch to continuous processing. An end-to-end bioprocessing platform offers a novel enabling solution to address the inefficiencies of the current non-standardized, large, inflexible, and low throughput facilities. Inherently, continuous flow processes are more conducive to process optimization and control, particularly with respect to steady-state product quality. Further, the end-to-end bioprocessing platform is lean, flexible, and portable, facilitating implementation in the emerging markets throughout the world.

Acknowledgments The authors would like to thank Yan Yao, Ph.D., Associate Director, Bristol-Myers Squibb, and Gregory Barker, Ph.D., Senior Engineer, Bristol-Myers Squibb for their critical review and feedback. The authors would also like to thank Professor Giorgio Carta, Ph.D., School of Engineering and Applied Sciences, University of Virginia, Arch Creasy, graduate student at the School of Engineering and Applied Sciences, University of Virginia, and aforementioned colleagues Gregory Barker and Yan Yao for the generous permission to reproduce Fig. 6, and Gregory Barker for the generous provision of Fig. 4.

References

1. Wilson ID, Allard ER, Cooke M, Poole CF (2000) Encyclopedia of separation science. Academic, San Diego
2. Cohn EJ (1947) The separation of blood into fractions of therapeutic value. *Ann Int Med* 26:341–352
3. Constantino P (2002) Basel, Switzerland Patent No 20050106181 A1
4. Hagen AJ, Oliver CN, Sitrin R (1997) Optimization and scale-up of solvent extraction in purification of hepatitis A virus (VAQTA™). *Biotechnol Bioeng* 56:83–88
5. Kniskern PJ, Miller WJ, Hagopian A, Charlotte C, Hennessey J, John P et al (1998) US Patent No 5,847,112
6. Merieux I (1980) Belgium Patent No B74899
7. Yavordios D, Cousin M (1983) France Patent No. 0071515A1
8. Stern SA, Noble RD (1995) Membrane separations technology. Principles and applications. Elsevier, Amsterdam
9. Peterson EA, Sober HA (1954) Chromatography of proteins. I Cellulose ion-exchange; adsorbents. *J Am Chem Soc* 78(4):751–755
10. Porath J, Flodin P (1959) Gel filtration: a method for desalting and group separation. *Nature* 183:1657
11. Lea DJ, Sehon AH (1961) Preparation of synthetic gels for chromatography of macromolecules. *Can J Chem* 40:159
12. Vaughan MF (1960) Fractionation of polystyrene by gel filtration. *Nature* 188:55
13. Hjerten S (1961) Agarose as an anticonvection agent in zone electrophoresis. *Biochim Biophys Acta* 53(3):514–517
14. Hjerten S (1964) The preparation of agarose spheres for the chromatography of molecules and particles. *Biochim Biophys Acta* 79:393–398

15. March SC, Parikh I, Cuatrecasas P (1974) Affinity chromatography — old problems and new approaches- in immobilized biochemicals and affinity chromatography- part of the series *Advances in Experimental Medicine and Biology* (Vol. 42): Springer
16. Williams KW, Smith RC (1975) Recent advances in column chromatography. *Prog Med Chem* 12:105–158
17. Hofstee BH (1973) Protein binding by agarose carrying hydrophobic groups in conjunction with charges. *Biochem Biophys Res Commun* 50(3):751–757
18. Er-EI Z, Zaidenzaig Y, Shaltiel S (1972) Hydrocarbon-coated sepharoses. Use in the purification of glycogen phosphorylase. *Biochem Biophys Res Commun* 49(2):383–390
19. Hjerten S (1973) Some general aspects of hydrophobic interaction chromatography. *J Chromatogr A*:325–331
20. Yon RJ (1972) Chromatography of lipophilic proteins on adsorbents containing mixed hydrophobic and ionic groups. *Biochem J* 126(3):765–767
21. Gagnon P (2012) Technology trends in antibody purification. *J Chromatogr A* 1221:57–70
22. Gottschalk U (2011) The future of downstream processing. *pharmtech.com*. Retrieved from <http://www.pharmtech.com/print/212418?page=full>
23. Jon HC, Zarbis-Papastoitis G (2011) Advances in the production and downstream processing of antibodies. *N Biotechnol* 28(5)
24. Low D, O’Leary R, Pujar NS (2007) Future of antibody purification. *J Chromatogr B* 848:48–63
25. Marichal-Gallardo PA, Alvarez MM (2012) State-of-the-art in downstream processing of monoclonal antibodies: process trends in design and validation. *Biotechnol Prog* 28(4):899–916
26. Zhou JX, Tressel T, Yang X, Seewoester T (2008) Implementation of advanced technologies in commercial monoclonal antibody production. *Biotechnol J* 3:1185–1200
27. Kelley B (2007) Very large scale monoclonal antibody purification: the case for conventional unit operations. *Biotechnol Prog* 23(5):995–1008. doi:[10.1021/bp070117s](https://doi.org/10.1021/bp070117s)
28. Shukla AA, Thoemmes J (2010) Recent advances in large-scale production of monoclonal antibodies and related proteins. *Trends Biotechnol* 28(5):253–261
29. Banholzer WF, Vosejka LJ (2011) Risk taking and effective R&D management. *Annu Rev Chem Biomol Eng* 2:173–188
30. Hueske A-K, Endrikat J, Guenther E (2015) External environment, the innovating organization, and its individuals: a multilevel model for identifying innovation barriers accounting for social uncertainties. *J Eng Technol Manage* 35:45–70
31. Rogers M (2012) Energy = Innovation: 10 disruptive technologies. *Sustainability & Resource Productivity Summer*
32. Saunila M, Ukko J (2014) Intangible aspects of innovation capability in SMEs: impacts of size and industry. *J Eng Technol Manage* 33:32–46
33. Story VM, Daniels K, Zolkiewski J, Dainty AJ (2014) The barriers and consequences of radical innovations: introduction to the issue. *Ind Mark Manag* 24:1271–1277
34. Weiss JC, Dale BC (1998) Diffusing against mature technology: Issues and Strategy. *Ind Mark Manag* 27:293–304
35. Tswett M (1903) O novoy kategorii adsorbtsionnykh yavleniy i o primenenii ikh k biokhimičeskomu analizu (A new category of adsorption phenomena and their use in biochemical analysis). *Trudy Varhavskego Obshchestva estevoispytatelei Otd B* 14:20–39
36. Curling J (2007) Process chromatography: five decades of innovation. *BioPharm Int* 13–18:48
37. Guiochon G, Felinger A, Shirazi DG, Katti AM (2006) Introduction. In: *Fundamentals of preparative and nonlinear chromatography*. Academic
38. Grace JR, Leckner B, Zhu J, Cheng Y (2005) Fluidized beds. In: *Multiphase flow handbook*. CRC Press

39. Bartels CR, Kleiman G, Irish DB, Korzun JN (1957) United States/New York Patent No. US 2786831 A
40. Buijs A, Wesselingh JA (1980) Batch fluidized ion-exchange columns for streams containing suspended particles. *J Chromatogr* 201:319–327
41. Burns MA, Graves DJ (1985) Continuous affinity chromatography using a magnetically stabilized fluidized bed. *Biotechnol Prog* 1:95–103
42. Draeger MN, Chase HA (1990) Liquid fluidized beds for protein purification. *Chem Eng Symp Ser No*, 1, 12.11–12.12
43. Nixon L, Koval CA, Xu L, Noble RD, Slaff GS (1991) The effects of magnetic stabilization on the structure and performance of fluidized beds. *Bioseparations* 2:217–230
44. Hjorth R (1997) Expanded-bed adsorption in industrial bioprocessing: recent developments. *Trends Biotechnol* 15:230–235
45. Noppe W, Van Damme U, Gent N, Geeraerts F, Vanhoorelbeke K, Deckmyn H (2003) Biology and life sciences. In: Downstream: EBA '02 abstracts: Extended reports from the 4th international conference on expanded bed adsorption, St Petersburg, 2002. Amersham Biosciences, Uppsala
46. Flickinger MC (2013) Downstream industrial biotechnology: recovery and purification. Wiley
47. Silver N (2012) The signal and the noise. The Penguin Press, New York
48. Fee CJ, Nawada S, Dimartino S (2014) 3D printed porous media columns with fine control of column packing geometry. *J Chromatogr A* 1333:18–24
49. Rathore AV, Velayudhan A (2003) In: Cazes J (ed) Scale-up and optimization in preparative chromatography. Marcel Dekker, Inc
50. Allmendinger R, Simaria AS, Turner R, Farid SS (2014) Closed-loop optimization of chromatography column sizing strategies in biopharmaceutical manufacture. *J Chem Technol Biotechnol* 89(10):1481–1490. doi:[10.1002/jctb.4267](https://doi.org/10.1002/jctb.4267)
51. Farid SS (2007) Process economics of industrial monoclonal antibody manufacture. *J Chromatogr B Analyt Technol Biomed Life Sci* 848(1):8–18. doi:[10.1016/j.jchromb.2006.07.037](https://doi.org/10.1016/j.jchromb.2006.07.037)
52. Hammerschmidt N, Tscheliessnig A, Sommer R, Helk B, Jungbauer A (2014) Economics of recombinant antibody production processes at various scales: industry-standard compared to continuous precipitation. *Biotechnol J* 9(6):766–775. doi:[10.1002/biot.201300480](https://doi.org/10.1002/biot.201300480)
53. Liu S, Simaria AS, Farid SS, Papageorgiou LG (2013) Designing cost-effective biopharmaceutical facilities using mixed-integer optimization. *Biotechnol Prog* 29(6):1472–1483. doi:[10.1002/btpr.1795](https://doi.org/10.1002/btpr.1795)
54. Schugerl K, Hubbuch J (2005) Integrated bioprocesses. *Curr Opin Microbiol* 8(3):294–300. doi:[10.1016/j.mib.2005.01.002](https://doi.org/10.1016/j.mib.2005.01.002)
55. Kelley B (2009) Industrialization of mAb production technology: the bioprocessing industry at a crossroads. *MAbs* 1(5):443–452
56. Werner RG (2004) Economic aspects of commercial manufacture of biopharmaceuticals. *J Biotechnol* 113(1-3):171–182. doi:[10.1016/j.jbiotec.2004.04.036](https://doi.org/10.1016/j.jbiotec.2004.04.036)
57. Liu HF, Ma J, Winter C, Bayer R (2010) Recovery and purification process development for monoclonal antibody production. *MAbs* 2(5):480–499
58. Lu Y, Williamson B, Gillespie R (2009) Recent advancement in application of hydrophobic interaction chromatography for aggregate removal in industrial purification process. *Curr Pharm Biotechnol* 10(4):427–433
59. Nfor BK, Verhaert PD, van der Wielen LA, Hubbuch J, Ottens M (2009) Rational and systematic protein purification process development: the next generation. *Trends Biotechnol* 27(12):673–679. doi:[10.1016/j.tibtech.2009.09.002](https://doi.org/10.1016/j.tibtech.2009.09.002)
60. Shukla AA, Hubbard B, Tressel T, Guhan S, Low D (2007) Downstream processing of monoclonal antibodies—application of platform approaches. *J Chromatogr B Analyt Technol Biomed Life Sci* 848(1):28–39. doi:[10.1016/j.jchromb.2006.09.026](https://doi.org/10.1016/j.jchromb.2006.09.026)

61. Singh N, Arunkumar A, Chollangi S, Tan ZG, Borys M, Li ZJ (2015) Clarification technologies for monoclonal antibody manufacturing processes: current state and future perspectives. *Biotechnol Bioeng*. doi:[10.1002/bit.25810](https://doi.org/10.1002/bit.25810)
62. Tao Y, Ibraheem A, Conley L, Cecchini D, Ghose S (2014) Evaluation of high-capacity cation exchange chromatography for direct capture of monoclonal antibodies from high-titer cell culture processes. *Biotechnol Bioeng* 111(7):1354–1364. doi:[10.1002/bit.25192](https://doi.org/10.1002/bit.25192)
63. Wang F, Yan X, Song L, Wang P, Lu D, Feng J, et al. (2013) A novel ‘pipeline’ system for downstream preparation of therapeutic monoclonal antibodies. *Biotechnol Lett* 35(9):1411–1418. doi:[10.1007/s10529-013-1234-2](https://doi.org/10.1007/s10529-013-1234-2)
64. Aaberg PM, Houshmand H, Ljungloef A, Van AJ (2005) A method for chromatographic purification, US20,070,213,513 A1
65. Agner E (2003) Method for displacement chromatography, US6,576,134 B1
66. Cramer SM, Moore JA, Kundu A, Li Y, Jayaraman G (1995) Displacement chromatography of proteins using low molecular weight displacers, US5,478,924 A
67. Cramer SM, Shukla AA, Sunasara KM (2001) Low molecular weight displacers for protein purification in hydrophobic interaction and reversed phase chromatographic systems, US6,239,262 B1
68. Eriksson K, Johansson HJ, Olsson U (2007) Method of separating monomeric protein(s), US20,090,264,630 A1
69. Godavarti R, Iskra T (2006) Methods of purifying fc region containing proteins, US20,070,082,367 A1
70. Poll DJ, Harding DRK, Hancock WS (1986) High performance liquid chromatography mobile phase, US4,909,941 A
71. Shujun S (2013) Arginine wash in protein purification using affinity chromatography. US Patents No. 8,350,013 B2
72. Staby A (2000) Ion exchange chromatography of proteins and peptides with an organic modifier in the elution step, US6,451,987 B1
73. Sun S, Gallo C (2011) Arginine derivative wash in protein purification using affinity chromatography. US Patent No. 7,714,111
74. Sundberg R, Hopfer R (2004) Removal of bacterial endotoxin in a protein solution by immobilized metal affinity chromatography, US20,040,112,832 A1
75. Gillespie R, Vunnum S, Nguyen T, Macneil S (2012). *J Chromatogr A* 1251:101–110
76. Van Alstine J, Houshmand H, Ljunglof A, Aberg PM (2007) Method for chromatographic purification, US20,070,213,513 A1
77. Wang C, Coppola G, Chumsae C (2015) Protein purification using displacement chromatography: Google Patents
78. Satzer P, Tscheliebnigg A, Sommer R, Jungbauer A (2014) Separation of recombinant antibodies from DNA using divalent cations. *Eng Life Sci* 14(5)
79. Tsumoto K, Ejima D, Senczuk AM, Kita Y, Arakawa T (2007) Effects of salts on protein–surface interactions: applications for column chromatography. *J Pharm Sci* 96(7):1677–1690. doi:[10.1002/jps.20821](https://doi.org/10.1002/jps.20821)
80. Johansson K, Frederiksen SS, Degerman M, Breil MP, Mollerup JM, Nilsson B (2015) Combined effects of potassium chloride and ethanol as mobile phase modulators on hydrophobic interaction and reversed-phase chromatography of three insulin variants. *J Chromatogr A* 1381:64–73. doi:[10.1016/j.chroma.2014.12.081](https://doi.org/10.1016/j.chroma.2014.12.081)
81. Ngo TT, Narinesingh D (2008) Kosmotropes enhance the yield of antibody purified by affinity chromatography using immobilized bacterial immunoglobulin binding proteins. *J Immunoassay Immunochem* 29(1):105–115. doi:[10.1080/15321810701735203](https://doi.org/10.1080/15321810701735203)
82. Arakawa T, Tsumoto K, Nagase K, Ejima D (2007) The effects of arginine on protein binding and elution in hydrophobic interaction and ion-exchange chromatography. *Protein Expr Purif* 54(1):110–116. doi:[10.1016/j.pep.2007.02.010](https://doi.org/10.1016/j.pep.2007.02.010)
83. Cochet S, Hasnaoui MH, Debbia M, Kroviarski Y, Lambin P, Cartron JP, Bertrand O (1994) Chromatography of human immunoglobulin G on immobilized drimarene rubine R/K-5BL. Study of mild, efficient elution procedures. *J Chromatogr A* 663(2):175–186

84. Lin M-F, Williams C, Murray MV, Ropp PA (2005) Removal of lipopolysaccharides from protein–lipopolysaccharide complexes by nonflammable solvents. *J Chromatogr B* 816(1–2):167–174. doi:[10.1016/j.jchromb.2004.11.029](https://doi.org/10.1016/j.jchromb.2004.11.029)
85. Hou Y, Cramer SM (2011) Evaluation of selectivity in multimodal anion exchange systems: a priori prediction of protein retention and examination of mobile phase modifier effects. *J Chromatogr A* 1218(43):7813–7820. doi:[10.1016/j.chroma.2011.08.080](https://doi.org/10.1016/j.chroma.2011.08.080)
86. Hirano A, Arakawa T, Kameda T (2014) Interaction of arginine with Canto MMC in multimodal chromatography. *J Chromatogr A* 1338:58–66. doi:[10.1016/j.chroma.2014.02.053](https://doi.org/10.1016/j.chroma.2014.02.053)
87. Hirano A, Maruyama T, Shiraki K, Arakawa T, Kameda T (2014) Mechanism of protein desorption from 4-mercaptoethylpyridine resins by arginine solutions. *J Chromatogr A* 1373:141–148. doi:[10.1016/j.chroma.2014.11.032](https://doi.org/10.1016/j.chroma.2014.11.032)
88. Holstein MA, Parimal S, McCallum SA, Cramer SM (2012) Mobile phase modifier effects in multimodal cation exchange chromatography. *Biotechnol Bioeng* 109(1):176–186. doi:[10.1002/bit.23318](https://doi.org/10.1002/bit.23318)
89. Herzer S, Bhargale A, Barker G, Chowdhary I, Conover M, O'Mara BW, et al. (2015) Development and scale-up of the recovery and purification of a domain antibody Fc fusion protein-comparison of a two and three-step approach. *Biotechnol Bioeng* 112(7):1417–1428. doi:[10.1002/bit.25561](https://doi.org/10.1002/bit.25561)
90. Liu Z, Gurgel PV, Carbonell RG (2012) Purification of human immunoglobulins A, G and M from Cohn fraction II/III by small peptide affinity chromatography. *J Chromatogr A* 1262:169–179. doi:[10.1016/j.chroma.2012.09.026](https://doi.org/10.1016/j.chroma.2012.09.026)
91. Ishihara T, Hosono M (2015) Improving impurities clearance by amino acids addition to buffer solutions for chromatographic purifications of monoclonal antibodies. *J Chromatogr B Analyt Technol Biomed Life Sci* 995–996:107–114. doi:[10.1016/j.jchromb.2015.05.018](https://doi.org/10.1016/j.jchromb.2015.05.018)
92. Bolton GR, Boesch AW, Basha J, Lacasse DP, Kelley BD, Acharya H (2011) Effect of protein and solution properties on the Donnan effect during the ultrafiltration of proteins. *Biotechnol Prog* 27(1):140–152. doi:[10.1002/btpr.523](https://doi.org/10.1002/btpr.523)
93. Miao F, Velayudhan A, DiBella E, Shervin J, Felo M, Teeters M, Alred P (2009) Theoretical analysis of excipient concentrations during the final ultrafiltration/diafiltration step of therapeutic antibody. *Biotechnol Prog* 25(4):964–972. doi:[10.1002/btpr.168](https://doi.org/10.1002/btpr.168)
94. Stoner MR, Fischer N, Nixon L, Buckel S, Benke M, Austin F, et al. (2004) Protein-solute interactions affect the outcome of ultrafiltration/diafiltration operations. *J Pharm Sci* 93(9):2332–2342. doi:[10.1002/jps.20145](https://doi.org/10.1002/jps.20145)
95. Shukla D, Zamolo L, Cavallotti C, Trout BL (2011) Understanding the role of arginine as an eluent in affinity chromatography via molecular computations. *J Phys Chem B* 115(11):2645–2654. doi:[10.1021/jp111156z](https://doi.org/10.1021/jp111156z)
96. Bolton GR, Selvitelli KR, Iliescu I, Cecchini DJ (2015) Inactivation of viruses using novel protein A wash buffers. *Biotechnol Prog* 31(2):406–413. doi:[10.1002/btpr.2024](https://doi.org/10.1002/btpr.2024)
97. Chollangi S, Parker R, Singh N, Li Y, Borys M, Li Z (2015) Development of robust antibody purification by optimizing protein-A chromatography in combination with precipitation methodologies. *Biotechnol Bioeng* 112(11):2292–2304. doi:[10.1002/bit.25639](https://doi.org/10.1002/bit.25639)
98. Frauenschuh A, Bill K (2011) Wash solution and method for affinity chromatography, US20,120,283,416 A1
99. Gillespie R, Vunnum S, Nguyen T, Macneil S (2012) Protein purification, US20,120,149,878 A1
100. Srajer Gajdosik M, Clifton J, Josic D (2012) Sample displacement chromatography as a method for purification of proteins and peptides from complex mixtures. *J Chromatogr A* 1239:1–9. doi:[10.1016/j.chroma.2012.03.046](https://doi.org/10.1016/j.chroma.2012.03.046)
101. Huang B, Liu FF, Dong XY, Sun Y (2011) Molecular mechanism of the affinity interactions between protein A and human immunoglobulin G1 revealed by molecular simulations. *J Phys Chem B* 115(14):4168–4176. doi:[10.1021/jp111216g](https://doi.org/10.1021/jp111216g)
102. Schuler G, Reinacher M (1991) Development and optimization of a single-step procedure using protein A affinity chromatography to isolate murine IgG1 monoclonal antibodies from hybridoma supernatants. *J Chromatogr* 587(1):61–70

103. Lund LN, Christensen T, Toone E, Houen G, Staby A, St Hilaire PM (2011) Exploring variation in binding of protein A and protein G to immunoglobulin type G by isothermal titration calorimetry. *J Mol Recognit* 24(6):945–952. doi:[10.1002/jmr.1140](https://doi.org/10.1002/jmr.1140)
104. Baumgartner K, Oelmeier SA, Hubbuch J (2015) The influence of mixed salts on the capacity of hic adsorbers: a predictive correlation to the surface tension and the aggregation temperature. *Biotechnol Prog*. doi:[10.1002/btpr.2166](https://doi.org/10.1002/btpr.2166)
105. Werner A, Hasse H (2013) Experimental study and modeling of the influence of mixed electrolytes on adsorption of macromolecules on a hydrophobic resin. *J Chromatogr A* 1315:135–144. doi:[10.1016/j.chroma.2013.09.071](https://doi.org/10.1016/j.chroma.2013.09.071)
106. Wolfe LS, Barringer CP, Mostafa SS, Shukla AA (2014) Multimodal chromatography: characterization of protein binding and selectivity enhancement through mobile phase modulators. *J Chromatogr A* 1340:151–156. doi:[10.1016/j.chroma.2014.02.086](https://doi.org/10.1016/j.chroma.2014.02.086)
107. Sipple et al., in preparation
108. Duhamel RC, Schur PH, Brendel K, Meezan E (1979) pH gradient elution of human IgG1, IgG2 and IgG4 from protein A-sepharose. *J Immunol Methods* 31(3–4):211–217
109. Gaza-Bulseco G, Hickman K, Sinicropi-Yao S, Hurkmans K, Chumsae C, Liu H (2009) Effect of the conserved oligosaccharides of recombinant monoclonal antibodies on the separation by protein A and protein G chromatography. *J Chromatogr A* 1216(12):2382–2387. doi:[10.1016/j.chroma.2009.01.014](https://doi.org/10.1016/j.chroma.2009.01.014)
110. Ejima D, Yumioka R, Tsumoto K, Arakawa T (2005) Effective elution of antibodies by arginine and arginine derivatives in affinity column chromatography. *Anal Biochem* 345(2):250–257. doi:[10.1016/j.ab.2005.07.004](https://doi.org/10.1016/j.ab.2005.07.004)
111. Roben PW, Salem AN, Silverman GJ (1995) VH3 family antibodies bind domain D of staphylococcal protein A. *J Immunol* 154(12):6437–6445
112. Hogwood CE, Ahmad SS, Tarrant RD, Bracewell DG, Smales CM (2015) An ultra scale-down approach identifies host cell protein differences across a panel of mAb producing CHO cell line variants. *Biotechnol J*. doi:[10.1002/biot.201500010](https://doi.org/10.1002/biot.201500010)
113. Levy NE, Valente KN, Choe LH, Lee KH, Lenhoff AM (2014) Identification and characterization of host cell protein product-associated impurities in monoclonal antibody bioprocessing. *Biotechnol Bioeng* 111(5):904–912. doi:[10.1002/bit.25158](https://doi.org/10.1002/bit.25158)
114. Levy NE, Valente KN, Lee KH, Lenhoff AM (2015) Host cell protein impurities in chromatographic polishing steps for monoclonal antibody purification. *Biotechnol Bioeng*. doi:[10.1002/bit.25882](https://doi.org/10.1002/bit.25882)
115. Lewus RA, Levy NE, Lenhoff AM, Sandler SI (2015) A comparative study of monoclonal antibodies. 1. Phase behavior and protein-protein interactions. *Biotechnol Prog* 31(1):268–276. doi:[10.1002/btpr.2011](https://doi.org/10.1002/btpr.2011)
116. Tarrant RD, Velez-Suberbie ML, Tait AS, Smales CM, Bracewell DG (2012) Host cell protein adsorption characteristics during protein A chromatography. *Biotechnol Prog* 28(4):1037–1044. doi:[10.1002/btpr.1581](https://doi.org/10.1002/btpr.1581)
117. Zhang S, Daniels W, Salm J, Glynn J, Martin J, Gallo C, et al. (2016) Nature of foulants and fouling mechanism in the protein A MabSelect resin cycled in a monoclonal antibody purification process. *Biotechnol Bioeng* 113(1):141–149. doi:[10.1002/bit.25706](https://doi.org/10.1002/bit.25706)
118. Liu Z, Mostafa SS, Shukla AA (2015) A comparison of protein A chromatographic stationary phases: performance characteristics for monoclonal antibody purification. *Biotechnol Appl Biochem* 62(1):37–47. doi:[10.1002/bab.1243](https://doi.org/10.1002/bab.1243)
119. Swinnen K, Krul A, Van Goidsenhoven I, Van Tichelt N, Roosen A, Van Houdt K (2007) Performance comparison of protein A affinity resins for the purification of monoclonal antibodies. *J Chromatogr B Analyt Technol Biomed Life Sci* 848(1):97–107. doi:[10.1016/j.jchromb.2006.04.050](https://doi.org/10.1016/j.jchromb.2006.04.050)
120. McCaw TR, Koepf EK, Conley L (2014) Evaluation of a novel methacrylate-based protein A resin for the purification of immunoglobulins and Fc-fusion proteins. *Biotechnol Prog* 30(5):1125–1136. doi:[10.1002/btpr.1951](https://doi.org/10.1002/btpr.1951)

121. Meininger DP, Rance M, Starovasnik MA, Fairbrother WJ, Skelton NJ (2000) Characterization of the binding interface between the E-domain of Staphylococcal protein A and an antibody Fv-fragment. *Biochemistry* 39(1):26–36
122. Moks T, Abrahmsen L, Nilsson B, Hellman U, Sjoquist J, Uhlen M (1986) Staphylococcal protein A consists of five IgG-binding domains. *Eur J Biochem* 156(3):637–643
123. Ghose S, Hubbard B, Cramer SM (2007) Binding capacity differences for antibodies and Fc-fusion proteins on protein A chromatographic materials. *Biotechnol Bioeng* 96(4):768–779. doi:[10.1002/bit.21044](https://doi.org/10.1002/bit.21044)
124. Sjoquist J, Wadso II (1971) A thermochemical study of the reaction between protein A from *S. aureus* and fragment Fc from immunoglobulin G. *FEBS Lett* 14(4):254–256
125. Yang L, Biswas ME, Chen P (2003) Study of binding between protein A and immunoglobulin G using a surface tension probe. *Biophys J* 84(1):509–522. doi:[10.1016/s0006-3495\(03\)74870-x](https://doi.org/10.1016/s0006-3495(03)74870-x)
126. Muller E, Vajda J (2016) Routes to improve binding capacities of affinity resins demonstrated for protein A chromatography. *J Chromatogr B Analyt Technol Biomed Life Sci*. doi:[10.1016/j.jchromb.2016.01.036](https://doi.org/10.1016/j.jchromb.2016.01.036)
127. Lifesciences G Application note (AA ed., Vol. 29190587). GE Lifesciences, Piscataway
128. Ghose S, Zhang J, Conley L, Caple R, Williams KP, Cecchini D (2014) Maximizing binding capacity for protein A chromatography. *Biotechnol Prog* 30(6):1335–1340. doi:[10.1002/btpr.1980](https://doi.org/10.1002/btpr.1980)
129. Gonzalez-Valdez J, Yoshikawa A, Weinberg J, Benavides J, Rito-Palomares M, Przybycien TM (2014) Toward improving selectivity in affinity chromatography with PEGylated affinity ligands: the performance of PEGylated protein A. *Biotechnol Prog* 30(6):1364–1379. doi:[10.1002/btpr.1994](https://doi.org/10.1002/btpr.1994)
130. Gulich S, Uhlen M, Hober S (2000) Protein engineering of an IgG-binding domain allows milder elution conditions during affinity chromatography. *J Biotechnol* 76(2-3):233–244
131. Pabst TM, Palmgren R, Forss A, Vasic J, Fonseca M, Thompson C, et al. (2014) Engineering of novel Staphylococcal protein A ligands to enable milder elution pH and high dynamic binding capacity. *J Chromatogr A* 1362:180–185. doi:[10.1016/j.chroma.2014.08.046](https://doi.org/10.1016/j.chroma.2014.08.046)
132. Watanabe H, Matsumaru H, Ooishi A, Honda S (2013) Structure-based histidine substitution for optimizing pH-sensitive Staphylococcus protein A. *J Chromatogr B Analyt Technol Biomed Life Sci* 929:155–160. doi:[10.1016/j.jchromb.2013.04.029](https://doi.org/10.1016/j.jchromb.2013.04.029)
133. Xia HF, Liang ZD, Wang SL, Wu PQ, Jin XH (2014) Molecular modification of protein A to improve the elution pH and alkali resistance in affinity chromatography. *Appl Biochem Biotechnol* 172(8):4002–4012. doi:[10.1007/s12010-014-0818-1](https://doi.org/10.1007/s12010-014-0818-1)
134. Gagnon P, Nian R (2016) Conformational plasticity of IgG during protein A affinity chromatography. *J Chromatogr A* 1433:98–105. doi:[10.1016/j.chroma.2016.01.022](https://doi.org/10.1016/j.chroma.2016.01.022)
135. Gagnon P, Nian R, Leong D, Hoi A (2015) Transient conformational modification of immunoglobulin G during purification by protein A affinity chromatography. *J Chromatogr A* 1395:136–142. doi:[10.1016/j.chroma.2015.03.080](https://doi.org/10.1016/j.chroma.2015.03.080)
136. Mazzer AR, Perraud X, Halley J, O'Hara J, Bracewell DG (2015) Protein A chromatography increases monoclonal antibody aggregation rate during subsequent low pH virus inactivation hold. *J Chromatogr A* 1415:83–90. doi:[10.1016/j.chroma.2015.08.068](https://doi.org/10.1016/j.chroma.2015.08.068)
137. Shukla AA, Gupta P, Han X (2007) Protein aggregation kinetics during protein A chromatography. Case study for an Fc fusion protein. *J Chromatogr A* 1171(1-2):22–28. doi:[10.1016/j.chroma.2007.09.040](https://doi.org/10.1016/j.chroma.2007.09.040)
138. Zhang S, Xu K, Daniels W, Salm J, Glynn J, Martin J, et al. (2016) Structural and functional characteristics of virgin and fouled Protein A MabSelect resin cycled in a monoclonal antibody purification process. *Biotechnol Bioeng* 113(2):367–375. doi:[10.1002/bit.25708](https://doi.org/10.1002/bit.25708)
139. Boulet-Audet M, Byrne B, Kazarian SG (2015) Cleaning-in-place of immunoaffinity resins monitored by in situ ATR-FTIR spectroscopy. *Anal Bioanal Chem* 407(23):7111–7122. doi:[10.1007/s00216-015-8871-3](https://doi.org/10.1007/s00216-015-8871-3)

140. Rogers M, Hiraoka-Sutow M, Mak P, Mann F, Lebreton B (2009) Development of a rapid sanitization solution for silica-based protein A affinity adsorbents. *J Chromatogr A* 1216 (21):4589–4596. doi:[10.1016/j.chroma.2009.03.065](https://doi.org/10.1016/j.chroma.2009.03.065)
141. Wang L, Dembecki J, Jaffe NE, O'Mara BW, Cai H, Sparks CN, et al. (2013) A safe, effective, and facility compatible cleaning in place procedure for affinity resin in large-scale monoclonal antibody purification. *J Chromatogr A* 1308:86–95. doi:[10.1016/j.chroma.2013.07.096](https://doi.org/10.1016/j.chroma.2013.07.096)
142. Gronberg A, Eriksson M, Ersoy M, Johansson HJ (2011) A tool for increasing the lifetime of chromatography resins. *MAbs* 3(2):192–202
143. Yang L, Harding JD, Ivanov AV, Ramasubramanyan N, Dong DD (2015) Effect of cleaning agents and additives on protein A ligand degradation and chromatography performance. *J Chromatogr A* 1385:63–68. doi:[10.1016/j.chroma.2015.01.068](https://doi.org/10.1016/j.chroma.2015.01.068)
144. Saraswat M, Musante L, Ravida A, Shortt B, Byrne B, Holthofer H (2013) Preparative purification of recombinant proteins: current status and future trends. *Biomed Res Int* 2013:312709. doi:[10.1155/2013/312709](https://doi.org/10.1155/2013/312709)
145. Stonier A, Simaria AS, Smith M, Farid SS (2012) Decisional tool to assess current and future process robustness in an antibody purification facility. *Biotechnol Prog* 28(4):1019–1028. doi:[10.1002/btpr.1569](https://doi.org/10.1002/btpr.1569)
146. Liu HF, McCooey B, Duarte T, Myers DE, Hudson T, Amanullah A, et al. (2011) Exploration of overloaded cation exchange chromatography for monoclonal antibody purification. *J Chromatogr A* 1218(39):6943–6952. doi:[10.1016/j.chroma.2011.08.008](https://doi.org/10.1016/j.chroma.2011.08.008)
147. Iskra T, Sacramo A, Gallo C, Godavarti R, Chen S, Lute S, Brorson K (2015) Development of a modular virus clearance package for anion exchange chromatography operated in weak partitioning mode. *Biotechnol Prog* 31(3):750–757. doi:[10.1002/btpr.2080](https://doi.org/10.1002/btpr.2080)
148. Kelley BD, Jakubik J, Vivic S (2008) Viral clearance studies on new and used chromatography resins: critical review of a large dataset. *Biologicals* 36(2):88–98. doi:[10.1016/j.biologicals.2007.08.001](https://doi.org/10.1016/j.biologicals.2007.08.001)
149. Miesegaes GR, Lute SC, Read EK, Brorson KA (2014) Viral clearance by flow-through mode ion exchange columns and membrane adsorbents. *Biotechnol Prog* 30(1):124–131. doi:[10.1002/btpr.1832](https://doi.org/10.1002/btpr.1832)
150. Norling L, Lute S, Emery R, Khuu W, Voisard M, Xu Y, et al. (2005) Impact of multiple re-use of anion-exchange chromatography media on virus removal. *J Chromatogr A* 1069 (1):79–89
151. Roush D (2014) Viral clearance using traditional, well-understood unit operations (session I): anion exchange chromatography (AEX). *PDA J Pharm Sci Technol* 68(1):23–29. doi:[10.5731/pdajpst.2014.00963](https://doi.org/10.5731/pdajpst.2014.00963)
152. Roush D (2015) Viral clearance using traditional, well-understood unit operations: session 1.2. Anion exchange chromatography; and session 1.3. Protein A chromatography. *PDA J Pharm Sci Technol* 69(1):154–162. doi:[10.5731/pdajpst.2015.01039](https://doi.org/10.5731/pdajpst.2015.01039)
153. Strauss DM, Cano T, Cai N, Delucchi H, Plancarte M, Coleman D, et al. (2010) Strategies for developing design spaces for viral clearance by anion exchange chromatography during monoclonal antibody production. *Biotechnol Prog* 26(3):750–755. doi:[10.1002/btpr.385](https://doi.org/10.1002/btpr.385)
154. Strauss DM, Gorrell J, Plancarte M, Blank GS, Chen Q, Yang B (2009) Anion exchange chromatography provides a robust, predictable process to ensure viral safety of biotechnology products. *Biotechnol Bioeng* 102(1):168–175. doi:[10.1002/bit.22051](https://doi.org/10.1002/bit.22051)
155. Zhou JX, Solamo F, Hong T, Shearer M, Tressel T (2008) Viral clearance using disposable systems in monoclonal antibody commercial downstream processing. *Biotechnol Bioeng* 100 (3):488–496. doi:[10.1002/bit.21781](https://doi.org/10.1002/bit.21781)
156. Yao Y, Lenhoff AM (2006) Pore size distributions of ion exchangers and relation to protein binding capacity. *J Chromatogr A* 1126(1-2):107–119. doi:[10.1016/j.chroma.2006.06.057](https://doi.org/10.1016/j.chroma.2006.06.057)
157. DePhillips P, Lenhoff AM (2000) Pore size distributions of cation-exchange adsorbents determined by inverse size-exclusion chromatography. *J Chromatogr A* 883(1-2):39–54

158. Tao Y, Carta G (2008) Rapid monoclonal antibody adsorption on dextran-grafted agarose media for ion-exchange chromatography. *J Chromatogr A* 1211(1-2):70–79. doi:[10.1016/j.chroma.2008.09.096](https://doi.org/10.1016/j.chroma.2008.09.096)
159. Bowes BD, Lenhoff AM (2011) Protein adsorption and transport in dextran-modified ion-exchange media. II Intraparticle uptake and column breakthrough. *J Chromatogr A* 1218(29):4698–4708. doi:[10.1016/j.chroma.2011.05.054](https://doi.org/10.1016/j.chroma.2011.05.054)
160. Lenhoff AM (2011) Protein adsorption and transport in polymer-functionalized ion-exchangers. *J Chromatogr A* 1218(49):8748–8759. doi:[10.1016/j.chroma.2011.06.061](https://doi.org/10.1016/j.chroma.2011.06.061)
161. Perez Almodovar EX, Glatz B, Carta G (2012) Counterion effects on protein adsorption equilibrium and kinetics in polymer-grafted cation exchangers. *J Chromatogr A* 1253:83–93. doi:[10.1016/j.chroma.2012.06.100](https://doi.org/10.1016/j.chroma.2012.06.100)
162. Perez-Almodovar EX, Wu Y, Carta G (2012) Multicomponent adsorption of monoclonal antibodies on macroporous and polymer grafted cation exchangers. *J Chromatogr A* 1264:48–56. doi:[10.1016/j.chroma.2012.09.064](https://doi.org/10.1016/j.chroma.2012.09.064)
163. Xu Z, Li J, Zhou JX (2012) Process development for robust removal of aggregates using cation exchange chromatography in monoclonal antibody purification with implementation of quality by design. *Prep Biochem Biotechnol* 42(2):183–202. doi:[10.1080/10826068.2012.654572](https://doi.org/10.1080/10826068.2012.654572)
164. Riordan WT, Heilmann SM, Brorson K, Seshadri K, Etzel MR (2009) Salt tolerant membrane adsorbers for robust impurity clearance. *Biotechnol Prog* 25(6):1695–1702. doi:[10.1002/btpr.256](https://doi.org/10.1002/btpr.256)
165. Yoshimoto N, Itoh D, Isakari Y, Podgornik A, Yamamoto S (2015) Salt tolerant chromatography provides salt tolerance and a better selectivity for protein monomer separations. *Biotechnol J* 10(12):1929–1934. doi:[10.1002/biot.201400550](https://doi.org/10.1002/biot.201400550)
166. Gu F, Chodavarapu K, McCreary D, Plitt TA, Tamoria E, Ni M, et al. (2015) Silica-based strong anion exchange media for protein purification. *J Chromatogr A* 1376:53–63. doi:[10.1016/j.chroma.2014.11.082](https://doi.org/10.1016/j.chroma.2014.11.082)
167. Fang F, Aguilar MI, Hearn MT (1996) Influence of temperature on the retention behaviour of proteins in cation-exchange chromatography. *J Chromatogr A* 729(1-2):49–66
168. Guo J, Carta G (2014) Unfolding and aggregation of a glycosylated monoclonal antibody on a cation exchange column. Part II. Protein structure effects by hydrogen deuterium exchange mass spectrometry. *J Chromatogr A* 1356:129–137. doi:[10.1016/j.chroma.2014.06.038](https://doi.org/10.1016/j.chroma.2014.06.038)
169. Guo J, Carta G (2015) Unfolding and aggregation of monoclonal antibodies on cation exchange columns: effects of resin type, load buffer, and protein stability. *J Chromatogr A* 1388:184–194. doi:[10.1016/j.chroma.2015.02.047](https://doi.org/10.1016/j.chroma.2015.02.047)
170. Gospodarek AM, Hiser DE, O'Connell JP, Fernandez EJ (2014) Unfolding of a model protein on ion exchange and mixed mode chromatography surfaces. *J Chromatogr A* 1355:238–252. doi:[10.1016/j.chroma.2014.06.024](https://doi.org/10.1016/j.chroma.2014.06.024)
171. Bracewell DG, Boychyn M, Baldascini H, Storey SA, Bulmer M, More J, Hoare M (2008) Impact of clarification strategy on chromatographic separations: pre-processing of cell homogenates. *Biotechnol Bioeng* 100(5):941–949. doi:[10.1002/bit.21823](https://doi.org/10.1002/bit.21823)
172. Kramarczyk JF, Kelley BD, Coffman JL (2008) High-throughput screening of chromatographic separations: II. Hydrophobic interaction. *Biotechnol Bioeng* 100(4):707–720. doi:[10.1002/bit.21907](https://doi.org/10.1002/bit.21907)
173. McCue JT (2009) Theory and use of hydrophobic interaction chromatography in protein purification applications. *Methods Enzymol* 463:405–414. doi:[10.1016/s0076-6879\(09\)63025-1](https://doi.org/10.1016/s0076-6879(09)63025-1)
174. McCue JT, Engel P, Ng A, Macniven R, Thommes J (2008) Modeling of protein monomer/aggregate purification and separation using hydrophobic interaction chromatography. *Bioprocess Biosyst Eng* 31(3):261–275. doi:[10.1007/s00449-008-0200-1](https://doi.org/10.1007/s00449-008-0200-1)
175. Deitcher RW, O'Connell JP, Fernandez EJ (2010) Changes in solvent exposure reveal the kinetics and equilibria of adsorbed protein unfolding in hydrophobic interaction chromatography. *J Chromatogr A* 1217(35):5571–5583. doi:[10.1016/j.chroma.2010.06.051](https://doi.org/10.1016/j.chroma.2010.06.051)

176. Deitcher RW, Xiao Y, O'Connell JP, Fernandez EJ (2009) Protein instability during HIC: evidence of unfolding reversibility, and apparent adsorption strength of disulfide bond-reduced alpha-lactalbumin variants. *Biotechnol Bioeng* 102(5):1416–1427. doi:[10.1002/bit.22171](https://doi.org/10.1002/bit.22171)
177. Gospodarek AM, Smatlak ME, O'Connell JP, Fernandez EJ (2011) Protein stability and structure in HIC: hydrogen exchange experiments and COREX calculations. *Langmuir* 27(1):286–295. doi:[10.1021/la103793r](https://doi.org/10.1021/la103793r)
178. Muca R, Marek W, Piatkowski W, Antos D (2010) Influence of the sample-solvent on protein retention, mass transfer and unfolding kinetics in hydrophobic interaction chromatography. *J Chromatogr A* 1217(17):2812–2820. doi:[10.1016/j.chroma.2010.02.043](https://doi.org/10.1016/j.chroma.2010.02.043)
179. Ueberbacher R, Rodler A, Hahn R, Jungbauer A (2010) Hydrophobic interaction chromatography of proteins: thermodynamic analysis of conformational changes. *J Chromatogr A* 1217(2):184–190. doi:[10.1016/j.chroma.2009.05.033](https://doi.org/10.1016/j.chroma.2009.05.033)
180. Eriksson KO, Belew M (2011) Hydrophobic interaction chromatography. *Methods Biochem Anal* 54:165–181
181. To BC, Lenhoff AM (2007) Hydrophobic interaction chromatography of proteins. I The effects of protein and adsorbent properties on retention and recovery. *J Chromatogr A* 1141(2):191–205. doi:[10.1016/j.chroma.2006.12.020](https://doi.org/10.1016/j.chroma.2006.12.020)
182. To BC, Lenhoff AM (2007) Hydrophobic interaction chromatography of proteins. II Solution thermodynamic properties as a determinant of retention. *J Chromatogr A* 1141(2):235–243. doi:[10.1016/j.chroma.2006.12.022](https://doi.org/10.1016/j.chroma.2006.12.022)
183. To BC, Lenhoff AM (2008) Hydrophobic interaction chromatography of proteins III. Transport and kinetic parameters in isocratic elution. *J Chromatogr A* 1205(1-2):46–59. doi:[10.1016/j.chroma.2008.07.079](https://doi.org/10.1016/j.chroma.2008.07.079)
184. To BC, Lenhoff AM (2011) Hydrophobic interaction chromatography of proteins. IV Protein adsorption capacity and transport in preparative mode. *J Chromatogr A* 1218(3):427–440. doi:[10.1016/j.chroma.2010.11.051](https://doi.org/10.1016/j.chroma.2010.11.051)
185. Mirani MR, Rahimpour F (2015) Thermodynamic modelling of hydrophobic interaction chromatography of biomolecules in the presence of salt. *J Chromatogr A* 1422:170–177. doi:[10.1016/j.chroma.2015.10.019](https://doi.org/10.1016/j.chroma.2015.10.019)
186. Nfor BK, Zuluaga DS, Verheijen PJ, Verhaert PD, van der Wielen LA, Ottens M (2011) Model-based rational strategy for chromatographic resin selection. *Biotechnol Prog* 27(6):1629–1643
187. Lemmens R, Olsson U, Nyhammar T, Stadler J (2003) Supercoiled plasmid DNA: selective purification by thiophilic/aromatic adsorption. *J Chromatogr B Analyt Technol Biomed Life Sci* 784(2):291–300
188. Senczuk AM, Klinker R, Arakawa T, Vedantham G, Yigzaw Y (2009) Hydrophobic interaction chromatography in dual salt system increases protein binding capacity. *Biotechnol Bioeng* 103(5):930–935. doi:[10.1002/bit.22313](https://doi.org/10.1002/bit.22313)
189. Melander W, Horvath C (1977) Salt effect on hydrophobic interactions in precipitation and chromatography of proteins: an interpretation of the lyotropic series. *Arch Biochem Biophys* 183(1):200–215
190. Kelley et al (2008) Weak partitioning chromatography for anion exchange purification of monoclonal antibodies. *Biotechnol Bioeng* 101:553–566
191. Johansson B-L, Belew M, Eriksson S, Glad G, Lind O, Maloisel J-L, Norrman N (2003) Preparation and characterization of prototypes for multi-modal separation aimed for capture of positively charged biomolecules at high-salt conditions. *J Chromatogr A* 1016(1):35–49. doi:[10.1016/S0021-9673\(03\)01141-5](https://doi.org/10.1016/S0021-9673(03)01141-5)
192. Yang T, Malmquist G, Johansson B-L, Maloisel J-L, Cramer S (2007) Evaluation of multi-modal high salt binding ion exchange materials. *J Chromatogr A* 1157(1-2):171–177. doi:[10.1016/j.chroma.2007.04.070](https://doi.org/10.1016/j.chroma.2007.04.070)

193. Chen J, Tetrault J, Zhang Y, Wasserman A, Conley G, Dileo M, et al. (2010) The distinctive separation attributes of mixed-mode resins and their application in monoclonal antibody downstream purification process. *J Chromatogr A* 1217(2):216–224. doi:[10.1016/j.chroma.2009.09.047](https://doi.org/10.1016/j.chroma.2009.09.047)
194. Kaleas KA, Tripodi M, Revelli S, Sharma V, Pizarro SA (2014) Evaluation of a multimodal resin for selective capture of CHO-derived monoclonal antibodies directly from harvested cell culture fluid. *J Chromatogr B Analyt Technol Biomed Life Sci* 969:256–263. doi:[10.1016/j.jchromb.2014.08.026](https://doi.org/10.1016/j.jchromb.2014.08.026)
195. Pezzini J, Joucla G, Gantier R, Toueille M, Lomenech A-M, Le Sénéchal C, et al. (2011) Antibody capture by mixed-mode chromatography: a comprehensive study from determination of optimal purification conditions to identification of contaminating host cell proteins. *J Chromatogr A* 1218(45):8197–8208. doi:[10.1016/j.chroma.2011.09.036](https://doi.org/10.1016/j.chroma.2011.09.036)
196. Pete G (2009) IgG aggregate removal by charged-hydrophobic mixed mode chromatography. *Curr Pharm Biotechnol* 10(4):434–439. doi:[10.2174/138920109788488888](https://doi.org/10.2174/138920109788488888)
197. Li P, Xiu G, Mata VG, Grande CA, Rodrigues AE (2006) Expanded bed adsorption/desorption of proteins with Streamline Direct CST I adsorbent. *Biotechnol Bioeng* 94(6):1155–1163. doi:[10.1002/bit.20952](https://doi.org/10.1002/bit.20952)
198. Mollerup JM, Hansen TB, Kidal S, Staby A (2008) Quality by design—thermodynamic modelling of chromatographic separation of proteins. *J Chromatogr A* 1177(2):200–206. doi:[10.1016/j.chroma.2007.08.059](https://doi.org/10.1016/j.chroma.2007.08.059)
199. Nfor BK, Noverraz M, Chilamkurthi S, Verhaert PDEM, van der Wielen LAM, Ottens M (2010) High-throughput isotherm determination and thermodynamic modeling of protein adsorption on mixed mode adsorbents. *J Chromatogr A* 1217(44):6829–6850. doi:[10.1016/j.chroma.2010.07.069](https://doi.org/10.1016/j.chroma.2010.07.069)
200. Pitiot O, Folley L, Vijayalakshmi MA (2001) Protein adsorption on histidyl-aminohexyl-sepharose 4B: I. Study of the mechanistic aspects of adsorption for the separation of human serum albumin from its non-enzymatic glycosylated isoforms (advanced glycosylated end products). *J Chromatogr B Biomed Sci Appl* 758(2):163–172. doi:[10.1016/S0378-4347\(01\)00176-1](https://doi.org/10.1016/S0378-4347(01)00176-1)
201. Wang J, Jenkins EW, Robinson JR, Wilson A, Husson SM (2015) A new multimodal membrane adsorber for monoclonal antibody purifications. *J Membr Sci* 492:137–146. doi:[10.1016/j.memsci.2015.05.013](https://doi.org/10.1016/j.memsci.2015.05.013)
202. Follman DK, Fahrner RL (2004) Factorial screening of antibody purification processes using three chromatography steps without protein A. *J Chromatogr A* 1024(1–2):79–85
203. Komkova EN, Honeyman CH (2014) Mixed-mode chromatography membranes, US20, sss140,238,935 A1
204. Wang J, Sproul RT, Anderson LS, Husson SM (2014) Development of multimodal membrane adsorbers for antibody purification using atom transfer radical polymerization. *Polymer* 55(6):1404–1411. doi:[10.1016/j.polymer.2013.12.023](https://doi.org/10.1016/j.polymer.2013.12.023)
205. Stone MT, Kozlov M (2014) Separating proteins with activated carbon. *Langmuir* 30(27):8046–8055. doi:[10.1021/la501005s](https://doi.org/10.1021/la501005s)
206. Amara J, Boyle J, Yavorsky D, Cacace B (2016) High surface area fiber media with nanofibrillated surface features, WO2,016,036,508 A1
207. Amara J, Cacace B, Yavorsky D, Boyle J (2014) Chromatography media for purifying vaccines and viruses, US20,150,352,465 A1
208. Yavorsky D, Amara J, Umana J, Cataldo W, Kozlov M, Stone M (2015) Chromatography media and method, US20,120,029,176 A1
209. Hardick O, Dods S, Stevens B, Bracewell DG (2015) Nanofiber adsorbents for high productivity continuous downstream processing. *J Biotechnol* 213:74–82. doi:[10.1016/j.jbiotec.2015.01.031](https://doi.org/10.1016/j.jbiotec.2015.01.031)
210. Baur D, Angarita M, Muller-Spath T, Steinebach F, Morbidelli M (2016) Comparison of batch and continuous multi-column protein A capture processes by optimal design. *Biotechnol J*. doi:[10.1002/biot.201500481](https://doi.org/10.1002/biot.201500481)

211. Konstantinov KB, Cooney CL (2015) White paper on continuous bioprocessing, May 20–21, 2014, Continuous manufacturing symposium. *J Pharm Sci* 104(3):813–820. doi:[10.1002/jps.24268](https://doi.org/10.1002/jps.24268)
212. Muller-Spath T, Aumann L, Strohlein G, Kormmann H, Valax P, Delegrange L, et al. (2010) Two step capture and purification of IgG2 using multicolumn countercurrent solvent gradient purification (MCSGP). *Biotechnol Bioeng* 107(6):974–984. doi:[10.1002/bit.22887](https://doi.org/10.1002/bit.22887)
213. Warikoo V, Godawat R, Brower K, Jain S, Cummings D, Simons E, et al. (2012) Integrated continuous production of recombinant therapeutic proteins. *Biotechnol Bioeng* 109(12):3018–3029. doi:[10.1002/bit.24584](https://doi.org/10.1002/bit.24584)
214. Anderson NG (2001) Practical use of continuous processing in developing and scaling up laboratory processes. *Org Process Res Dev* 5(6):613–621. doi:[10.1021/op0100605](https://doi.org/10.1021/op0100605)
215. Fletcher N (2010) Turn batch to continuous processing. *Manufacturing Chemist Pharma*
216. Laird T (2007) Continuous processes in small-scale manufacture. *Org Process Res Dev* 11(6):927–927. doi:[10.1021/op700233e](https://doi.org/10.1021/op700233e)
217. Laird T (2014) Process intensification: engineering for efficiency, sustainability and flexibility. *Org Process Res Dev* 18(1):276–276. doi:[10.1021/op400341e](https://doi.org/10.1021/op400341e)
218. Mazumdar S, Ray SK (2001) Solidification control in continuous casting of steel. *Sadhana* 26(1):179–198. doi:[10.1007/bf02728485](https://doi.org/10.1007/bf02728485)
219. Reay DA, Ramshaw C, Harvey A (2013) Process intensification engineering for efficiency, sustainability and flexibility. Retrieved from <http://search.ebscohost.com/login.aspx?direct=true&scope=site&db=nlebk&db=nlabk&AN=486200>
220. Thomas H (2008) Coming of age. *Chem Eng* (805):38–40. Retrieved from <https://www.scopus.com/inward/record.url?eid=2-s2.0-46949110144&partnerID=40&md5=127c830aa56e3a069687fa3af8cc731e>
221. Ruthven DM, Ching CB (1989) Counter-current and simulated counter-current adsorption separation processes. *Chem Eng Sci* 44(5):1011–1038. doi:[10.1016/0009-2509\(89\)87002-2](https://doi.org/10.1016/0009-2509(89)87002-2)
222. Godawat R, Brower K, Jain S, Konstantinov K, Riske F, Warikoo V (2012) Periodic counter-current chromatography – design and operational considerations for integrated and continuous purification of proteins. *Biotechnol J* 7(12):1496–1508. doi:[10.1002/biot.201200068](https://doi.org/10.1002/biot.201200068)
223. Gjoka X, Rogler K, Martino RA, Gantier R, Schofield M (2015) A straightforward methodology for designing continuous monoclonal antibody capture multi-column chromatography processes. *J Chromatogr A* 1416:38–46. doi:[10.1016/j.chroma.2015.09.005](https://doi.org/10.1016/j.chroma.2015.09.005)
224. Dutta AK, Tran T, Napadensky B, Teella A, Brookhart G, Ropp PA, et al. (2015) Purification of monoclonal antibodies from clarified cell culture fluid using protein A capture continuous countercurrent tangential chromatography. *J Biotechnol* 213:54–64. doi:[10.1016/j.jbiotec.2015.02.026](https://doi.org/10.1016/j.jbiotec.2015.02.026)
225. Napadensky B, Shinkazh O, Teella A, Zydne AL (2013) Continuous countercurrent tangential chromatography for monoclonal antibody purification. *Sep Sci Technol* 48(9):1289–1297. doi:[10.1080/01496395.2013.767837](https://doi.org/10.1080/01496395.2013.767837)
226. Shinkazh O, Kanani D, Barth M, Long M, Hussain D, Zydne AL (2011) Countercurrent tangential chromatography for large-scale protein purification. *Biotechnol Bioeng* 108(3):582–591. doi:[10.1002/bit.22960](https://doi.org/10.1002/bit.22960)
227. Casey C, Gallos T, Alekseev Y, Ayturk E, Pearl S (2011) Protein concentration with single-pass tangential flow filtration (SPTFF). *J Membr Sci* 384(1–2):82–88. doi:[10.1016/j.memsci.2011.09.004](https://doi.org/10.1016/j.memsci.2011.09.004)
228. Dizon-Maspat J, Bourret J, D’Agostini A, Li F (2012) Single pass tangential flow filtration to debottleneck downstream processing for therapeutic antibody production. *Biotechnol Bioeng* 109(4):962–970. doi:[10.1002/bit.24377](https://doi.org/10.1002/bit.24377)
229. Chenette HCS, Robinson JR, Hobley E, Husson SM (2012) Development of high-productivity, strong cation-exchange adsorbers for protein capture by graft polymerization from membranes with different pore sizes. *J Membr Sci* 432–424:43–52. doi:[10.1016/j.memsci.2012.07.040](https://doi.org/10.1016/j.memsci.2012.07.040)

230. Kuczewski M, Schirmer E, Lain B, Zarbis-Papastoitsis G (2011) A single-use purification process for the production of a monoclonal antibody produced in a PER.C6 human cell line. *Biotechnol J* 6(1):56–65. doi:[10.1002/biot.201000292](https://doi.org/10.1002/biot.201000292)
231. Orr V, Zhong L, Moo-Young M, Chou CP (2013) Recent advances in bioprocessing application of membrane chromatography. *Biotechnol Adv* 31(4):450–465. doi:[10.1016/j.biotechadv.2013.01.007](https://doi.org/10.1016/j.biotechadv.2013.01.007)
232. Klutz S, Lobedann M, Bramsiepe C, Schembecker G (2016) Continuous viral inactivation at low pH value in antibody manufacturing. *Chem Eng Process Process Intensification* 102:88–101. doi:[10.1016/j.cep.2016.01.002](https://doi.org/10.1016/j.cep.2016.01.002)
233. Klutz S, Magnus J, Lobedann M, Schwan P, Mäiser B, Niklas J, et al. (2015) Developing the biofacility of the future based on continuous processing and single-use technology. *J Biotechnol* 213:120–130. doi:[10.1016/j.jbiotec.2015.06.388](https://doi.org/10.1016/j.jbiotec.2015.06.388)
234. Pollock J, Bolton G, Coffman J, Ho SV, Bracewell DG, Farid SS (2013) Optimising the design and operation of semi-continuous affinity chromatography for clinical and commercial manufacture. *J Chromatogr A* 1284:17–27. doi:[10.1016/j.chroma.2013.01.082](https://doi.org/10.1016/j.chroma.2013.01.082)
235. Dutta AK, Tan J, Napadensky B, Zydney AL, Shinkazh O (2016) Performance optimization of continuous countercurrent tangential chromatography for antibody capture. *Biotechnol Prog* 32(2):430–439. doi:[10.1002/btpr.2250](https://doi.org/10.1002/btpr.2250)
236. Brower KP, Ryakala VK, Bird R, Godawat R, Riske FJ, Konstantinov K, et al. (2014) Single-step affinity purification of enzyme biotherapeutics: a platform methodology for accelerated process development. *Biotechnol Prog* 30(3):708–717. doi:[10.1002/btpr.1870](https://doi.org/10.1002/btpr.1870)
237. Read EK, Park JT, Shah RB, Riley BS, Brorson KA, Rathore AS (2010) Process analytical technology (PAT) for biopharmaceutical products: Part I. Concepts and applications. *Biotechnol Bioeng* 105(2):276–284. doi:[10.1002/bit.22528](https://doi.org/10.1002/bit.22528)
238. Tharmalingam T, Wu CH, Callahan S, T Goudar C (2015) A framework for real-time glycosylation monitoring (RT-GM) in mammalian cell culture. *Biotechnol Bioeng* 112(6):1146–1154. doi:[10.1002/bit.25520](https://doi.org/10.1002/bit.25520)
239. Box GEP, Hunter JS, Hunter WG (2005) *Statistics for experimenters: design, innovation, and discovery*. Wiley-Interscience
240. Montgomery DC (2012) *Design and analysis of experiments*. Wiley
241. Begley CG (2013) An unappreciated challenge to oncology drug discovery: pitfalls in preclinical research. *Am Soc Clin Oncol Educ Book*, 466–468. doi:[10.1200/EdBook_AM.2013.33.466](https://doi.org/10.1200/EdBook_AM.2013.33.466)
242. Cook DA, Beckman TJ, Bordage G (2007) Quality of reporting of experimental studies in medical education: a systematic review. *Med Educ* 41(8):737–745. doi:[10.1111/j.1365-2923.2007.02777.x](https://doi.org/10.1111/j.1365-2923.2007.02777.x)
243. Deming SN (1986) Chemometrics: an overview. *Clin Chem* 32(9):1702–1706
244. Ilzarbe L, Álvarez MJ, Viles E, Tanco M (2008) Practical applications of design of experiments in the field of engineering: a bibliographical review. *Qual Reliab Eng Int* 24(4):417–428. doi:[10.1002/qre.909](https://doi.org/10.1002/qre.909)
245. Tanco M, Viles E, Ilzarbe L, Alvarez MJ (2007) Manufacturing industries need design of experiments (DoE). *Proceedings of the World Congress on Engineering, II*
246. Nfor BK, Ripic J, van der Padt A, Jacobs M, Ottens M (2012) Model-based high-throughput process development for chromatographic whey proteins separation. *Biotechnol J* 7(10):1221–1232. doi:[10.1002/biot.201200191](https://doi.org/10.1002/biot.201200191)
247. Welsh J (2015) Pushing the limits of high-throughput chromatography process development: current state and future directions. *Pharm Bioprocess* 3(1):1–3
248. Hussain M (2015) A direct qPCR method for residual DNA quantification in monoclonal antibody drugs produced in CHO cells. *J Pharm Biomed Anal* 115:603–606. doi:[10.1016/j.jpba.2015.03.005](https://doi.org/10.1016/j.jpba.2015.03.005)
249. Nissom PM (2007) Specific detection of residual CHO host cell DNA by real-time PCR. *Biologicals* 35(3):211–215. doi:[10.1016/j.biologicals.2006.09.001](https://doi.org/10.1016/j.biologicals.2006.09.001)

250. Diederich P, Hoffmann M, Hubbuch J (2015) High-throughput process development of purification alternatives for the protein avidin. *Biotechnol Prog* 31(4):957–973. doi:[10.1002/btpr.2104](https://doi.org/10.1002/btpr.2104)
251. Van Cleave VH (2003) Validation of immunoassays for anti-drug antibodies. *Dev Biol (Basel)* 112:107–112
252. Antony J (2003) 2 – Fundamentals of design of experiments. In: *Design of experiments for engineers and scientists* (pp. 6–16). Butterworth-Heinemann, Oxford
253. Tye H (2004) Application of statistical ‘design of experiments’ methods in drug discovery. *Drug Discov Today* 9(11):485–491. doi:[10.1016/S1359-6446\(04\)03086-7](https://doi.org/10.1016/S1359-6446(04)03086-7)
254. Donev AN (2004) Design of experiments in the presence of errors in factor levels. *J Stat Plann Inference* 126(2):569–585. doi:[10.1016/j.jspi.2003.09.002](https://doi.org/10.1016/j.jspi.2003.09.002)
255. Franceschini G, Macchietto S (2008) Model-based design of experiments for parameter precision: state of the art. *Chem Eng Sci* 63(19):4846–4872. doi:[10.1016/j.ces.2007.11.034](https://doi.org/10.1016/j.ces.2007.11.034)
256. Rathore AS, Mittal S, Pathak M, Arora A (2014) Guidance for performing multivariate data analysis of bioprocessing data: pitfalls and recommendations. *Biotechnol Prog* 30(4):967–973. doi:[10.1002/btpr.1922](https://doi.org/10.1002/btpr.1922)
257. Shen X, Huang H-C (2006) Optimal model assessment, selection, and combination. *J Am Stat Assoc* 101(474):554–568. doi:[10.1198/016214505000001078](https://doi.org/10.1198/016214505000001078)
258. Shen X, Ye J (2002) Adaptive model selection. *J Am Stat Assoc* 97(457):210–221. doi:[10.1198/016214502753479356](https://doi.org/10.1198/016214502753479356)
259. Zhang B, Shen X, Mumford SL (2012) Generalized degrees of freedom and adaptive model selection in linear mixed-effects models. *Comput Stat Data Anal* 56(3):574–586. doi:[10.1016/j.csda.2011.09.001](https://doi.org/10.1016/j.csda.2011.09.001)
260. Cortina JM (1993) Interaction, nonlinearity, and multicollinearity: implications for multiple regression. *J Manag* 19(4):915–922. doi:[10.1177/014920639301900411](https://doi.org/10.1177/014920639301900411)
261. Strobl C, Malley J, Tutz G (2009) An introduction to recursive partitioning: rationale, application and characteristics of classification and regression trees, bagging and random forests. *Psychol Methods* 14(4):323–348. doi:[10.1037/a0016973](https://doi.org/10.1037/a0016973)
262. Luo L, Yao Y, Gao F (2015) Bayesian improved model migration methodology for fast process modeling by incorporating prior information. *Chem Eng Sci* 134:23–35. doi:[10.1016/j.ces.2015.04.045](https://doi.org/10.1016/j.ces.2015.04.045)
263. Sainani KL (2013) Multivariate regression: the pitfalls of automated variable selection. *PM R* 5(9):791–794. doi:[10.1016/j.pmrj.2013.07.007](https://doi.org/10.1016/j.pmrj.2013.07.007)
264. Berry EM, Coustere-Yakir C, Grover NB (1998) The significance of non-significance. *QJM* 91(9):647–653
265. Gerss J (2006) Not significant—what now? *Zentralbl Gynakol* 128(6):307–310. doi:[10.1055/s-2006-942088](https://doi.org/10.1055/s-2006-942088)
266. Yan W, Hu S, Yang Y, Gao F, Chen T (2011) Bayesian migration of Gaussian process regression for rapid process modeling and optimization. *Chem Eng J* 166(3):1095–1103. doi:[10.1016/j.cej.2010.11.097](https://doi.org/10.1016/j.cej.2010.11.097)
267. Barker GA, Calzada J, Herzer S, Rieble S (2015) Adaptation to high throughput batch chromatography enhances multivariate screening. *Biotechnol J* 10(9):1493–1498. doi:[10.1002/biot.201400671](https://doi.org/10.1002/biot.201400671)
268. Ryan TP (2006) *Modern experimental design*. Wiley-Interscience
269. Eriksson L (2008) *Design of experiments: principles and applications*. Umetrics
270. Harring JR, Weiss BA, Li M (2015) Assessing spurious interaction effects in structural equation modeling: a cautionary note. *Educ Psychol Meas* 75(5):721–738. doi:[10.1177/0013164414565007](https://doi.org/10.1177/0013164414565007)
271. Bedeian AG, Mossholder KW (1994) Simple question, not so simple answer: interpreting interaction terms in moderated multiple regression. *J Manag* 20(1):159–165
272. Myers RH, Montgomery DC, Anderson-Cook CM (2016) *Response surface methodology: process and product optimization using designed experiments*, 3rd edn. Wiley

273. Dougherty S, Simpson JR, Hill RR, Pignatiello JJ, White ED (2014) Augmentation of definitive screening designs (DSD+). *Int J Exp Des Process Optimisation* 4(2):91–115. doi:[10.1504/IJEDPO.2014.066465](https://doi.org/10.1504/IJEDPO.2014.066465)
274. Hecht ES, McCord JP, Muddiman DC (2015) Definitive screening design optimization of mass spectrometry parameters for sensitive comparison of filter and solid phase extraction purified, INLIGHT plasma N-glycans. *Anal Chem* 87(14):7305–7312. doi:[10.1021/acs.analchem.5b01609](https://doi.org/10.1021/acs.analchem.5b01609)
275. Tai M, Ly A, Leung I, Nayar G (2015) Efficient high-throughput biological process characterization: definitive screening design with the Ambr250 bioreactor system. *Biotechnol Prog* 31(5):1388–1395. doi:[10.1002/btpr.2142](https://doi.org/10.1002/btpr.2142)
276. Yang Y-P, D'Amore T (2014) Protein subunit vaccine purification. In: Wen EP, Ellis R, Pujar NS (eds) *Vaccine development and manufacturing* 1st edn. Wiley, Hoboken
277. Goos P (2002) *The optimal design of blocked and split-plot experiments*. Springer
278. Barker TB (2005) *Quality by experimental design*, 3rd edn. CRC Press
279. Fisher RA (1966) *The design of experiments* 8th edn. Oliver and Boyd, Edinburgh
280. Lau CY, Zahidi AAA, Liew OW, Ng TW (2015) A direct heating model to overcome the edge effect in microplates. *J Pharm Biomed Anal* 102:199–202. doi:[10.1016/j.jpba.2014.09.021](https://doi.org/10.1016/j.jpba.2014.09.021)
281. Close EJ, Salm JR, Bracewell DG, Sorensen E (2014) Modelling of industrial biopharmaceutical multicomponent chromatography. *Chem Eng Res Des* 92(7):1304–1314. doi:[10.1016/j.cherd.2013.10.022](https://doi.org/10.1016/j.cherd.2013.10.022)
282. Chhatre S, Bracewell DG, Titchener-Hooker NJ (2009) A microscale approach for predicting the performance of chromatography columns used to recover therapeutic polyclonal antibodies. *J Chromatogr A* 1216(45):7806–7815. doi:[10.1016/j.chroma.2009.09.038](https://doi.org/10.1016/j.chroma.2009.09.038)
283. Hutchinson N, Chhatre S, Baldascini H, Davies JL, Bracewell DG, Hoare M (2009) Ultra scale-down approach to correct dispersive and retentive effects in small-scale columns when predicting larger scale elution profiles. *Biotechnol Prog* 25(4):1103–1110. doi:[10.1002/btpr.172](https://doi.org/10.1002/btpr.172)
284. Lacki KM (2012) High throughput process development of chromatography steps: advantages and limitations of different formats used. *Biotechnol J* 7(10):1192–1202
285. Kelley BD, Switzer M, Bastek P, Kramarczyk JF, Molnar K, Yu T, Coffman J (2008) High-throughput screening of chromatographic separations: IV. Ion-exchange. *Biotechnol Bioeng* 100(5):950–963. doi:[10.1002/bit.21905](https://doi.org/10.1002/bit.21905)
286. Creasy A, Barker G, Yao Y, Carta G (2015) Systematic interpolation method predicts protein chromatographic elution from batch isotherm data without a detailed mechanistic isotherm model. *Biotechnol J* 10:1400–1411
287. Coffman JL, Kramarczyk JF, Kelley BD (2008) High-throughput screening of chromatographic separations: I. Method development and column modeling. *Biotechnol Bioeng* 100(4):605–618. doi:[10.1002/bit.21904](https://doi.org/10.1002/bit.21904)
288. Boning DS, Mozumder PK (1994) DOE/Opt: a system for design of experiments, response surface modeling, and optimization using process and device simulation. *IEEE Trans Semiconductor Manuf* 7(2):233–244. doi:[10.1109/66.286858](https://doi.org/10.1109/66.286858)
289. Wu JCW, Hamada MS (2009) *Experiments: planning, analysis, and optimization*, 2nd edn. Wiley
290. Ladd Effio C, Baumann P, Weigel C, Vormittag P, Middelberg A, Hubbuch J (2016) High-throughput process development of an alternative platform for the production of virus-like particles in *Escherichia coli*. *J Biotechnol* 219:7–19. doi:[10.1016/j.jbiotec.2015.12.018](https://doi.org/10.1016/j.jbiotec.2015.12.018)
291. Staby A, Jensen RH, Bensch M, Hubbuch J, Dunweber DL, Krarup J, et al. (2007) Comparison of chromatographic ion-exchange resins VI. Weak anion-exchange resins. *J Chromatogr A* 1164(1–2):82–94. doi:[10.1016/j.chroma.2007.06.048](https://doi.org/10.1016/j.chroma.2007.06.048)
292. Wensel DL, Kelley BD, Coffman JL (2008) High-throughput screening of chromatographic separations: III. Monoclonal antibodies on ceramic hydroxyapatite. *Biotechnol Bioeng* 100(5):839–854. doi:[10.1002/bit.21906](https://doi.org/10.1002/bit.21906)

293. Traylor SJ, Xu X, Li Y, Jin M, Li ZJ (2014) Adaptation of the pore diffusion model to describe multi-addition batch uptake high-throughput screening experiments. *J Chromatogr A* 1368:100–106. doi:[10.1016/j.chroma.2014.09.058](https://doi.org/10.1016/j.chroma.2014.09.058)
294. Carta G (2012) Predicting protein dynamic binding capacity from batch adsorption tests. *Biotechnol J* 7(10):1216–1220. doi:[10.1002/biot.201200136](https://doi.org/10.1002/biot.201200136)
295. Luo H, Cao M, Newell K, Afdahl C, Wang J, Wang WK, Li Y (2015) Double-peak elution profile of a monoclonal antibody in cation exchange chromatography is caused by histidine-protonation-based charge variants. *J Chromatogr A* 1424:92–101. doi:[10.1016/j.chroma.2015.11.008](https://doi.org/10.1016/j.chroma.2015.11.008)
296. Ho SV, McLaughlin JM, Cue BW, Dunn PJ (2010) Environmental considerations in biologics manufacturing. *Green Chem* 12(5):755–766. doi:[10.1039/B927443J](https://doi.org/10.1039/B927443J)
297. Lopes AG (2015) Single-use in the biopharmaceutical industry: a review of current technology impact, challenges and limitations. *Food Bioprod Process* 93:98–114
298. Shukla AA, Gottschalk U (2013) Single-use disposable technologies for biopharmaceutical manufacturing. *Trends Biotechnol* 31(3):147–154. doi:[10.1016/j.tibtech.2012.10.004](https://doi.org/10.1016/j.tibtech.2012.10.004)
299. Langer ES, Rader R (2015) Future proofing biopharmaceutical manufacturing: the industry seeks a leaner version of itself. *Pharm Bioprocess* 1(5):415–418
300. McNerney T, Thomas A, Senczuk A, Petty K, Zhao X, Piper R, et al. (2015) PDADMAC flocculation of Chinese hamster ovary cells: enabling a centrifuge-less harvest process for monoclonal antibodies. *MABs* 7(2):413–428. doi:[10.1080/19420862.2015.1007824](https://doi.org/10.1080/19420862.2015.1007824)
301. Weaver J, Husson SM, Murphy L, Wickramasinghe SR (2013) Anion exchange membrane adsorbers for flow-through polishing steps: Part I. Clearance of minute virus of mice. *Biotechnol Bioeng* 110(2):491–499. doi:[10.1002/bit.24720](https://doi.org/10.1002/bit.24720)
302. Weaver J, Husson SM, Murphy L, Wickramasinghe SR (2013) Anion exchange membrane adsorbers for flow-through polishing steps: Part II. Virus, host cell protein, DNA clearance, and antibody recovery. *Biotechnol Bioeng* 110(2):500–510. doi:[10.1002/bit.24724](https://doi.org/10.1002/bit.24724)
303. Smrekar V, Smrekar F, Strancar A, Podgornik A (2013) Single step plasmid DNA purification using methacrylate monolith bearing combination of ion-exchange and hydrophobic groups. *J Chromatogr A* 1276:58–64. doi:[10.1016/j.chroma.2012.12.029](https://doi.org/10.1016/j.chroma.2012.12.029)
304. Sousa A, Almeida AM, Cernigoi U, Sousa F, Queiroz JA (2014) Histamine monolith versatility to purify supercoiled plasmid deoxyribonucleic acid from *Escherichia coli* lysate. *J Chromatogr A* 1355:125–133. doi:[10.1016/j.chroma.2014.06.003](https://doi.org/10.1016/j.chroma.2014.06.003)
305. Teeters MA, Conrardy SE, Thomas BL, Root TW, Lightfoot EN (2003) Adsorptive membrane chromatography for purification of plasmid DNA. *J Chromatogr A* 989(1):165–173
306. Ladd Effio C, Hahn T, Seiler J, Oelmeier SA, Asen I, Silberer C, et al. (2016) Modeling and simulation of anion-exchange membrane chromatography for purification of Sf9 insect cell-derived virus-like particles. *J Chromatogr A* 1429:142–154. doi:[10.1016/j.chroma.2015.12.006](https://doi.org/10.1016/j.chroma.2015.12.006)
307. Nestola P, Peixoto C, Villain L, Alves PM, Carrondo MJ, Mota JP (2015) Rational development of two flowthrough purification strategies for adenovirus type 5 and retro virus-like particles. *J Chromatogr A* 1426:91–101. doi:[10.1016/j.chroma.2015.11.037](https://doi.org/10.1016/j.chroma.2015.11.037)
308. Banjac M, Roethl E, Gelhart F, Kramberger P, Jarc BL, Jarc M, Peterka M (2014) Purification of Vero cell derived live replication deficient influenza A and B virus by ion exchange monolith chromatography. *Vaccine* 32(21):2487–2492. doi:[10.1016/j.vaccine.2014.02.086](https://doi.org/10.1016/j.vaccine.2014.02.086)
309. Mundle ST, Giel-Moloney M, Kleanthous H, Pugachev KV, Anderson SF (2015) Preparation of pure, high titer, pseudoinfectious Flavivirus particles by hollow fiber tangential flow filtration and anion exchange chromatography. *Vaccine* 33(35):4255–4260. doi:[10.1016/j.vaccine.2014.09.074](https://doi.org/10.1016/j.vaccine.2014.09.074)
310. Thömmes J, Kula MR (1995) Membrane chromatography—an integrative concept in the downstream processing of proteins. *Biotechnol Prog* 11(4):357–367. doi:[10.1021/bp00034a001](https://doi.org/10.1021/bp00034a001)

311. Francis P, von Lieres E, Haynes CA (2011) Zonal rate model for stacked membrane chromatography. I. Characterizing solute dispersion under flow-through conditions. *J Chromatogr A* 1218(31):5071–5078. doi:[10.1016/j.chroma.2011.05.017](https://doi.org/10.1016/j.chroma.2011.05.017)
312. Zhou JX, Tresselt T, Gottschalk U, Solamo F, Pastor A, Dermawan S, et al. (2006) New Q membrane scale-down model for process-scale antibody purification. *J Chromatogr A* 1134 (1–2):66–73. doi:[10.1016/j.chroma.2006.08.064](https://doi.org/10.1016/j.chroma.2006.08.064)
313. Ghosh P, Vahedipour K, Leuthold M, von Lieres E (2014) Model-based analysis and quantitative prediction of membrane chromatography: extreme scale-up from 0.08 ml to 1200 ml. *J Chromatogr A* 1332:8–13. doi:[10.1016/j.chroma.2014.01.047](https://doi.org/10.1016/j.chroma.2014.01.047)
314. Tatarova I, Faber R, Denoyel R, Polakovic M (2009) Characterization of pore structure of a strong anion-exchange membrane adsorbent under different buffer and salt concentration conditions. *J Chromatogr A* 1216(6):941–947. doi:[10.1016/j.chroma.2008.12.018](https://doi.org/10.1016/j.chroma.2008.12.018)
315. Borsoi-Ribeiro M, Bresolin IT, Vijayalakshmi M, Bueno SM (2013) Behavior of human immunoglobulin G adsorption onto immobilized Cu(II) affinity hollow-fiber membranes. *J Mol Recognit* 26(10):514–520. doi:[10.1002/jmr.2296](https://doi.org/10.1002/jmr.2296)
316. Yavuz H, Bereli N, Yilmaz F, Armutcu C, Denizli A (2015) Antibody purification from human plasma by metal-chelated affinity membranes. *Methods Mol Biol* 1286:43–46. doi:[10.1007/978-1-4939-2447-9_4](https://doi.org/10.1007/978-1-4939-2447-9_4)
317. Francis P, Haynes CA (2009) Scale-up of controlled-shear affinity filtration using computational fluid dynamics. *Biotechnol J* 4(5):665–673. doi:[10.1002/biot.200800331](https://doi.org/10.1002/biot.200800331)
318. Francis P, Martinez DM, Taghipour F, Bowen BD, Haynes CA (2006) Optimizing the rotor design for controlled-shear affinity filtration using computational fluid dynamics. *Biotechnol Bioeng* 95(6):1207–1217. doi:[10.1002/bit.21090](https://doi.org/10.1002/bit.21090)
319. Hou Y, Brower M, Pollard D, Kanani D, Jacquemart R, Kachuik B, Stout J (2015) Advective hydrogel membrane chromatography for monoclonal antibody purification in bioprocessing. *Biotechnol Prog* 31(4):974–982. doi:[10.1002/btpr.2113](https://doi.org/10.1002/btpr.2113)
320. Nachman M, Azad AR, Bailon P (1992) Efficient recovery of recombinant proteins using membrane-based immunoaffinity chromatography (MIC). *Biotechnol Bioeng* 40(5):564–571. doi:[10.1002/bit.260400503](https://doi.org/10.1002/bit.260400503)
321. Kuczewski M, Fraud N, Faber R, Zarbis-Papastoitis G (2010) Development of a polishing step using a hydrophobic interaction membrane adsorber with a PER.C6-derived recombinant antibody. *Biotechnol Bioeng* 105(2):296–305. doi:[10.1002/bit.22538](https://doi.org/10.1002/bit.22538)
322. Kanavova N, Kosior A, Antosova M, Faber R, Polakovic M (2012) Application of a micromembrane chromatography module to the examination of protein adsorption equilibrium. *J Sep Sci* 35(22):3177–3183. doi:[10.1002/jssc.201200396](https://doi.org/10.1002/jssc.201200396)
323. Rathore AS, Muthukumar S (2014) High-throughput process development: II. Membrane chromatography. *Methods Mol Biol* 1129:39–44. doi:[10.1007/978-1-62703-977-2_4](https://doi.org/10.1007/978-1-62703-977-2_4)
324. Close EJ, Salm JR, Iskra T, Sorensen E, Bracewell DG (2013) Fouling of an anion exchange chromatography operation in a monoclonal antibody process: visualization and kinetic studies. *Biotechnol Bioeng* 110(9):2425–2435. doi:[10.1002/bit.24898](https://doi.org/10.1002/bit.24898)
325. Corbett R, Carta G, Iskra T, Gallo C, Godavarti R, Salm JR (2013) Structure and protein adsorption mechanisms of clean and fouled tentacle-type anion exchangers used in a monoclonal antibody polishing step. *J Chromatogr A* 1278:116–125. doi:[10.1016/j.chroma.2013.01.006](https://doi.org/10.1016/j.chroma.2013.01.006)
326. Cheong WJ, Yang SH, Ali F (2013) Molecular imprinted polymers for separation science: a review of reviews. *J Sep Sci* 36(3):609–628. doi:[10.1002/jssc.201200784](https://doi.org/10.1002/jssc.201200784)
327. Bhut BV, Christensen KA, Husson SM (2010) Membrane chromatography: protein purification from *E. coli* lysate using newly designed and commercial anion-exchange stationary phases. *J Chromatogr A* 1217(30):4946–4957. doi:[10.1016/j.chroma.2010.05.049](https://doi.org/10.1016/j.chroma.2010.05.049)
328. Boi C (2007) Membrane adsorbents as purification tools for monoclonal antibody purification. *J Chromatogr B Analyt Technol Biomed Life Sci* 848(1):19–27. doi:[10.1016/j.jchromb.2006.08.044](https://doi.org/10.1016/j.jchromb.2006.08.044)

329. Herigstad MO, Gurgel PV, Carbonell RG (2011) Transport and binding characterization of a novel hybrid particle impregnated membrane material for bioseparations. *Biotechnol Prog* 27 (1):129–139. doi:[10.1002/btpr.502](https://doi.org/10.1002/btpr.502)
330. Nestola P, Villain L, Peixoto C, Martins DL, Alves PM, Carrondo MJ, Mota JP (2014) Impact of grafting on the design of new membrane adsorbers for adenovirus purification. *J Biotechnol* 181:1–11. doi:[10.1016/j.jbiotec.2014.04.003](https://doi.org/10.1016/j.jbiotec.2014.04.003)
331. Sum CH, Chong JY, Wettig S, Slavcev RA (2014) Separation and purification of linear covalently closed deoxyribonucleic acid by Q-anion exchange membrane chromatography. *J Chromatogr A* 1339:214–218. doi:[10.1016/j.chroma.2014.03.016](https://doi.org/10.1016/j.chroma.2014.03.016)
332. Zhong L, Scharer J, Moo-Young M, Fenner D, Crossley L, Honeyman CH, et al. (2011) Potential application of hydrogel-based strong anion-exchange membrane for plasmid DNA purification. *J Chromatogr B Analyt Technol Biomed Life Sci* 879(9–10):564–572. doi:[10.1016/j.jchromb.2011.01.017](https://doi.org/10.1016/j.jchromb.2011.01.017)
333. Mould DL, Syngé RLM (1952) Electrokinetic ultrafiltration analysis of polysaccharides. A new approach to the chromatography of large molecules. *Analyst* 77(921):964–969. doi:[10.1039/AN9527700964](https://doi.org/10.1039/AN9527700964)
334. Hahn R, Jungbauer A (2001) Control method for integrity of continuous beds. *J Chromatogr A* 908(1–2):179–184
335. Xie S, Allington RW, Frechet JM, Svec F (2002) Porous polymer monoliths: an alternative to classical beads. *Adv Biochem Eng Biotechnol* 76:87–125
336. Barroso T, Hussain A, Roque AC, Aguiar-Ricardo A (2013) Functional monolithic platforms: chromatographic tools for antibody purification. *Biotechnol J* 8(6):671–681. doi:[10.1002/biot.201200328](https://doi.org/10.1002/biot.201200328)
337. Rajamanickam V, Herwig C, Spadiut O (2015) Monoliths in bioprocess technology. *Chromatography* 2(2):195–212
338. Trilisky EI, Lenhoff AM (2009) Flow-dependent entrapment of large bioparticles in porous process media. *Biotechnol Bioeng* 104(1):127–133. doi:[10.1002/bit.22370](https://doi.org/10.1002/bit.22370)
339. Trilisky EI, Lenhoff AM (2010) Effect of bioparticle size on dispersion and retention in monolithic and perfusive beds. *J Chromatogr A* 1217(47):7372–7384. doi:[10.1016/j.chroma.2010.09.040](https://doi.org/10.1016/j.chroma.2010.09.040)
340. Herigstad MO, Dimartino S, Boi C, Sarti GC (2015) Experimental characterization of the transport phenomena, adsorption, and elution in a protein A affinity monolithic medium. *J Chromatogr A* 1407:130–138. doi:[10.1016/j.chroma.2015.06.045](https://doi.org/10.1016/j.chroma.2015.06.045)
341. Barroso T, Branco RJ, Aguiar-Ricardo A, Roque AC (2014) Structural evaluation of an alternative Protein A biomimetic ligand for antibody purification. *J Comput Aided Mol Des* 28(1):25–34. doi:[10.1007/s10822-013-9703-1](https://doi.org/10.1007/s10822-013-9703-1)
342. Dean PD, Watson DH (1979) *J Chromatogr* 165:301–319
343. Regnault V, Rivat C, Vallar L, Geschier C, Stolz JF (1992) Purification of biologically active human plasma transthyretin by dye-affinity chromatography: studies on dye leakage and possibility of heat treatment for virus inactivation. *J Chromatogr* 584(1):93–99
344. Ongkudon CM, Kansil T, Wong C (2014) Challenges and strategies in the preparation of large-volume polymer-based monolithic chromatography adsorbents. *J Sep Sci* 37(5):455–464. doi:[10.1002/jssc.201300995](https://doi.org/10.1002/jssc.201300995)
345. Arrua RD, Haddad PR, Hilder EF (2013) Monolithic cryopolymers with embedded nanoparticles. II Capillary liquid chromatography of proteins using charged embedded nanoparticles. *J Chromatogr A* 1311:121–126. doi:[10.1016/j.chroma.2013.08.077](https://doi.org/10.1016/j.chroma.2013.08.077)
346. Guo SZ, Yang X, Heuzey MC, Therriault D (2015) 3D printing of a multifunctional nanocomposite helical liquid sensor. *Nanoscale* 7(15):6451–6456. doi:[10.1039/c5nr00278h](https://doi.org/10.1039/c5nr00278h)
347. Krejčová L, Nejdil L, Rodrigo MA, Zurek M, Matousek M, Hynek D, et al. (2014) 3D printed chip for electrochemical detection of influenza virus labeled with CdS quantum dots. *Biosens Bioelectron* 54:421–427. doi:[10.1016/j.bios.2013.10.031](https://doi.org/10.1016/j.bios.2013.10.031)
348. Lee W, Kwon D, Choi W, Jung GY, Jeon S (2015) 3D-printed microfluidic device for the detection of pathogenic bacteria using size-based separation in helical channel with trapezoid cross-section. *Sci Rep* 5:7717. doi:[10.1038/srep07717](https://doi.org/10.1038/srep07717)

349. Xu L, Gutbrod SR, Bonifas AP, Su Y, Sulkin MS, Lu N, et al. (2014) 3D multifunctional integumentary membranes for spatiotemporal cardiac measurements and stimulation across the entire epicardium. *Nat Commun* 5:3329. doi:[10.1038/ncomms4329](https://doi.org/10.1038/ncomms4329)
350. Guo SZ, Heuzey MC, Theriault D (2014) Properties of polylactide inks for solvent-cast printing of three-dimensional freeform microstructures. *Langmuir* 30(4):1142–1150. doi:[10.1021/la4036425](https://doi.org/10.1021/la4036425)
351. Liu W, Li Y, Feng S, Ning J, Wang J, Gou M, et al. (2014) Magnetically controllable 3D microtissues based on magnetic microcryogels. *Lab Chip* 14(15):2614–2625. doi:[10.1039/c4lc00081a](https://doi.org/10.1039/c4lc00081a)
352. Wang X, Schroder HC, Feng Q, Draenert F, Muller WE (2013) The deep-sea natural products, biogenic polyphosphate (Bio-PolyP) and biogenic silica (Bio-Silica), as biomimetic scaffolds for bone tissue engineering: fabrication of a morphogenetically-active polymer. *Mar Drugs* 11(3):718–746. doi:[10.3390/md11030718](https://doi.org/10.3390/md11030718)
353. Wang X, Schroder HC, Grebenjuk V, Diehl-Seifert B, Mailander V, Steffen R, et al. (2014) The marine sponge-derived inorganic polymers, biosilica and polyphosphate, as morphogenetically active matrices/scaffolds for the differentiation of human multipotent stromal cells: potential application in 3D printing and distraction osteogenesis. *Mar Drugs* 12(2):1131–1147. doi:[10.3390/md12021131](https://doi.org/10.3390/md12021131)
354. Yao Q, Wei B, Liu N, Li C, Guo Y, Shamie AN, et al. (2015) Chondrogenic regeneration using bone marrow clots and a porous polycaprolactone-hydroxyapatite scaffold by three-dimensional printing. *Tissue Eng Part A* 21(7-8):1388–1397. doi:[10.1089/ten.TEA.2014.0280](https://doi.org/10.1089/ten.TEA.2014.0280)
355. Farahani RD, Chizari K, Theriault D (2014) Three-dimensional printing of freeform helical microstructures: a review. *Nanoscale* 6(18):10470–10485. doi:[10.1039/c4nr02041c](https://doi.org/10.1039/c4nr02041c)
356. Guo SZ, Gosselin F, Guerin N, Lanouette AM, Heuzey MC, Theriault D (2013) Solvent-cast three-dimensional printing of multifunctional microsystems. *Small* 9(24):4118–4122. doi:[10.1002/smll.201300975](https://doi.org/10.1002/smll.201300975)
357. Lee H, Fang NX (2012) Micro 3D printing using a digital projector and its application in the study of soft materials mechanics. *J Vis Exp* 69:e4457. doi:[10.3791/4457](https://doi.org/10.3791/4457)
358. Qin Z, Compton BG, Lewis JA, Buehler MJ (2015) Structural optimization of 3D-printed synthetic spider webs for high strength. *Nat Commun* 6:7038. doi:[10.1038/ncomms8038](https://doi.org/10.1038/ncomms8038)
359. Shin D, Kim J, Yoo DS, Kim K (2015) Design of 3D isotropic metamaterial device using smart transformation optics. *Opt Express* 23(17):21892–21898. doi:[10.1364/oe.23.021892](https://doi.org/10.1364/oe.23.021892)
360. Amor-Coarasa A, Kelly JM, Babich JW (2015) Synthesis of [11C]palmitic acid for PET imaging using a single molecular sieve 13X cartridge for reagent trapping, radiolabeling and selective purification. *Nucl Med Biol* 42(8):685–690. doi:[10.1016/j.nucmedbio.2015.03.008](https://doi.org/10.1016/j.nucmedbio.2015.03.008)
361. Gross BC, Anderson KB, Meisel JE, McNitt MI, Spence DM (2015) Polymer coatings in 3D-printed fluidic device channels for improved cellular adherence prior to electrical lysis. *Anal Chem* 87(12):6335–6341. doi:[10.1021/acs.analchem.5b01202](https://doi.org/10.1021/acs.analchem.5b01202)
362. Liu X, Lei Z, Liu F, Liu D, Wang Z (2014) Fabricating three-dimensional carbohydrate hydrogel microarray for lectin-mediated bacterium capturing. *Biosens Bioelectron* 58:92–100. doi:[10.1016/j.bios.2014.02.056](https://doi.org/10.1016/j.bios.2014.02.056)
363. Chae MP, Rozen WM, McMenamin PG, Findlay MW, Spychal RT, Hunter-Smith DJ (2015) Emerging applications of bedside 3D printing in plastic surgery. *Front Surg* 2:25. doi:[10.3389/fsurg.2015.00025](https://doi.org/10.3389/fsurg.2015.00025)
364. Truskett VN, Watts MP (2006) Trends in imprint lithography for biological applications. *Trends Biotechnol* 24(7):312–317. doi:[10.1016/j.tibtech.2006.05.005](https://doi.org/10.1016/j.tibtech.2006.05.005)
365. Tseng P, Murray C, Kim D, Di Carlo D (2014) Research highlights: printing the future of microfabrication. *Lab Chip* 14(9):1491–1495. doi:[10.1039/c4lc90023e](https://doi.org/10.1039/c4lc90023e)
366. Cheong WJ, Ali F, Kim YS, Lee JW (2013) Comprehensive overview of recent preparation and application trends of various open tubular capillary columns in separation science. *J Chromatogr A* 1308:1–24. doi:[10.1016/j.chroma.2013.07.107](https://doi.org/10.1016/j.chroma.2013.07.107)

367. Thayer JR, Flook KJ, Woodruff A, Rao S, Pohl CA (2010) New monolith technology for automated anion-exchange purification of nucleic acids. *J Chromatogr B Analyt Technol Biomed Life Sci* 878(13-14):933–941. doi:[10.1016/j.jchromb.2010.01.030](https://doi.org/10.1016/j.jchromb.2010.01.030)
368. Dinh NP, Cam QM, Nguyen AM, Shchukarev A, Irgum K (2009) Functionalization of epoxy-based monoliths for ion exchange chromatography of proteins. *J Sep Sci* 32(15-16):2556–2564. doi:[10.1002/jssc.200900243](https://doi.org/10.1002/jssc.200900243)
369. Du K (2014) Peptide immobilized monolith containing tentacle-type functionalized polymer chains for high-capacity binding of immunoglobulin G. *J Chromatogr A* 1374:164–170. doi:[10.1016/j.chroma.2014.11.060](https://doi.org/10.1016/j.chroma.2014.11.060)
370. Hanora A, Savina I, Plieva FM, Izumrudov VA, Mattiasson B, Galaev IY (2006) Direct capture of plasmid DNA from non-clarified bacterial lysate using polycation-grafted monoliths. *J Biotechnol* 123(3):343–355. doi:[10.1016/j.jbiotec.2005.11.017](https://doi.org/10.1016/j.jbiotec.2005.11.017)
371. Savina IN, Galaev IY, Mattiasson B (2006) Ion-exchange macroporous hydrophilic gel monolith with grafted polymer brushes. *J Mol Recognit* 19(4):313–321. doi:[10.1002/jmr.774](https://doi.org/10.1002/jmr.774)
372. Singh NK, Dsouza RN, Grasselli M, Fernandez-Lahore M (2014) High capacity cryogel-type adsorbents for protein purification. *J Chromatogr A* 1355:143–148. doi:[10.1016/j.chroma.2014.06.008](https://doi.org/10.1016/j.chroma.2014.06.008)
373. Tao SP, Zheng J, Sun Y (2015) Grafting zwitterionic polymer onto cryogel surface enhances protein retention in steric exclusion chromatography on cryogel monolith. *J Chromatogr A* 1389:104–111. doi:[10.1016/j.chroma.2015.02.051](https://doi.org/10.1016/j.chroma.2015.02.051)
374. Gagnon P (2010) Monoliths open the door to key growth sectors. *Bioprocess Int*
375. Martin C, Coyne J, Carta G (2005) Properties and performance of novel high-resolution/highpermeability ion-exchange media for protein chromatography. *J Chromatogr A* 1069(1):43–52
376. Hoshino Y, Kodama T, Okahata Y, Shea KJ (2008) Peptide imprinted polymer nanoparticles: a plastic antibody. *J Am Chem Soc* 130:15242–15243
377. Hoshino Y, Urakami T, Kodama T, Koide H, Oku N, Okahata Y, Shea KJ (2009) Design of synthetic polymer nanoparticles that capture and neutralize a toxic peptide. *Small* 5(13):1562–1568
378. Zeng Z, Hoshino Y, Rodriguez A, Yoo H, Shea KJ (2010) Synthetic polymer nanoparticles with antibody-like affinity for a hydrophilic peptide. *ACS Nano* 4(1):199–204
379. Wong G (2009) Biotech scientists bank on big pharma's biologics push. *Nat Biotechnol* 27(3):293–295. doi:[10.1038/nbt0309-293](https://doi.org/10.1038/nbt0309-293)
380. Dinon F, Salvalaglio M, Gallotta A, Beneduce L, Pengo P, Cavallotti C, Fassina G (2011) Structural refinement of protein A mimetic peptide. *J Mol Recognit* 24(6):1087–1094. doi:[10.1002/jmr.1157](https://doi.org/10.1002/jmr.1157)
381. Thompson AD, Dugan A, Gestwicki JE, Mapp AK (2012) Fine-tuning multiprotein complexes using small molecules. *ACS Chem Biol* 7(8):1311–1320. doi:[10.1021/cb300255p](https://doi.org/10.1021/cb300255p)
382. Ulucan O, Eyrich S, Helms V (2012) Druggability of dynamic protein-protein interfaces. *Curr Pharm Des* 18(30):4599–4606
383. Lee S-H, Hoshino Y, Randall AJ, Zeng Z, Baldi PJ, Doong R-a, Shea KJ (2012) Engineered synthetic polymer nanoparticles as IgG affinity ligands. *J Am Chem Soc* 134(38):15765–15772
384. Hoshino Y, Arata Y, Yonamine Y, Lee S-H, Yamasaki A, Tshura R, et al. (2015) Preparation of nanogel-immobilized porous gel beads for affinity separation of proteins: fusion of nano and micro gel materials. *Polym J* 47(2):220–225. doi:[10.1038/pj.2014.101](https://doi.org/10.1038/pj.2014.101)
385. Box GEP (1976) Science and statistics. *J Am Stat Assoc* 71(356):791–799
386. Whitehead AN, Russell B (1963) *Principia mathematica*, vol III. 2nd edn. Cambridge University Press, New York
387. Hillestad M, Nesvik GO (1994) A comparison of deductive and inductive models for product quality estimation. In: Bonvin D (ed) *IFAC advanced control of chemical processes*. Pergamon, Kyoto, pp 327–332

388. Hurford A (2012) Mechanistic models: what is the value of understanding? Just simple enough: the art of mathematical modelling, vol 2016
389. Burden F, Winkler D (2008) Bayesian regularization of neural networks. *Methods Mol Biol* 458:25–44
390. Insaído FK, Rauscher MA, Smithline SJ, Kaarsholm NC, Feuston BP, Ortigosa AD, et al. (2015) Targeted purification development enabled by computational biophysical modeling. *Biotechnol Prog* 31(1):154–164. doi:[10.1002/btpr.2023](https://doi.org/10.1002/btpr.2023)
391. Kayala MA, Azencott CA, Chen JH, Baldi P (2011) Learning to predict chemical reactions. *J Chem Inf Model* 51(9):2209–2222. doi:[10.1021/ci200207y](https://doi.org/10.1021/ci200207y)
392. Kayala MA, Baldi P (2012) ReactionPredictor: prediction of complex chemical reactions at the mechanistic level using machine learning. *J Chem Inf Model* 52(10):2526–2540. doi:[10.1021/ci3003039](https://doi.org/10.1021/ci3003039)
393. Osberghaus A, Hepbildikler S, Nath S, Haindl M, von Lieres E, Hubbuch J (2012) Optimizing a chromatographic three component separation: a comparison of mechanistic and empiric modeling approaches. *J Chromatogr A* 1237:86–95. doi:[10.1016/j.chroma.2012.03.029](https://doi.org/10.1016/j.chroma.2012.03.029)
394. Hanke AT, Ottens M (2014) Purifying biopharmaceuticals: knowledge-based chromatographic process development. *Trends Biotechnol* 32(4):210–220. doi:[10.1016/j.tibtech.2014.02.001](https://doi.org/10.1016/j.tibtech.2014.02.001)
395. Heinonen J, Kukkonen S, Sainio T (2014) Evolutionary multi-objective optimization based comparison of multi-column chromatographic separation processes for a ternary separation. *J Chromatogr A* 1358:181–191. doi:[10.1016/j.chroma.2014.07.004](https://doi.org/10.1016/j.chroma.2014.07.004)
396. Korifi R, Le Dreau Y, Dupuy N (2014) Comparative study of the alignment method on experimental and simulated chromatographic data. *J Sep Sci* 37(22):3276–3291. doi:[10.1002/jssc.201400700](https://doi.org/10.1002/jssc.201400700)
397. Marks RE, Schnabl H (1999) Genetic algorithms and neural networks: a comparison based on the repeated prisoners dilemma. In: Brenner T (ed) *Computational techniques for modelling learning in economics*, vol 11. Springer, pp 197–219
398. Brooks CA, Cramer SM (1992) Steric mass-action ion exchange: displacement profiles and induced salt gradients. *AIChE J* 38(12):1969–1978. doi:[10.1002/aic.690381212](https://doi.org/10.1002/aic.690381212)
399. Jungbauer A, Carta G (2010) *Protein chromatography: process development and scale-up*, 1st edn. Wiley
400. Guiochon GF, Felinger A, Shirazi DG (2006) *Fundamentals of preparative and nonlinear chromatography*, 2nd edn. Academic
401. Guiochon GL, Lin B (2003) *Modeling for preparative chromatography*, 1st edn. Academic
402. Lopez ZK, Tejeda A, Montesinos RM, Magana I, Guzman R (1997) Modeling column regeneration effects on ion-exchange chromatography. *J Chromatogr A* 791(1-2):99–107
403. Karlsson D, Jakobsson N, Axelsson A, Nilsson B (2004) Model-based optimization of a preparative ion-exchange step for antibody purification. *J Chromatogr A* 1055(1-2):29–39
404. Edwards-Parton S, Thornhill NF, Bracewell DG, Liddell JM, Titchener-Hooker NJ (2008) Principal component score modeling for the rapid description of chromatographic separations. *Biotechnol Prog* 24(1):202–208. doi:[10.1021/bp070240j](https://doi.org/10.1021/bp070240j)
405. Susanto A, Herrmann T, von Lieres E, Hubbuch J (2007) Investigation of pore diffusion hindrance of monoclonal antibody in hydrophobic interaction chromatography using confocal laser scanning microscopy. *J Chromatogr A* 1149(2):178–188. doi:[10.1016/j.chroma.2007.03.002](https://doi.org/10.1016/j.chroma.2007.03.002)
406. Ishihara T, Kadota T, Endo N, Yamamoto S (2006) Optimization of elution salt concentration in stepwise elution of protein chromatography using linear gradient elution data. Reducing residual protein A by cation-exchange chromatography in monoclonal antibody purification. *J Chromatogr A* 1114(1):97–101. doi:[10.1016/j.chroma.2006.02.042](https://doi.org/10.1016/j.chroma.2006.02.042)
407. Ishihara T, Kadota T, Yamamoto S (2007) Application of a chromatography model with linear gradient elution experimental data to the rapid scale-up in ion-exchange process chromatography of proteins. *J Chromatogr A* 1162(1):34–40. doi:[10.1016/j.chroma.2007.03.016](https://doi.org/10.1016/j.chroma.2007.03.016)

408. Ishihara T, Yamamoto S (2005) Optimization of monoclonal antibody purification by ion-exchange chromatography. Application of simple methods with linear gradient elution experimental data. *J Chromatogr A* 1069(1):99–106
409. Muller-Spath T, Strohhlein G, Aumann L, Kormmann H, Valax P, Delegrange L, et al. (2011) Model simulation and experimental verification of a cation-exchange IgG capture step in batch and continuous chromatography. *J Chromatogr A* 1218(31):5195–5204. doi:[10.1016/j.chroma.2011.05.103](https://doi.org/10.1016/j.chroma.2011.05.103)
410. Osberghaus A, Hepbildikler S, Nath S, Haindl M, von Lieres E, Hubbuch J (2012) Determination of parameters for the steric mass action model—a comparison between two approaches. *J Chromatogr A* 1233:54–65. doi:[10.1016/j.chroma.2012.02.004](https://doi.org/10.1016/j.chroma.2012.02.004)
411. Westerberg K, Broberg-Hansen E, Sejergaard L, Nilsson B (2013) Model-based risk analysis of coupled process steps. *Biotechnol Bioeng* 110(9):2462–2470. doi:[10.1002/bit.24909](https://doi.org/10.1002/bit.24909)
412. Lapelosa M, Patapoff TW, Zarraga IE (2015) Modeling of protein-anion exchange resin interaction for the human growth hormone charge variants. *Biophys Chem* 207:1–6. doi:[10.1016/j.bpc.2015.07.004](https://doi.org/10.1016/j.bpc.2015.07.004)
413. Kluters S, Wittkopp F, Johnck M, Frech C (2015) Application of linear pH gradients for the modeling of ion exchange chromatography: separation of monoclonal antibody monomer from aggregates. *J Sep Sci*. doi:[10.1002/jssc.201500994](https://doi.org/10.1002/jssc.201500994)
414. Mazumder J, Zhu J, Bassi AS, Ray AK (2009) Modeling and simulation of liquid-solid circulating fluidized bed ion exchange system for continuous protein recovery. *Biotechnol Bioeng* 104(1):111–126. doi:[10.1002/bit.22368](https://doi.org/10.1002/bit.22368)
415. Xenopoulos A (2015) A new, integrated, continuous purification process template for monoclonal antibodies: process modeling and cost of goods studies. *J Biotechnol* 213:42–53. doi:[10.1016/j.jbiotec.2015.04.020](https://doi.org/10.1016/j.jbiotec.2015.04.020)
416. Teeters M, Benner T, Bezila D, Shen H, Velayudhan A, Alred P (2009) Predictive chromatographic simulations for the optimization of recovery and aggregate clearance during the capture of monoclonal antibodies. *J Chromatogr A* 1216(33):6134–6140. doi:[10.1016/j.chroma.2009.06.066](https://doi.org/10.1016/j.chroma.2009.06.066)
417. Ladiwala A, Rege K, Breneman CM, Cramer SM (2005) A priori prediction of adsorption isotherm parameters and chromatographic behavior in ion-exchange systems. *Proc Natl Acad Sci U S A* 102(33):11710–11715. doi:[10.1073/pnas.0408769102](https://doi.org/10.1073/pnas.0408769102)
418. Guelat B, Strohhlein G, Lattuada M, Delegrange L, Valax P, Morbidelli M (2012) Simulation model for overloaded monoclonal antibody variants separations in ion-exchange chromatography. *J Chromatogr A* 1253:32–43. doi:[10.1016/j.chroma.2012.06.081](https://doi.org/10.1016/j.chroma.2012.06.081)
419. Sarangapani PS, Hudson SD, Jones RL, Douglas JF, Pathak JA (2015) Critical examination of the colloidal particle model of globular proteins. *Biophys J* 108(3):724–737. doi:[10.1016/j.bpj.2014.11.3483](https://doi.org/10.1016/j.bpj.2014.11.3483)
420. Lang KM, Kittelmann J, Durr C, Osberghaus A, Hubbuch J (2015) A comprehensive molecular dynamics approach to protein retention modeling in ion exchange chromatography. *J Chromatogr A* 1381:184–193. doi:[10.1016/j.chroma.2015.01.018](https://doi.org/10.1016/j.chroma.2015.01.018)
421. Paloni M, Cavallotti C (2015) Molecular modeling of the affinity chromatography of monoclonal antibodies. *Methods Mol Biol* 1286:321–335. doi:[10.1007/978-1-4939-2447-9_25](https://doi.org/10.1007/978-1-4939-2447-9_25)
422. Kisley L, Chen J, Mansur AP, Dominguez-Medina S, Kulla E, Kang MK, et al. (2014) High ionic strength narrows the population of sites participating in protein ion-exchange adsorption: a single-molecule study. *J Chromatogr A* 1343:135–142. doi:[10.1016/j.chroma.2014.03.075](https://doi.org/10.1016/j.chroma.2014.03.075)
423. Kisley L, Poongavanam M-V, Kourentzi K, Willson RC, Landes CF (2015) pH-dependence of single-protein adsorption and diffusion at a liquid chromatographic interface. *J Separation Sci*. doi:[10.1002/jssc.201500809](https://doi.org/10.1002/jssc.201500809)
424. Marek W, Muca R, Wos S, Piatkowski W, Antos D (2013) Isolation of monoclonal antibody from a Chinese hamster ovary supernatant. II: Dynamics of the integrated separation on ion exchange and hydrophobic interaction chromatography media. *J Chromatogr A* 1305:64–75. doi:[10.1016/j.chroma.2013.06.076](https://doi.org/10.1016/j.chroma.2013.06.076)

425. Baumann P, Hahn T, Hubbuch J (2015) High-throughput micro-scale cultivations and chromatography modeling: powerful tools for integrated process development. *Biotechnol Bioeng* 112(10):2123–2133. doi:[10.1002/bit.25630](https://doi.org/10.1002/bit.25630)
426. Huuk TC, Hahn T, Osbergerhaus A, Hubbuch J (2014) Model-based integrated optimization and evaluation of a multi-step ion exchange chromatography. *Separation and Purification Technology* 136:207–222. doi:[10.1016/j.seppur.2014.09.012](https://doi.org/10.1016/j.seppur.2014.09.012)
427. Sharma C, Malhotra D, Rathore AS (2011) Review of computational fluid dynamics applications in biotechnology processes. *Biotechnol Prog* 27(6):1497–1510
428. Joshi V, Shivach T, Kumar V, Yadav N, Rathore A (2014) Avoiding antibody aggregation during processing: establishing hold times. *Biotechnol J* 9(9):1195–1205. doi:[10.1002/biot.201400052](https://doi.org/10.1002/biot.201400052)
429. Lapelosa M, Patapoff TW, Zarraga IE (2014) Molecular simulations of the pairwise interaction of monoclonal antibodies. *J Phys Chem B* 118(46):13132–13141. doi:[10.1021/jp508729z](https://doi.org/10.1021/jp508729z)
430. Helling C, Borrmann C, Strube J (2012) Optimal integration of directly combined hydrophobic interaction and ion exchange chromatography purification processes. *Chem Eng Technol* 35(10):1786–1796. doi:[10.1002/ceat.201200043](https://doi.org/10.1002/ceat.201200043)
431. Buyel JF, Woo JA, Cramer SM, Fischer R (2013) The use of quantitative structure–activity relationship models to develop optimized processes for the removal of tobacco host cell proteins during biopharmaceutical production. *J Chromatogr A* 1322:18–28. doi:[10.1016/j.chroma.2013.10.076](https://doi.org/10.1016/j.chroma.2013.10.076)
432. Kruhlak NL, Benz RD, Zhou H, Colatsky TJ (2012) (Q)SAR modeling and safety assessment in regulatory review. *Clin Pharmacol Ther* 91(3):529–534. doi:[10.1038/clpt.2011.300](https://doi.org/10.1038/clpt.2011.300)